A Coarse-to-Fine Semantic Conflict Detection System for Ex-Ante Davao City Ordinances Using Information Retrieval and Natural Language Inference

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Chapter 1

Introduction

1.1 Background of the Study

A functional legal system requires strict consistency so that citizens can clearly understand their rights and obligations [43]. However, the sheer volume of modern legislation has grown so rapidly that human practitioners can no longer track every active rule [123]. As governing bodies continuously pass new regulations to address emerging issues, they frequently enact laws that inadvertently contradict older ones [72]. Because the total number of documents is overwhelming, identifying these hidden conflicts through manual review is computationally impossible for legal staff. This issue is particularly severe in local government units, where historical ordinances are often scattered across fragmented files rather than maintained in a structured digital database [39].

Within the Philippine legal framework, local governments like Davao City operate under delegated authority and can only enact laws permitted by the national government [50]. Due to this structural limitation, the Magtajas v. Pryce Doctrine strictly prohibits any local municipal ordinance from contradicting established national statutes [140]. Consequently, in the local context, city councils must rigorously audit every newly proposed draft ordinance to ensure it logically aligns with existing laws before final approval. If reviewers overlook a normative conflict, such as a local ordinance prohibiting an activity that a national law explicitly permits, the resulting local legislation is legally void. Passing invalid laws creates severe jurisdictional disputes, wastes administrative resources, and ultimately compromises the constitutional rights of the public.

To address these legal risks, the legislative process must utilize an ex-ante evaluation: a term derived from Latin meaning 'before the event.' Unlike an 'ex-post' review, which happens after a law is already in effect and often after legal damage has occurred, an ex-ante audit acts as a first line of defense. By checking a proposed ordinance for conflicts while it is still in the drafting phase, the Sangguniang Panlungsod can proactively align local rules with national statutes. This preventative approach is consistent with modern Philippine standards like Republic Act No. 11032 or the Ease of Doing Business and Efficient Government Service Delivery Act of 2018, which encourages government agencies to perform regulatory assessments to ensure that new rules do not create unnecessary or conflicting burdens on the public.

The decentralized drafting process of local ordinances further complicates manual review because the resulting texts are often highly unstructured [126]. City councils frequently splice and amend different drafts, producing verbose documents that blend formal legal English with regional dialects like Cebuano and administrative ”Taglish” [40]. Furthermore, because municipal archives heavily rely on digitized PDF scans, the text suffers from severe Optical Character Recognition (OCR) errors. While OCR is typically 99 percent accurate for clean English text, studies show accuracy can drop to 78 percent for low-resource languages, with Cebuano specifically seeing word error rates around 6.9 percent [107]. These errors cause an “error cascade” that shatters standard vocabulary, making it difficult for traditional search engines to semantically differentiate between a statutory obligation and a strict prohibition.

In light of these challenges, the proponents seek to develop a specialized application capable of retrieving relevant legal documents and comparing them against proposed ordinances to detect potential conflicts.

1.2 Problem Statement

This study is conducted to investigate the use of a multi-stage architecture to automate the manual auditing of proposed local legislation in Davao City.

Specifically, the study aims to answer the following research questions:

- What are the specific linguistic parameters and structural markers that define a ”semantic conflict” between a proposed local ordinance and existing local or national laws?
- How can a Coarse-to-Fine system be designed to efficiently retrieve relevant national and local statutes before performing deep logical inference on constrained LGU hardware?
- Which combination of a hybrid Information Retrieval (IR) algorithms and Natural Language Inference (NLI) models provides the highest accuracy when processing the specific hierarchy of Philippine legal texts?
- What is the empirical performance of the proposed pipeline when evaluated against a manually curated Ground Truth dataset of local and national conflict pairs?
- How usable and viable is the final system interface for key stakeholders within the Davao City Sangguniang Panlungsod?

1.3 Objectives of the Study

The primary goal of this research is to develop a computationally efficient, two-stage semantic conflict detection system for proposed Davao City ordinances.

Specifically, this study aims to:

- Identify the specific linguistic characteristics and structural constraints of Davao City's legislative archive that contribute to semantic ambiguity and retrieval failure;
- Design a Coarse-to-Fine computational pipeline that decouples document retrieval from logical inference to optimize resource usage for local deployment;
- Implement a hybrid Information Retrieval (IR) retrieval algorithm and Natural Language Inference (NLI) system specifically calibrated for the Philippine legal hierarchy;
- Evaluate the system's reliability by comparing its automated detections against a verified set of known legal conflicts to measure how accurately it identifies contradictions; and
- Develop a user-friendly web interface for the Davao City Sangguniang Panlungsod that primarily functions as accepting uploading drafts and providing real-time conflict alerts.

1.4 Significance of the Study

The Davao City Sangguniang Panlungsod stands as the primary beneficiary of this automated consistency checking system. By transitioning from manual review to computational auditing, the tool saves legislative staff hundreds of hours of reading time. This technological intervention actively prevents the city from enacting logically flawed laws that invite future litigation or administrative deadlocks. It empowers the local government to draft regulatory measures with a significantly higher degree of legal certainty and operational speed. Ultimately, the system modernizes the legislative lifecycle by embedding rigorous logical checks directly into the initial drafting phase.

This study also provides substantial value to legal researchers and the local Natural Language Processing community. It establishes an empirical baseline for processing messy, localized Philippine legal corpora that frequently contain "Taglish" and archaic Spanish loanwords. The research demonstrates that efficient, edge-deployable encoder models can successfully parse OCR-degraded documents without relying on expensive, cloud-based supercomputers. Beyond academic and governmental applications, the general public benefits directly through the consistent implementation of coherent public policy. The tool acts as a procedural safety net, protecting citizens from the legal traps and frustrations of contradictory municipal obligations.

1.5 Scope and Limitations

The scope of this study focuses on developing a semantic conflict detection system to evaluate draft ordinances from Davao City against a static database of 25,432 national statutes. This national statutory knowledge base is compiled from the Lawphil Project, encompassing congressional enactments such as Republic Acts and Batas Pambansa alongside major presidential issuances like Executive Orders and Presidential Decrees. To keep the computational workload manageable on standard local computers, judicial case law from Supreme Court decisions and departmental administrative circulars are excluded from the archive. The system focuses strictly on statutory preemption, auditing local legislative proposals against enacted national laws before municipal enactment. By bounding the national archive to verified statutory texts, the system establishes a dependable baseline for automated legal consistency checks.

A significant limitation is the incompleteness of the municipal archive. For ordinances enacted earlier than 2021, the collection consists primarily of "Landmark Ordinances" rather than the full set of laws for those years. While the proponents utilized full access to post-2021 records via the LISSP database, gaps still exist in the numerical sequence of the documents, resulting in a final knowledge base of approximately 1,660 ordinances. These records represent the most complete dataset currently available after exhausting all retrieval efforts through public repositories and direct coordination with the Sangguniang Panlungsod; consequently, the system's ability to detect conflicts is limited to the records present in this compiled database [127].

Furthermore, the system is restricted to English-language texts to avoid the complexities of translation and regional dialects. This choice is necessary because local municipal records are currently under-represented in legal AI research; while national laws are well-studied in central databases, local ordinances remain neglected because they are unorganized and often "messy" [127, 8, 94]. This has led to a lack of a computational footprint, meaning there are no high-quality, standardized datasets for local ordinances that an AI can follow to learn legal reasoning [17]. Additionally, focusing only on English avoids "semantic drift," where the specific legal meaning of a rule may get lost or accidentally changed when moved into a local dialect [75]. The database is compiled from existing public government repositories, meaning the tool will not perform live web scraping during its operation.

Operationally, the system uses a Natural Language Inference (NLI) framework to perform a single-premise comparison, evaluating a draft ordinance against a single national provision at a time. While the system uses Asymmetric Span-Level Pairing to compare a multi-sentence block from a local draft against a specific national rule, it does not perform "multi-hop" reasoning. This means it cannot identify conflicts that only appear when combining two or more separate national laws, as such complex processing would exceed the hardware limits of a standard office computer. The system classifies these relationships as Entailment, Neutral, or Contradiction, with a specific optimization for flagging contradictions.

To ensure transparency, the study uses a "white box" approach to Explainable AI (XAI), providing mathematical data like confidence scores and attention weights instead of generated text summaries. To conserve resources and ensure practical deployment, the framework deliberately excludes generative Large Language Models (LLMs); it cannot write, rewrite, or automatically correct legal text. Finally, all development and testing are strictly limited to a single consumer-grade device. This ensures the final product is a practical diagnostic tool that can be deployed by a standard Local Government Unit (LGU) without the need for expensive, enterprise-grade cloud computing resources.

1.6 Definition of Terms

Coarse-to-Fine Architecture. A two-step computer process. First, the system quickly searches through thousands of documents to find "candidates" (coarse), then it carefully analyzes those specific candidates for logical conflicts (fine).

Contradiction. A relationship in NLI where the second sentence is false if the first sentence is true, indicating a logical conflict.

Davao City Ordinances. These are written local laws created by the Davao City government that are intended to be permanent and enforceable.

Entailment. A relationship in NLI where the first sentence (premise) proves that the second sentence (hypothesis) is true.

Ex-Ante. Derived from Latin meaning "before the event". Operationally, this refers to the evaluation or checking of a proposed city ordinance before it is officially signed into law.

Information Retrieval (IR). The process of searching for and finding relevant documents (like specific laws) within a massive collection of text.

Local Government Code of 1991 (RA 7160). The national law that gives local governments the power to create their own ordinances for the "General Welfare" of their citizens.

Magtajas v. Pryce Doctrine. A legal rule from the Supreme Court stating that local ordinances are inferior to national laws and cannot go against any Act of Congress.

Natural Language Inference (NLI). A task where the computer compares two sentences—a "premise" and a "hypothesis"—to see if they agree or disagree with each other.

Neutral. A relationship in NLI where the two sentences are unrelated and do not affect each other's truth.

OCR Noise. Text errors (like misspellings or broken words) that happen when physical paper documents are scanned into digital files.

Sangguniang Panlungsod (SP). The local legislative body or "City Council" of Davao City, responsible for creating and approving local laws.

Semantic Conflict. A logical clash where two different laws give opposite instructions. For example, a conflict occurs if one law permits an activity while another strictly prohibits it.

Chapter 2

Literature Review

2.1 Legal Informatics Conflict Detection

For a legal system to work properly, its laws must be consistent. This means that citizens and the government should be able to understand their rights and duties without running into laws that give opposite instructions [43]. Semantic Conflict Detection—also called the detection of statutory or normative conflicts in recent studies—is the computer's task of finding these logical clashes between different legal rules [83, 3]. Specifically, it identifies cases where two or more valid laws give opposite orders to the same person under the same situation, such as when one law says you "must" do an act while another law "forbids" it [3, 66]. It is also important to distinguish between a new law that is meant to change an old one (a valid amendment) and a law that accidentally goes against an old one without saying so (a legal error) [61]. In the world of computer science and law, the move from manual reading to automated conflict detection is necessary because there are simply too many laws for humans to track. Ruhl and Katz [123] found that the number of new regulations has grown so fast that human experts can no longer keep up with all of them. As Shao et al. [131] points out, while lawyers are good at resolving known conflicts, it is physically impossible for them to cross-reference thousands of old laws at the same time to find hidden contradictions. Therefore, Semantic Conflict Detection acts like a "health check" for laws, using math to map out where different rules overlap so that problems can be fixed before they lead to court cases or government confusion. To understand how a computer can detect these conflicts, we must look at the factors that make it difficult. This section will examine the challenges of "legal inflation" (the growing number of laws), the limits of manual auditing, and the confusion caused by legal language. This section will also look at conflict detection research within the Philippines, specifically focusing on the hurdles found in local government archives.

2.1.1 The Unstructured Text Problem in Legal Corpora

Dealing with "unstructured" legal text—meaning text that isn't organized into a neat database—is not just a data problem; it is a failure in how laws are maintained. To build a system that works, it must be able to handle three specific issues: the massive amount of text, the slow pace of manual review, and the confusing language used in drafting. These three factors are why the field is moving away from human oversight and toward deep-learning tools for searching and checking laws.

2.1.1.1 Legal Inflation

We refer to ””Legal inflation” as to how the number of laws and their connections keep growing. Katz et al. argue that the sheer volume of laws has made it too difficult for humans to check for consistency manually [72]. As new rules are made to keep up with technology, they create ”regulatory debt,” where new laws might conflict with older ones without anyone noticing. This is a big problem for local governments, because unlike national laws, local ordinances are often not organized or scanned properly. Dias et al. [39] notes that turning these laws into machine-readable data is hard because they are often scattered across different files, from formal resolutions to quick executive orders.

The ”search space” for finding conflicts is therefore a messy collection of different texts rather than a neat database. Zhong et al. [169] describe this as a ”dynamic consistency problem” where facts and rules are always changing. Because this data is not organized, government agencies cannot be proactive. They only realize there is a conflict after something goes wrong. This is why the field has moved from old AI systems that use simple rules to new ”neural models” that can learn from data. These systems are needed to process huge collections of laws that are too big for the human brain to handle [24].

2.1.1.2 The Manual Review Bottleneck

Scanning documents into a computer is no longer the main challenge; the real ”bottleneck” is now checking if those documents agree with each other. As legislative bodies pass laws faster, they create a massive pile of unorganized data. The main difficulty is telling the difference between a ”good” overlap (where a law is meant to cover multiple areas) and a ”bad” conflict (where a law is written poorly and contradicts another). Recent surveys show that standard search engines usually fail to understand these nuances [10]. As a result, many conflicts stay hidden until someone gets sued. This moves the burden of finding conflicts from the people writing the laws to the judges in court [145].

2.1.1.3 Semantic Ambiguity

Old computer search systems look for exact word matches. However, this fails to catch the real meaning of human language. Legal language is like a specialized code. Words like ”shall,” ”must,” or ”may” have very strict meanings that keyword searches often mix up. Chalkidis et al. [24] show that this ”ambiguity” makes exact-match techniques fail at complex legal tasks. For example, a law banning ”motor vehicles” might conflict with a law allowing ”electric scooters.” A keyword search for ”vehicle” will not find the problem if the words do not match exactly.

Legal consistency often depends on ”implied” rules—things the law suggests but doesn't say directly. Rigano et al. [119] argue that finding these contradictions requires a system that can understand logic, not just compare strings of text. ”Transformer” AI models are an example of a solution that can solve this gap by creating mathematical representations (vectors) that capture deep meaning [168], which shall be discussed in later sections. However, there is a trade-off. Sovrano et al. [134] argue that while these models are very accurate, their "black box" nature is a problem because they don't explain *why* they found a conflict. In a legal setting, the reasoning is just as important as the detection [129].

2.1.2 Semantic Conflict Detection in the Local Context

While researchers around the world are building conflict detection tools using benchmarks like LexGLUE, these tools are still in the early stages in less prominent countries like the Philippines [24]. Since the Local Government Code of 1991, cities like Davao have created thousands of ordinances that are not well-organized [50]. Although the math for detecting conflicts exists in general AI research, it has rarely been applied to Philippine laws.

2.1.2.1 Legal-Related Semantic Conflict Detection in the Philippines

When Philippine AI research does focus on law, it is usually about organizing data rather than deep reasoning. Early work by Virtucio et al. [153] used basic classifiers to predict Supreme Court decisions, while Sagum et al. [125] used AI just to summarize national court rulings.

More recently, studies have started using "Transformer" models, a method that shall be discussed in later sections, but they are still used mostly for search. Roque et al. showed in 2024 [121] that "dense embedding" models are much better than keyword searches at finding specific tax laws. However, these embedding models are basically just advanced search engines. They can find two laws about "business taxes" because they are topically similar, but they lack the logic layers needed to tell if one law actually contradicts the other.

Looking at these trends, there is a clear gap in the research. The Philippine AI community has successfully used computers to categorize, summarize, and search national laws. They have also proven that NLI models can understand logical contradictions in local dialects for fact-checking [153, 125, 121, 105]. However, actually using these advanced NLI models to find conflicting rules in the messy, unorganized ordinances of a local city like Davao remains almost entirely unexplored.

2.1.2.2 The Landscape of Logic and Conflict Detection in the Philippines

In the Philippine AI community, the tools needed for conflict detection, specifically Natural Language Inference (NLI), are already being used, but mostly for other tasks. Recent local studies have used NLI to find logical contradictions in things like online news and misinformation. For example, Nuñez et al. proved that these AI models can handle the logic needed for Tagalog and "Taglish" (a mix of Tagalog and English) [105].

However, when researchers try to analyze local government records, they usually go back to traditional methods. Most document analysis in the Philippines still relies on "Boolean" search (keyword matching) and simple rule-based AI. As recent Philippine studies show, these old systems are ”lexically fragile” [99]. They fail if two laws use different words for the same thing (like ”fireworks” vs. ”pyrotechnics”) and they break down when there are typos or scanning errors (OCR noise) in the digital files. Trying to use manual rules to find clashes in thousands of messy city ordinances is simply too hard for a computer or a person to handle alone.

2.1.2.3 Gaps in Local Legislative Conflict Detection

While the Philippine AI community has successfully automated the search and summarization of national laws, a significant gap remains in using deep logic models (NLI) to audit local municipal ordinances. Research suggests this gap exists because local legislation is far more difficult for a computer to process than national statutes. This section outlines the specific reasons why this area of research remains largely unexplored.

Dataset Gap. The primary barrier to research is the lack of “ground truth” datasets for Philippine ordinances. Unlike national laws, which are well-documented, local ordinances are considered “low-resource” data [115]. Standard AI models are trained on massive Western datasets like LexGLUE, but there are no equivalent datasets that provide annotated pairs of conflicting Philippine ordinances to train an NLI model [8, 101]. Without a standardized benchmark to measure progress, local researchers have found it difficult to build systems that move beyond simple keyword matching to actual logical reasoning [8].

Resource Gap. Deductive logic models like Cross-Encoders require significant computing power that many Local Government Units (LGUs) simply do not have. Research on Philippine public administration shows that AI adoption is still in its early stages due to limited funding, a lack of skilled IT personnel, and outdated computer hardware [44]. Studies have shown that in some municipal councils, only 20 percent of legislative staff have received formal ICT training, and most still rely heavily on printed documents rather than structured databases [44, 8]. This lack of digital infrastructure makes it difficult for researchers to justify developing complex AI models that would be too “heavy” to run on actual LGU workstations [63, 8].

Linguistic Gap. Local ordinances are linguistically more complex than national laws because they frequently blend English with regional dialects like Cebuano and administrative “Taglish” [40]. Most Natural Language Processing (NLP) tools are designed for standard English and often break down when they encounter local terms or Spanish legal loanwords common in the Philippines [94]. Research on Philippine social media AI has proven that NLI can handle local dialects for simple fact-checking, but applying this same logic to the dense, multi-generational language of local ordinances is a much harder task that requires specialized domain knowledge that has not yet been fully researched [27, 53].

Trust Gap. There is also a trust gap within the legal profession. As noted by Supreme Court Senior Associate Justice Marvic Leonen, the legal community has not yet fully embraced AI because current models are often “black boxes” that do not explain their reasoning [81]. Lawyers are cautious about using automated tools because inaccuracies could lead to legal errors or the passage of void laws [5]. Consequently, research has focused more on simple organization and retrieval tasks—where the computer acts as a simple search engine—rather than deep reasoning tasks where the computer makes a judgment about a legal conflict [11].

2.2 The Philippine Legislative Domain

The Philippine legal system is a unitary state. This means that Local Government Units (LGUs) only have powers that are given or delegated to them by the national government [50]. This setup creates a strict top-down hierarchy that limits how much freedom cities like Davao City have when making laws. Any computer system built to check these laws must understand this specific legal environment. This section explains the barriers in Philippine municipal law. First, it looks at the strict rules that make local ordinances follow national laws. Second, it examines the difficulties and language issues in the local drafting process. Finally, it reviews the current methods used to check for consistency, which shows why we need to move toward automated auditing.

2.2.1 Hierarchical Constraints

The legal system in the Philippines depends on the rule that national laws always override local ones. To understand the limits of writing local laws, we must look at the specific rules that put ordinances below national statutes. This part describes the centralized framework of the government, the powers given by the Local Government Code, and the tests used by the Supreme Court to cancel local laws that conflict with national ones.

Unitary Principle in the Philippines. Research on Philippine political law shows a rigid pyramid of power: the 1987 Constitution is at the very top, followed by national laws (Republic Acts) passed by Congress, international treaties, executive orders, and finally, local ordinances [50]. Under this setup, no LGU can ignore or contradict an Act of Congress. The rule of "national preemption" is a major constraint; when a national law already covers an entire subject—such as firearms or immigration—local governments are not allowed to make different rules [50].

The Local Government Code of 1991. The main law for LGUs is the Local Government Code (LGC) of 1991. Section 16, known as the General Welfare Clause, gives LGUs the power to create ordinances for the safety and success of their citizens, but only if these rules stay "within the framework of the law" [50]. Since Davao City is a "Highly Urbanized City" (HUC), it is independent of any province and its ordinances are not checked by a provincial board. This makes the city more responsible for checking its own laws for conflicts before they are passed [50].

Magtajas vs. Pryce Doctrine. Perhaps the most important test for whether an ordinance is valid comes from the Supreme Court case *Magtajas v. Pryce Properties Corporation, Inc.* The Court listed six requirements that a local law must meet to be valid [140]:

- It must not contravene the constitution or any statute.
- It must not be unfair or oppressive.

- It must not be partial or discriminatory.
- It must not prohibit but may regulate trade.
- It must be general and match public policy.
- It must not be unreasonable.

Local ordinances are considered lower in rank than state laws. Therefore, LGUs cannot try to regulate activities that the national government has already permitted or regulated [50].

2.2.2 Procedural and Linguistic Friction

Besides the hierarchy of laws, the actual way local laws are written and passed creates a lot of inconsistencies in language and format. National laws are usually written by specialized committees using standard legal terms. However, local ordinances go through a localized process that often breaks up formal language. To detect conflicts accurately, the computer system must be adjusted to handle the "noise" and confusion caused by using two languages in the Davao City Council (Sangguniang Panlungsod).

Ordinance Creation Process. In Davao City, the life of a proposed law is guided by the Internal Rules of Procedure, found in Resolution No. 001-22 [126]. The drafting process is decentralized, meaning different council members or their staff write their own laws rather than having one main office do all the writing [50, 126]. Because of this, the style and words used across different ordinances can be very inconsistent. For example, a transportation law might use completely different terms than an environmental law, making it hard for computer models to see how they relate.

Textual Fragmentation and the Three-Reading Rule. According to the law, no ordinance can be passed unless it goes through three separate readings [50, 126]. This process is democratic, but it often breaks the text apart. During the First Reading, the draft is sent to a committee. The Second Reading is where the most intense checking happens: the text is put into a period of interpolation, debates, and amendments [126]. This constant changing of the text can lead to disjointed rules. For instance, a new exception added in one section might accidentally contradict a general rule in another section, which may contain a hidden conflict that stays buried in the text after final approval on the Third Reading.

Bilingual Mandates and the Risks of Translation. A unique challenge within the Philippines is the legal requirement to make laws accessible in two languages. Under Section 469 of the Local Government Code, the Secretary must translate all approved ordinances into the local dialect (e.g. Cebuano for Davao City) [50, 126, 40]. However, this creates a legal risk. The Administrative Code of 1987 (Executive Order No. 292) says that if the English and local versions disagree, the English version is the one that counts [117]. Consequently, if a translation error happens, a citizen following the local language could be punished for breaking the English version of the law. This relates to the Supreme Court ruling in *Tañada v. Tuvera*, which says laws are void without proper public notice [138]. A mistranslated law is essentially "false notice." Despite new efforts like the *Batas sa Sariling Wika Act* to mandate more regional translations [110], there is still no technical system to check if these translations are actually correct. As researchers have noted, the small details in local languages are often ignored, leading to "semantic drifts" where the original legal meaning in English is lost in regional translation [94].

2.2.3 Ex-Ante in Legislation

The term "ex-ante" comes from Latin and means "before the event." In the context of lawmaking, it refers to the process of evaluating a proposed law or regulation *before* it is officially enacted [42]. This part looks at how the Philippines uses ex-ante reviews to prevent bad laws from being passed and how this study fits into that process.

Ex-Ante in the Philippine Context. In the Philippines, ex-ante review has become more important since the passage of Republic Act No. 11032, or the "Ease of Doing Business and Efficient Government Service Delivery Act of 2018" [51]. Under this law, the Anti-Red Tape Authority (ARTA) and the Department of the Interior and Local Government (DILG) mandates that government agencies and local government units (LGUs) conduct a Regulatory Impact Assessment (RIA) for any proposed regulation that might significantly affect the public [51]. The goal is to ensure that new rules have a solid legal basis and to avoid creating overlapping or conflicting regulations that could make it harder for citizens and businesses to operate.

Ex-Ante vs Ex-Post. The main difference between the two is timing and purpose. Ex-ante evaluation is predictive; it looks forward to stop problems before they happen. It acts as a "first line of defense" against poor lawmaking [42]. In contrast, ex-post evaluation happens *after* the law is already active, focusing on how the law has actually performed or whether it has caused damage [147]. While ex-post reviews are useful for fixing existing mistakes, they are often difficult because it is much harder to revoke a law once it is already being enforced [147]. Ex-ante detection is considered more efficient because it saves the government from the cost and trouble of litigation later on.

Conflict Detection in Local Ex-Ante Legislation. Using automated tools for the ex-ante Conflict Detection can help the local legislative branch identify "normative collisions" during the drafting phase. Because legislative bodies now produce a massive amount of text, manual review often fails to catch hidden clashes between a new ordinance and older statutes [72]. By treating Conflict Detection as a diagnostic triage tool, the system can flag potential contradictions while the ordinance is still just a draft. This ensures that the final law is consistent with the national legal hierarchy, protecting the city from passing ordinances that are legally void from the start [140].

2.2.4 Consistency Checking in Legislative Processes.

Because local governments are lower in rank and their drafting process is fragmented, checking for consistency is a critical duty. However, checking ordinances for conflicts is not as simple as matching titles, as it requires understanding complex legal meaning and identifying subtle oversteps in authority. This part examines current methods of consistency checking, highlighting how confusing language and the reliance on simple tracking systems slow down the legislative review process.

Legalese Linguistic Ambiguity. Discourse analysis of Philippine laws reveals many ambiguities in how words and sentences are used [139]. Specifically, the word ”shall” is often confusing because it can be interpreted as either a mandatory command or a simple suggestion depending on the sentence. Studies show that these language issues make it hard to interpret laws correctly, even when they follow standard formats [14].

Legislative Gaps and Implied Preemption. The Philippine legal framework recognizes that conflicts between laws are rarely obvious [33]. A local law almost never says, ”This law contradicts a national law.” Instead, conflicts often appear as ”implied preemption.” This happens when a national law already covers a subject, but a local government tries to add extra hurdles—like requiring a local fee to do something already allowed by the state [139]. In these cases, the Supreme Court rules that the local ordinance is unlawfully trying to ”amend” a national law under the guise of local regulation [139]. For a human reviewer or a basic keyword search, this is very hard to find because the laws might use different words. The conflict is in the actual effect of the rules, which requires deeper semantic checking to identify.

Local Context. The Davao City Sangguniang Panlungsod passed nearly 400 ordinances and over 2,000 resolutions in a single three-year term [108]. To manage this volume, the city uses digital platforms: the Resolution Ordinance Tracking System (ROTS) is used for encoding and routing documents, while the Legislative Information Support System Program (LISSP) lets council members track items and read ordinances on their devices [27]. However, while these systems help with logistics, the actual checking of content for ”dueling provisions” is still done manually. Furthermore, scanned PDFs often have OCR noise, which makes keyword searches fail [86].

2.3 Philippine Digital Legislative Archives

Building a functional Semantic Conflict Detection system requires access to comprehensive and machine-readable legal corpora. In the Philippines, the digital landscape of legislative data is highly fragmented, split between national statutory repositories and localized municipal archives. This structural division dictates how the initial data gathering phase must be engineered to capture all relevant textual evidence. Before any deep logical inference can occur, the architecture must first secure reliable data streams from these disparate legal environments. Consequently, understanding the specific characteristics of these sources informs the design of the retrieval and preprocessing pipelines.

2.3.1 National Sources

National legal repositories represent the authoritative apex of the Philippine legislative hierarchy, housing laws that universally preempt local ordinances. These databases are maintained by centralized government agencies and contain highly formalized texts such as Republic Acts, Commonwealth Acts, and Batas Pambansa. They serve as the primary knowledge base for the system's external legal context, providing the baseline rules against which municipal drafts are evaluated. Because national legislation undergoes a rigorous, standardized drafting process, the vocabulary within these repositories is generally consistent and adheres to strict legal English paradigms. Accessing these national databases provides the computational framework with the necessary macroscopic view of state-level mandates and prohibitions.

Official Gazette. The Official Gazette serves as the mandated repository for Philippine statutes, rendering it the official journal of the country for legislative text [52]. It acts as the primary publisher for all newly enacted national laws, executive orders, and administrative issuances, as mandated by decree of Republic Act No. 453 [109] and Commonwealth Act No. 638 [100]. Because it represents the final, legally binding version of a statute, any computational system evaluating municipal preemption must anchor its baseline rules to this repository. This archive effectively acts as the definitive historical ledger for state-level mandates, preserving the exact phrasing approved by the legislative and executive branches. Consequently, it remains the most critical target for establishing the system's foundational legal knowledge base.

Open Congress API. To programmatically map the complex hierarchy of national laws, developers utilize the Open Congress API provided by the House of Representatives' Legislative Information System (LEGIS) to access structured, machine-readable metadata regarding legislative routing and amendments [62]. This interface efficiently tracks a statute's formal lifecycle, charting its progression from initial committee filing to final executive approval. By providing standardized output formats, the API allows computational pipelines to establish relational dependencies between different legislative acts without manually parsing unstructured text.

The LawPhil Project. The Lawphil Project functions as a massive, privately maintained database of statutory records and jurisprudential decisions [9]. Unlike official government channels, this repository provides a comprehensive, pre-digitized plaintext corpus of historical and contemporary Philippine law. This accessibility makes it mathematically ideal for the unsupervised pre-training of foundational language models tailored to the local legal domain. By ingesting this vast corpus of clean text, NLP architectures can learn the highly specific syntactic structures and deontic operators inherent in Philippine jurisprudence. Utilizing this database effectively establishes the core vocabulary required for advanced Transformer models to process specialized statutory drafting [168].

2.3.2 Local Sources

In contrast to the centralized national databases, local legal repositories are maintained independently by individual Local Government Units (LGUs). These archives house municipal ordinances, resolutions, and executive orders that govern specific geographic jurisdictions. The legislative output at this tier is entirely decentralized, resulting in highly variable archiving practices depending on the technological capacity of the specific city council. These local databases form the immediate target corpus for the conflict detection system, containing the actual regulatory constraints imposed on city residents. Consequently, they represent a high-volume, dynamic data environment that continuously expands as the local council holds regular drafting sessions.

Davao City Sangguniang Panlungsod. The locality's primary corpus consists of the Davao City Sangguniang Panlungsod (City Council) website, which serves as the official digital repository for the city council's legislative outputs [28]. This database acts as the primary historical record for all municipal regulations, resolutions, and zoning policies enacted within the city's jurisdiction. It represents a dynamic and continuously expanding text environment that directly reflects the decentralized nature of local lawmaking. The texts housed here encapsulate the specific regulatory constraints and policy directions of the city government, providing the exact ordinance text required to evaluate new drafts for logical consistency against the existing legislative framework.

Legislative Information Support System. To manage the sheer volume of local legislation, the city introduced the Legislative Information Support System Program (LISSP) as its primary administrative backbone [27]. This platform serves as a digital tracking system designed to maintain metadata regarding ordinance routing, committee assignments, and plenary approval statuses. According to official reports, the system is openly available to the public, allowing citizens and stakeholders to monitor ordinances and resolutions in real-time [96]. By digitizing the legislative workflow, LISSP effectively functions as a centralized database of ordinances, providing the structured tracking and contextual data necessary for evaluating the lifecycle of local laws. This public transparency and administrative organization address the interoperability challenges identified in recent regional e-governance assessments [5].

2.3.3 Data Extraction Challenges

While the aforementioned repositories contain the necessary legal knowledge, extracting the needed data for computational analysis presents severe technical bottlenecks. Scraping and formatting these diverse sources exposes the pipeline to significant data degradation and structural inconsistencies.

Optical Character Recognition. A preliminary survey of the Official Gazette and the Davao City SP archive reveals a significant barrier to automated auditing: the lack of native, machine-readable text. Both repositories predominantly host scanned, image-based PDFs rather than "searchable" or "selectable" digital documents. During manual inspection, these files were observed to contain varying degrees of visual noise, including slanted scans, faded ink, and compression artifacts. As Lopresti [86] documents, such "noisy" visual data often results in severe Optical Character Recognition (OCR) degradation, where standard extraction tools produce broken alphanumeric strings and nonsensical tokens. These observed irregularities suggest that a direct pipeline would suffer from semantic fragmentation, requiring an additional, high-precision OCR layer and extensive text-cleaning heuristics to ensure the data is viable for program analysis.

Archival Fragmentation. Another critical challenge is archival fragmentation, a state where repositories suffer from systemic gaps in their digital records. This is a significant problem for high-fidelity auditing because the absence of even a single related ordinance can lead to "false negatives" in Conflict Detection, where a system declares a draft valid simply because it cannot see the contradicting record. One primary example is the Davao City Sangguniang Panlungsod public archive, which features intermittently missing ordinance files and unorganized chronological listings that actively hinder systematic data collection [127]. Furthermore, even relatively complete national databases like the Lawphil Project can lack specific Republic Acts or historical amendments. As Westermann et al. [57] observe, legal databases are frequently "unstructured" and "fragmented," forcing developers to move beyond simple scraping toward complex data engineering to ensure a "complete" dataset. This requires developing specialized, fault-tolerant scripts to account for missing files and broken links, significantly complicating the data-gathering phase required to build a reliable local knowledge base.

Database Accessibility. A significant challenge in data acquisition is the discrepancy between the theoretical and actual accessibility of local government databases. One notable example of this is with LISSP, because while the Program is promoted in institutional reports as a tool for public transparency [27, 96], empirical verification reveals a strict authentication barrier requiring login credentials for entry. Upon technical inspection by the proponents, the LISSP portal remains computationally isolated behind a login interface, effectively restricting access to internal government personnel and preventing programmatic extraction by public scrapers as well as access to the general public. This "gatekept" nature of the database was further confirmed through direct correspondence with system administrators by the proponents, where it was discovered that the archive is not open in a digital sense; access for this study was only secured through formal requests and manual persuasion via email. As Alfiani et al. [5] observe in similar administrative systems, such isolated data often exists as "digitally orphaned" artifacts, where metadata about an ordinance is entirely disconnected from its full-text searchable content. This structural disconnection forces an automated consistency-checking pipeline to bypass the LISSP's restrictive front-end and instead engineer custom preprocessing logic that attempts to mathematically re-bind any available public metadata to the continuous vector representations of the offline, full-text documents gathered from alternative, non-authenticated archives.

Textual Scarcity. Perhaps the most significant challenge in developing an automated legal auditing system is the pervasive systematic textual scarcity within the Philippine legal informatics landscape. Despite the existence of authoritative sources like the Official Gazette and the Davao City SP website, their web architectures remain inherently hostile to modern programmatic extraction, characterized by unindexed entries and "deep-linked" PDFs that cause automated scrapers to fail. Even community-driven initiatives or secondary repositories do not offer a single, straightforward solution. For instance, while the Open Congress API provides structured metadata for legislative routing, it lacks the full-text bodies of the laws themselves. Conversely, while the Lawphil Project offers a more consistent link format and a vast plaintext corpus, it is not an exhaustive archive and contains occasional missing Republic Acts or amendments. As Shao et al. [131] observe, such fragmentation prevents the construction of a unified, high-fidelity knowledge base from a single origin. Consequently, researchers are forced into a series of technical ”roundabouts” and compromises—engineering a patchwork pipeline that pulls metadata from one source and text from another—to circumvent the absence of a centralized, machine-readable repository for both national and local legislation.

2.4 Exact and Symbolic Methods

Before the adoption of neural networks, computational legal analysis relied heavily on deterministic systems to process text. Deterministic systems operate on absolute rules, meaning that a specific input will invariably produce the exact same output without any probabilistic guessing. Traditional methods for searching and evaluating legal information assumed that legislative ideas could be cleanly mapped either to specific vocabulary strings or to rigid, hand-coded logic formulas. While highly transparent and easy to debug, these foundational approaches frequently fail when forced to detect hidden conflicts in complex, unstructured legal texts. The core issue is that human language is inherently messy, and rigid systems break down when faced with the ambiguous phrasing typical of municipal ordinances.

2.4.1 Boolean Information Retrieval

To understand why traditional legal databases struggle with semantic conflict detection, we first need to look at how they fundamentally process text at the machine level. Boolean Information Retrieval (BIR) represents the foundational search architecture for almost all commercial law databases currently in use. Instead of reading a document like a human, BIR treats every file simply as a “bag of words,” stripping away all grammar and sentence structure. It relies on an inverted index, which is essentially a massive lookup table that tracks exactly which specific words appear in which files across the entire database. Because it only registers the presence or absence of characters, the system is completely blind to the actual context or legal intent behind those words.

At its core, the retrieval mechanism in BIR is strictly based on set theory and formal Boolean logic. A user constructs a query using logical operators like `AND` ($\wedge$), `OR` ($\vee$), and `NOT` ($\neg$) to create intersecting sets of documents. For example, a search for a specific ordinance regarding explosives might be formulated mathematically as:

$$Q = (\text{fireworks} \vee \text{pyrotechnics}) \wedge \neg \text{exempt}$$

The system then retrieves documents by calculating the exact mathematical intersection of these sets, pulling only the files that satisfy every condition simultaneously. If a document contains the specific string of characters requested, it is retrieved and scored as a 1; if it misses even a single required term, it is completely ignored and scored as a 0. It evaluates the raw presence of strings, entirely ignoring the conceptual meaning or legal nuance behind them [16].

2.4.1.1 Challenges with BIR

The primary limitation of BIR in legal auditing is its absolute, uncompromising reliance on exact string matching. This rigidity creates a severe “vocabulary mismatch” problem when processing natural language. Human language naturally features synonymy, where entirely different words possess the exact same meaning, and polysemy, where the identical word takes on drastically different meanings depending on the context. In statutory drafting, one author might use “prohibited” while another uses “unlawful,” referring to the exact same legal concept. Because BIR operates on character-level matching rather than conceptual mapping, it fundamentally cannot connect these two terms, leading to massive blind spots in the search results.

A landmark empirical evaluation of these systems demonstrated this vocabulary gap in a highly practical setting. Legal researchers conducting queries using standard BIR systems confidently estimated they were finding over 75 percent of the relevant documents for a given case. They experienced an illusion of effectiveness because the documents they did retrieve looked highly accurate, masking the massive volume of relevant files they missed. In reality, the system retrieved less than 20 percent of the actual relevant files contained within the database [16]. The researchers missed the remaining 80 percent simply because the authors of those hidden documents used slightly different, completely valid phrasing that didn’t perfectly align with the strict search query [149].

In a local government context like Davao City, this specific margin of error is administratively unacceptable. Missing even one existing ordinance during an ex-ante review due to a simple vocabulary mismatch could directly result in the passage of an invalid, conflicting law. Furthermore, legislative vocabulary inherently evolves over time as drafting conventions change. An ordinance passed in the 1990s and a modern draft written in 2026 might strictly regulate the exact same municipal subject but share absolutely zero common keywords. Recent studies emphasize that resolving this requires the implementation of “Transformer” models, which abandon exact matching entirely in favor of continuous vector spaces that recognize when different vocabulary points to the same underlying concept [75].

2.4.2 Symbolic AI

Because BIR fundamentally fails to capture underlying meaning, early artificial intelligence researchers attempted to automate legal reasoning by directly modeling the rules of law themselves. This approach, widely known as Symbolic AI or “Good Old-Fashioned AI” (GOFAI), operates on the premise that human cognition can be replicated by manipulating formal symbols [151]. It moves beyond simple word matching by teaching computers actual legal logic through explicit, human-coded “IF-THEN” formulas. Instead of searching a database for raw text strings, Symbolic AI requires human engineers to manually translate the constraints of a statute into formal logic structures. This theoretically allows the computer to process the actual regulatory mechanics of a law rather than just reading its surface vocabulary.

To execute this logic, these systems specifically utilize propositional or first-order logic combined with deontic operators. Deontic logic is a highly specialized branch of mathematical logic that deals specifically with obligations, permissions, and prohibitions. For example, a system might represent a local zoning prohibition mathematically as:

$$\forall x(\text{Business}(x) \land \text{ResidentialZone}(x) \rightarrow \text{Prohibited}(x))$$

In this formula, the universal quantifier ($\forall$) asserts that for any entity x, if it is classified as a business ($\land$) and located in a residential zone, it is strictly prohibited ($\rightarrow$). By converting ordinances into these rigid formulas, a system can theoretically use deductive reasoning to spot a conflict. If Rule A dictates $\mathcal{O}(X)$ (an absolute obligation to do X) and Rule B dictates $\mathcal{F}(X)$ (an absolute prohibition against doing X), the system successfully flags a formal logical contradiction ($\perp$) [12].

2.4.2.1 Challenges with Symbolic AI

While Symbolic AI provides perfectly transparent and mathematically sound reasoning, its absolute reliance on manual programming makes it practically impossible to scale for real-world governance. The primary architectural barrier is known as the "knowledge acquisition bottleneck." Human domain experts and software engineers must manually read, interpret, and translate thousands of unstructured city ordinances into rigid computer rules. This manual translation process is unacceptably slow, extremely expensive, and highly prone to human error during the coding phase [90]. Furthermore, every time the city council amends an existing law, the underlying symbolic formula must be manually rewritten, making the system incredibly difficult to maintain over time.

Beyond the logistical nightmare of manual coding, the rigid 1-and-0 architecture of Symbolic AI fundamentally clashes with the "open texture" of natural legal language. Laws frequently rely on subjective, open-ended standards—such as requiring a citizen to exercise "reasonable care" or negotiate in "good faith." These inherently subjective human concepts cannot be easily defined within a strict, binary rule-based formula because their definitions change depending on the specific context [11]. While these symbolic systems are highly precise in tightly controlled, hypothetical environments, they immediately shatter when exposed to the real world. Ultimately, they lack the computational flexibility required to handle the thousands of messy, unorganized, and ambiguously worded documents actually found in local government archives [169].

2.5 Graph-Based Architectures

Because exact word matching can be unreliable, some researchers use "graph-based" systems. Instead of looking at laws as just lists of words, these systems treat them like points in a large web. By looking at how laws quote each other and relate to the same topics, the computer can better understand their meaning.

2.5.1 Legal Knowledge Graphs and Graph Neural Networks

Legal Knowledge Graphs (LKGs) are a way to model how different laws depend on each other. Unlike a simple list, these graphs can use Graph Neural Networks (GNNs) to find hidden links and clashes. Researchers have created frameworks like GNN-ERE to automatically build these legal webs from messy text, which works much better than just searching for keywords [92]. Studies also show that these networks can find relevant court cases even if they do not use the exact same words [58].

The advantage of GNNs is that they can share information across the entire web structure. By treating laws as points in a network, the computer can learn the "purpose" of a law—for example, telling the difference between a rule and a procedure [15]. These systems can also perform "multi-hop reasoning." This means they can find "indirect conflicts" where two laws do not quote each other directly but still clash because of their relationship with a third law [163]. This is a big step up from old search engines that can only look at one thing at a time.

2.5.1.1 Challenges in GNNs

While Graph Neural Networks are powerful, they only work if the web of connections is high-quality. A GNN needs a clearly built index or list of links to follow.

Dependency on Clear Connections. GNNs are limited because they need clear links between laws to "see" the meaning. As researchers have pointed out, creating these links by hand is very slow and expensive [65]. In a city with thousands of ordinances, it is almost impossible for people to keep this list of links up to date. If the system fails to find a link between two conflicting laws at the start, the GNN will be "blind" to that problem.

Disconnection in Digital Archives. Even though systems like the Legislative Information Support System Program (LISSP) have helped move Davao City's archives online, a gap remains for graph-based AI. Most digital repositories in local government are designed for storing and viewing files rather than linking ideas together. Recent studies on Philippine digital governance show that local ordinances often act as "digitally orphaned" texts; they are available online but lack the machine-readable links needed for an AI to jump from one law to another [5]. This makes the digital archive a collection of separate records rather than a connected network, making it hard for GNNs to see how they overlap.

Weakness Against Messy Text and OCR Noise. Building these legal webs automatically is also very difficult because of how local government data looks. AI frameworks usually assume the text is "clean" and easy to read [92]. However, scanned PDFs in Philippine archives often have "OCR noise" (typos from scanning), which confuses the computer [86]. When there are typos or local "Taglish" words, the computer creates broken or "hallucinated" links. This makes the whole web unreliable for finding actual legal clashes.

2.5.2 Abstract Meaning Representation

While GNNs look at links between documents, Abstract Meaning Representation (AMR) looks at the logic *inside* a single sentence. AMR is a graph system that focuses on the "who is doing what to whom" logic. Researchers argue that this provides a more stable way to understand laws than just looking at grammar [154]. By mapping out the logic, AMR can find conflicts even if laws use different wording (like "The city bans X" vs "X is banned by the city").

AMR is also useful for identifying which rules are "obligations" (must do) and which are "prohibitions" (cannot do) [143]. Unlike old keyword searches, AMR understands the role of each word. This helps find subtle clashes, such as a local law allowing "business use" when a national law forbids "non-residential activity." New legal datasets have proven that AMR is a good way to combine human-like logic with computer power [154].

2.5.2.1 Challenges in AMR

Even though AMR is very precise, using it for Philippine city ordinances is difficult. It relies on very strict rules that don't always match the "messy" way local laws are written.

Limited to Single Sentences. The biggest flaw of AMR is that it only looks at one sentence at a time [143]. However, legal conflicts in city ordinances rarely happen in the same sentence; they are usually found in different sections or paragraphs. For example, a ban might be in Section 2, while an exception is buried in Section 5. Because AMR cannot "connect the dots" between different sentences easily, it is often "blind" to these types of conflicts.

Difficulty with Complex Legal Language. AMR relies on a static or fixed dictionary of meanings to work. The long, "run-on" sentences often found in Philippine laws are too complex for most AMR tools. For example, in RA 7160 within Section 378, such sentence can be found as follows:

"When a loss of government property occurs while the same is in transit or is caused by fire, theft, force majeure, or other casualty, the officer accountable therefor or having custody thereof shall immediately notify the provincial or city auditor concerned within thirty (30) days from the date the loss occurred or for such longer period as the provincial, city or municipal auditor, as the case may be, may in the particular case allow, and he shall present his application for relief, with the available evidence in support thereof."

Such a sentence causes the AI often fails or produces the wrong logic map, as studies show that as sentences get longer and more confusing, AMR becomes much less accurate [144].

Problems with Local Words and Typographical Errors. AMR is also very sensitive to typographical errors and unknown words. When the computer runs into a scanning error from an old PDF, or a local word that isn't in its dictionary, it fails to map the logic correctly. Research shows that this "lexical noise" often results in broken logic maps that lose the actual meaning of the law [148]. Without a tool specifically made for Philippine Legal English, relying on AMR for Davao City's archives would be unreliable.

2.6 Deep Learning and Transformer Models

The application of Natural Language Processing (NLP) in the legal domain has shifted from rigid, rule-based systems and basic word-matching toward Deep Learning [73]. Legacy algorithms treat legal texts as simple character strings, which frequently fail to capture the dense and interconnected nature of statutory drafting. Deep Learning, conversely, utilizes neural networks to map out the actual semantic meaning of legal language. This allows systems to process complex syntax and perform the deductive reasoning required to spot conflicts. This section breaks down the Transformer architecture, explaining its underlying mechanics, computational challenges, and why it remains the most practical framework for auditing unstructured municipal archives.

2.6.1 Transformer Models and Self-Attention Mechanism

A major breakthrough was the Transformer architecture, introduced by Vaswani et al. [150] in 2017. Prior models, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks, processed text sequentially. This step-by-step processing was computationally slow and made it difficult for the model to retain information from earlier in a long document. Transformers bypass this sequential bottleneck entirely by relying on a Self-Attention mechanism.

Self-Attention allows a model to analyze all words in a sequence simultaneously to determine their contextual relationships. This enables a specific term to connect with any other word in the document, regardless of the physical distance between them. Because it processes data in parallel, Transformers are significantly faster to train, directly enabling the creation of Large Language Models (LLMs). Modern iterations have introduced refinements like Disentangled Attention to better parse complex syntactic structures, alongside Windowed Attention to process long-context legal documents efficiently. However, while Generative LLMs like ChatGPT excel at text summarization, they are prone to 'hallucinations'—often inventing plausible but entirely fake legal citations [36]. To ensure legal validity, a system must be discriminative: specifically classifying whether statements exhibit a Conflict or Entailment, rather than simply generating new text.

2.6.2 Challenges in Transformer Models

Despite these architectural advantages, deploying Transformer models within the hardware limitations of Philippine Local Government Units (LGUs) introduces specific computational hurdles.

The Quadratic Bottleneck and Context Limits. The Transformers' core processing engine scales quadratically ($O(n^2)$), meaning it requires an exponentially larger amount of memory as the input text grows [150]. Because of this "quadratic bottleneck," standard models like BERT are generally capped at reading 512 tokens at a time [38]. Philippine ordinances are frequently verbose, often spreading rules and exceptions across multiple pages. To manage this, developers use a **Sliding Window** technique. This involves taking a fixed-size "window" (e.g., 512 words) and moving it across the document with a specific **Stride** [7]. For example, if the window moves 256 words at a time, the windows will overlap, ensuring that the end of one chunk is also the start of the next to try and keep context together [60]. However, this is like reading a long book through a small slot; if a general rule is on Page 1 and its specific exception is on Page 5, the AI can never see them at the same time [129]. This physically cuts the "semantic link" between related clauses, creating logical blind spots where the system might flag a conflict simply because it cannot see the full picture [129].

Morphological Noise and Domain Adaptation. Transformers require massive corpora of clean text for proper pre-training. While specialized models like Legal-BERT [22] perform well for US or European law, there is no equivalent foundation model trained specifically on Philippine local legislation. Applying Western-trained tokenizers to local government archives—which are saturated with uncorrected OCR errors, archaic colonial phrasing, and administrative 'Taglish'—causes severe tokenization fragmentation. The model fails to recognize these localized terms and shatters them into meaningless sub-tokens, effectively ruining its ability to comprehend the law.

Edge Deployment and Hardware Constraints. Executing deep logical checks across thousands of document pairs demands significant computing power, typically requiring expensive GPU resources [116]. Most municipal governments simply lack this caliber of server infrastructure. Consequently, deploying these heavy models on standard LGU workstations requires aggressive optimization and decoupled pipeline architectures to ensure the system does not freeze during inference.

2.6.3 Transformer Models as the Ideal Method

Notwithstanding these hardware limitations, Transformer models remain the most viable paradigm for Semantic Conflict Detection in local ordinances. General-domain language models typically struggle with legal text because common words frequently take on entirely different legal definitions. For instance, "party" refers to a litigant rather than a celebration, and "suit" signifies a legal action rather than clothing. Greco et al. [53] emphasize that this high degree of polysemy is exactly why domain-specific models are required. While older static systems assign a single, fixed meaning to a word [53], a Transformer dynamically adjusts its reading of a word based on the surrounding sentence context [22]. Chalkidis et al. [23] empirically showed that this contextual awareness is what allows a model to distinguish between amending a law and violating it—a nuance that simple keyword searches invariably miss.

The transition to Deep Learning necessitates explicitly moving away from legacy methodologies. Boolean keyword searches fundamentally fail because legislative drafting vocabulary constantly evolves over the decades. Conversely, manually mapping thousands of unlinked Republic Acts and local ordinances into deterministic graph structures creates an impossible administrative bottleneck. As Li et al. [82] observed, hand-crafting these legal knowledge graphs simply does not scale for volatile, unstructured municipal records. Ultimately, Transformers represent the ideal method: they possess the capacity to map the deep, latent semantic relationships in unstructured legal text without requiring an intractable amount of manual data entry.

2.7 Transformer Pipeline

The appeal of an end-to-end, single-stage neural architecture is its operational simplicity: funneling raw text directly into final outputs without intermediate steps. For general Natural Language Processing (NLP) tasks like sentiment analysis, a single Transformer model is usually enough. However, legal informatics, specifically Semantic Conflict Detection, requires a level of deductive precision that a single monolithic model structurally cannot provide. Forcing one architecture to act simultaneously as a high-volume search engine and a strict logical judge creates impossible mathematical trade-offs.

2.7.1 Challenges in Single-Stage Pipeline

Deploying a single Transformer model to scan decades of uncodified municipal ordinances introduces severe architectural friction. The system inevitably compromises either its recall (missing conflicting laws) or its precision (flagging false positives). This limitation changes depending on the specific type of single-stage model used.

Generative LLM Constraints. A common but flawed approach feeds a document pair directly into a Generative Large Language Model (Decoder-only architectures like GPT or LLaMA) and prompts it to declare a conflict. These models are auto-regressive text predictors, not discriminative logical classifiers. As Linna [85] notes, applying them to strict statutory auditing invites 'hallucinations'—the fabrication of plausible but legally fake citations. Furthermore, their massive parameter counts make offline Edge Deployment on standard Local Government Unit (LGU) hardware impossible without aggressive quantization. Dettmers et al. [37] demonstrated that this quantization process severely damages zero-shot reasoning fidelity in specialized domains.

Bi-Encoders and their Limitations. A Bi-Encoder, also known as a "Dual-Tower" or "Two-Tower" model, is an AI architecture designed for high-speed searching [116]. It works by processing the user's search query and the documents in the archive completely separately from each other. The model turns every ordinance into a mathematical coordinate (a vector) and stores them in a database [71]. Because it does not have to read the documents at the same time a person is searching, it can compare a query against more than 25,000 national statutes in just a few milliseconds [26]. This makes it the perfect tool for the "Coarse" stage of the pipeline, where the goal is to narrow down the entire legal archive to just a few relevant provisions [47].

Current municipal search systems predominantly rely on Dense Retrievers (Bi-Encoders). While highly efficient for filtering millions of documents, Thakur et al. [142] explain that Bi-Encoders suffer from a severe 'bottleneck of compression' because they collapse the semantic meaning of an entire document into a single, fixed-size vector. This architecture optimizes for topical similarity rather than logical consistency. Consequently, a Bi-Encoder has a critical logical blindspot: it will retrieve an ordinance "allowing fireworks" and one "banning fireworks" and score them as highly relevant because their vocabulary distributions are nearly identical. Kim et al. [75] observed that Bi-Encoders fundamentally lack the computational architecture to detect the logical contradiction operating beneath the shared syntax.

Overcoming this blindspot requires a secondary phase using a Cross-Encoder architecture. Instead of independently compressing documents into isolated vectors, a Cross-Encoder computes token-level attention across both documents simultaneously ([CLS] Ordinance A [SEP] Ordinance B). This allows the model's attention heads to directly compare negators (e.g., "banning" vs. "allowing") in their immediate contexts. Nogueira and Cho [104] demonstrated that this cross-attention mechanism shifts the task from basic topical retrieval to rigorous Natural Language Inference (NLI), enabling the precise Semantic Conflict Detection required for legal texts.

Quadratic Memory Constraints If a single-stage Bi-Encoder lacks logical depth, the theoretical alternative is running a single-stage Cross-Encoder across the entire database. However, this approach is computationally impossible on standard office workstations. Because the Self-Attention matrix scales quadratically ($O(n^2)$) with sequence length, Beltagy et al. [13] highlight that cross-encoding a target ordinance against the 25,432 national statutes simultaneously instantly exceeds available graphics memory. The model would be forced to split the database into tiny, fragmented sliding windows, severing the critical semantic links between general rules and remote exceptions. This memory bottleneck proves that isolating a small candidate shortlist through coarse retrieval is essential before running deep sequence classification.

2.7.2 Single-Stage vs. Dual-Stage Pipeline

Given these hardware and logical constraints, the literature discourages single-stage monoliths for legal analysis. Expanding to a dual-stage pipeline is not just an optional optimization, as it is a structural necessity for processing high-volume, unstructured text. Several critical considerations justify this architectural split:

Retrieve-Then-Entail Paradigm. Legal auditing requires a shift from topical similarity to logical inference. A Retrieve-Then-Entail pipeline serves as a Coarse-to-Fine guardrail: an efficient retriever isolates candidates, and a deep-reasoning NLI Cross-Encoder identifies specific policy violations. Williams et al. [161] indicate that this approach reduces manual review time by up to 70 percent while maintaining conceptual coherence.

Hardware Feasibility. A dual-stage approach segregates the computational heavy lifting. Stage one relies on Information Retrieval via pre-computed vector indexes that use minimal active memory. This offloads the intense $O(n^2)$ matrix multiplications of the Cross-Encoder exclusively to the top-K retrieved pairs. Khattab and Zaharia [74] demonstrated that this allows advanced NLP to run locally on standard servers without requiring enterprise GPU clusters.

Pragmatic Solution to Compression. Splitting the pipeline lets each model specialize. The first stage accepts the bottleneck of compression to maximize search speed. As Humeau et al. [64] observed, the second stage avoids compression entirely, reading raw, uncompressed tokens across both texts to achieve high-precision logic.

2.7.3 Competition on Legal Information Extraction and Entailment

The necessity of architectural decoupling is highlighted by over a decade of results from the Competition on Legal Information Extraction and Entailment (COLIEE), moving beyond mere theoretical preference. As Goebel et al. [47] detail, COLIEE serves as the premier international proving ground for legal NLP, challenging research teams to solve complex jurisprudential problems using standardized datasets. Crucially, the competition structurally enforces the separation of search and logic by bifurcating statutory analysis into distinct, sequential tasks, which includes two relevant ”Tasks”:

Statutory Information Retrieval. Systems must scan a vast civil code corpus to retrieve all candidate provisions relevant to a specific legal query.

Statutory Textual Entailment. Systems must perform rigorous logical deduction to determine whether the retrieved provisions logically entail or contradict the given query [6, 47].

Year after year, architectures deployed by winning teams consistently reject single-stage monoliths in favor of pipelined approaches. The consensus among top-performing systems is that Information Retrieval requires high-recall Dense Retrievers (e.g., BM25 paired with Bi-Encoders) to handle corpus volume, while the entailment task strongly demands deep-reasoning Cross-Encoders. For instance, Althammer et al. [6] demonstrated during COLIEE 2021 that delegating the ’reading’ phase to a disentangled Cross-Encoder completely prevents the semantic collapse inherent in long-document vectorization. This sustained global validation underscores that separating Information Retrieval (IR) from Natural Language Inference (NLI) is not just a hardware workaround, but the fundamental mathematical blueprint required to solve Semantic Conflict Detection as demonstrated in the Competition.

2.7.4 Pipeline Synthesis

Addressing context limits, logical blindspots, and morphological noise simultaneously is computationally impossible for a single monolithic model operating on consumer-grade LGU hardware. Therefore, this study adopts a dual-stage Coarse-to-Fine pipeline. Rather than a speculative architectural gamble, Reimers and Gurevych [116] note that this decoupled Retrieve-Then-Entail paradigm represents the empirical consensus in contemporary computational law.

The mathematical realities of Transformer scaling dictate this functional separation. Foundational work by Karpukhin et al. [71] established that Dense Retrievers must be independently optimized for Maximum Inner Product Search (MIPS) to handle corpus-level volume effectively. However, because Bi-Encoders lack the cross-attention necessary for deductive reasoning, they must feed into a secondary mechanism. Louis and Spanakis [88] validated this architectural split specifically for statutory law. They proved that utilizing a high-capacity vector retriever purely for filtering, followed by a localized Cross-Encoder for Semantic Conflict Detection, dramatically reduces false positives compared to single-stage retrieval systems.

Crucially, the necessity of this architectural decoupling is implemented by over a decade of evaluations from the Competition on Legal Information Extraction and Entailment (COLIEE). As the premier international benchmark for legal NLP, COLIEE structurally enforces the separation of search and logic by bifurcating statutory analysis into distinct, sequential operations: Statutory Information Retrieval and Statutory Textual Entailment [47]. Operating within this rigid competitive framework, Althammer et al. [6] demonstrated that in legal entailment tasks, delegating the "reading" phase to a disentangled Cross-Encoder prevents the semantic collapse inherent in long-document vectorization. Prior studies by Shao et al. [131] consistently corroborate this, showing that isolating the computational heavy-lifting of document filtering from the granular scrutiny of logical evaluation successfully bypasses the hardware bottlenecks that cause single-model architectures to fail.

Synthesizing these specific architectural components provides a rigorously tested blueprint: Stage One (Information Retrieval) absorbs the corpus volume and context length, while Stage Two (Textual Entailment) resolves the logical and morphological noise. This deliberate division of labor ensures semantic consistency across the unstructured archives of Philippine local legislation.

2.8 Information Retrieval

Information Retrieval (IR) is the process of extracting relevant documents from massive, unstructured datasets based on a specific query. In legal informatics, IR acts as the foundational filtering mechanism. It serves as the "coarse" stage of a dual-pipeline system, isolating candidate statutes from thousands of irrelevant municipal records before any intensive logical analysis takes place.

2.8.1 Nature of 'Relevance' in Retrieval

To understand how computational systems extract documents, we must first examine the human cognitive process that defines relevance in the physical world. In cognitive science, relevance is not a static property but a dynamic relationship between information and a specific task. According to Relevance Theory, humans naturally filter information by seeking the greatest cognitive benefit for the least mental effort [135]. When a legal researcher checks for conflicts, they do not simply count matching words; they instinctively evaluate the "intent" and "applicability" of a rule based on their background knowledge of the legal system. This human-centric view suggests that a document is relevant only if it provides a new piece of information that changes the researcher's understanding of the legal landscape [128].

Beyond the cognitive baseline, the definition of relevance in this study is anchored in the philosophical requirements of a functional legal system. According to the jurisprudence established by Lon Fuller in *The Morality of Law*, a primary requirement for the internal morality of law is consistency; a system that issues contradictory rules fails to be a legal system at all [43]. Within this philosophical framework, relevance is not merely about finding documents that talk about the same subject matter, but about identifying provisions that share a "normative overlap." This means that a retrieved document is relevant if it occupies the same jurisdictional or behavioral space as the proposed ordinance, potentially creating a situation where a citizen cannot obey one law without violating another. By grounding the retrieval stage in this quest for consistency, the system acts as a guardian of the legal system's internal logic, treating relevance as the identification of potential threats to the rule of law [43].

However, traditional computational methods struggle to replicate this human judgment because they rely on a rigid, mathematical definition of relevance. In standard deterministic systems, relevance is treated as a binary mathematical property based entirely on set intersection [16]. If we denote a candidate document as D and a user query as Q, this lexical relevance is formally evaluated as a binary function $R(D, Q) \in \{0, 1\}$. Under this strict lexical definition, a document is deemed fully relevant (i.e. 1) if and only if it contains the exact sequence of characters requested by the user [149]. Consequently, this mechanism establishes a purely topical baseline, effectively grouping documents together simply because they share a surface-level vocabulary [16]. While this mathematical simplicity allows for rapid indexing across massive databases, it completely strips away the contextual nuance required for complex legislative analysis.

This stark discrepancy between superficial lexical matching and actual legal consequence represents a major stumbling block for legacy search architectures. In statutory law, two documents discussing the exact same topic might be lexically relevant to one another, but they are only *legally* relevant if one provision actively modifies, permits, restricts, or preempts the other [149]. For instance, a retrieved document might use identical vocabulary but apply to a completely different jurisdiction or regulatory scope, rendering it completely useless for an ex-ante conflict check. Conversely, a document that governs the exact same civic behavior but uses entirely different phrasing—such as substituting "motor vehicle" for "automobile"—will be scored as mathematically irrelevant (i.e. 0) by a strict Boolean system [16]. Therefore, to accurately audit proposed Davao City ordinances, the retrieval system must abandon this binary, keyword-driven definition of relevance in favor of a dense, semantic approach capable of measuring actual jurisprudential relationships [75].

To resolve these limitations, modern computational pipelines fundamentally redefine relevance spatially rather than lexically. Instead of scanning a query as a raw string of text, the proposed system utilizes a Bi-Encoder architecture to compress entire legal provisions into high-dimensional mathematical coordinates, commonly referred to as dense vectors [116]. Under this advanced semantic framework, relevance is no longer determined by counting shared characters, but by calculating the geometric distance—such as cosine similarity—between these coordinate points [75]. If a newly drafted local ordinance and an existing national statute govern the same underlying civic behavior, the neural network learns to map their respective vectors closely together in the mathematical space, completely bypassing the vocabulary mismatch problem [162]. Consequently, the program evaluates relevance as a continuous proximity score rather than a rigid binary output, allowing it to successfully retrieve conceptually intertwined laws that a basic keyword search would otherwise ignore.

2.8.2 Long-Context Information Retrieval

Relying on standard BERT-based architectures introduces an artificial constraint on legal analysis: the 512-token limit. Because standard Self-Attention mechanisms scale quadratically with sequence length, processing long texts requires severe truncation. To get around this, researchers like Chalkidis et al. [24] routinely use Sliding Window techniques to segment documents into overlapping chunks. However, applying this to Philippine ordinances—which are historically wordy and often bury exceptions deep within 1,000-token provisions—is structurally destructive. It severs the crucial semantic link between a primary rule and its distant exception.

Consequently, the model operates with a logical blindspot. Beltagy et al. [13] note that this causes logical hallucinations, where a rule might be flagged as an absolute prohibition simply because its exempting clause was pushed into a separate, unconnected processing window. Furthermore, these systems must navigate the trade-off between Sparse Retrieval (e.g., BM25) for exact keyword matching, and Dense Retrieval for capturing conceptual meaning even without direct word overlap.

Resolving this fragmentation means abandoning chunking protocols entirely. The architectural solution requires an initial retrieval mechanism capable of native long-context retention. Recent Transformer models leverage advanced positional encoding, such as Rotary Positional Embeddings (RoPE), alongside the sparse attention matrices. Warner et al. [158] demonstrated that these upgrades extend context windows up to 8,192 tokens without triggering memory failures. Deploying this class of architecture allows the system to read entire local ordinances as cohesive semantic units, preserving the law's internal structure and eliminating the Sliding Window fallacy.

2.8.3 IR Benchmarks for Candidate Selection

Selecting the optimal retrieval architecture from a crowded field of open-weight models requires objective filtering. Rather than relying on architectural novelty, this study evaluates established empirical leaderboards to select candidates. While these testing frameworks mostly evaluate highly curated, Global North datasets, their rigorous zero-shot and clustering parameters provide a mathematically sound baseline. This helps identify models theoretically capable of handling the unstructured noise of Philippine municipal archives.

Benchmarking Information Retrieval (BEIR) The Benchmarking Information Retrieval (BEIR) framework evaluates a Dense Retriever's capacity for zero-shot retrieval across diverse and specialized datasets [142]. Zero-shot retrieval measures a model's ability to operate in a specific domain without prior fine-tuning. Because Philippine local legislation completely lacks annotated training sets, a candidate model must inherently possess robust zero-shot generalization. Thakur et al. [142] establish that a dominant position on the BEIR leaderboard is a critical proxy for this. It indicates the model's underlying capacity to map novel, localized legal vocabulary, including archaic syntax and uncodified administrative terms, into a coherent vector space without previous exposure.

Massive Text Embedding Benchmark (MTEB) Expanding on BEIR, Muennighoff et al. [98] introduced the Massive Text Embedding Benchmark (MTEB), which serves as the global standard for ranking embedding systems across over 130 datasets and multiple languages. For candidate selection, this study prioritizes top-performers within MTEB's "Retrieval" and "Clustering" subsets. Models excelling on these specific leaderboards have a proven capacity to group overlapping concepts and retrieve relevant texts based purely on semantic proximity. Therefore, MTEB acts as a foundational filter, ensuring that only architectures with empirically validated retrieval logic enter the noisy environment of Philippine statutory auditing.

2.8.4 Candidate Model Architectures for IR

The following retrieval models were selected based on their architectural advantages and performance on the MTEB leaderboard:

BAAI/bge-m3 Chen et al. [26] highlight BGE-M3 for its versatility in simultaneously performing Dense, Sparse, and Multi-Vector (ColBERT-style) retrieval. It provides a unified model foundation capable of processing varying inputs, from short sentences to long documents up to 8,192 tokens, across more than 100 languages [26]. Crucially, it obtains token weights similar to the BM25 algorithm without additional computational costs during dense embedding generation, making it a robust, production-ready baseline for large-scale municipal codes.

Qwen3-Embedding-8B Built on the Qwen3 foundation, the Qwen Team [113] positioned this model as the state-of-the-art upper bound on the MTEB multilingual leaderboard with a score of 70.58. Unlike traditional BERT-style encoders, it utilizes an auto-regressive language model architecture specifically adapted for embedding tasks. This allows it to excel at long-text understanding and instruction-following retrieval [113]. By successfully following complex relevance definitions, it delivers superior retrieval and clustering performance compared to earlier open foundation models.

Nomic-Embed-v1.5 Nussbaum et al. [106] introduced Nomic-Embed-v1.5 as the first fully auditable, long-context embedder to exceed OpenAI's text-embedding-ada-002 on the MTEB benchmark. It implements Matryoshka Representation Learning (MRL), a technique that produces resizable vectors ranging from 64 to 768 dimensions. As Nussbaum et al. [106] note, this allows developers to trade off embedding size for speed and memory savings with negligible performance loss. This flexibility is critical for scaling high-volume legal databases under tight computational constraints.

GTE-Multilingual Zhang et al. [167] designed this distilled encoder model to provide a joint semantic space for multiple languages while significantly reducing the computational burden of deployment. Distilled models typically retain over 95 percent of their teacher's performance, making them highly practical for real-world systems requiring high throughput and low latency [167]. Its specific ability to align different languages into a common vector space is essential for detecting semantic contradictions between the English and Cebuano versions of Davao City ordinances.

GTE-ModernBERT Warner et al. [159] present ModernBERT as a modernization of the traditional encoder-only Transformer. It integrates architectural upgrades like Rotary Positional Embeddings (RoPE) and Flash Attention to maintain high performance across an 8,192-token context window [159]. The model leverages an alternating attention mechanism that switches between local windowed attention for efficiency and global attention for long-range dependencies. According to Warner et al. [159], this achieves inference speeds over three times faster than standard BERT variants, making it ideal for time-sensitive legal applications.

2.9 Natural Language Processing

Natural Language Processing (NLP) provides the computational techniques needed to analyze and represent text at various linguistic levels, aiming for human-like comprehension [84]. As Jurafsky and Martin [69] explain, it acts as the bridge between human language and machine logic, allowing algorithms to parse and interpret unstructured text. In the legal field, NLP shifts document auditing from slow manual reviews to scalable, automated systems. However, applying standard NLP to law requires more than basic grammar checks; it demands systems that can navigate dense sentence structures, deontic operators (obligations, permissions, prohibitions), and the highly specific vocabulary found in statutory law.

Early Legal NLP relied heavily on Symbolic Artificial Intelligence (AI) [73]. These systems used manually built ontologies and strict Boolean logic to extract legal information. However, because municipal drafting language changes constantly over decades, these exact-match frameworks proved far too brittle. Contemporary NLP architectures solve this by moving to continuous vector representations. By mapping textual provisions into high-dimensional semantic spaces, modern Transformers bypass superficial vocabulary mismatches to mathematically quantify the deeper concepts connecting different legal terms. This allows the system to act as an active diagnostic engine that structurally maps legal conflicts, rather than just a passive search index.

This section breaks down how NLP operates within legislative analysis. It covers the systemic challenges of processing localized legal text, the structural need for Natural Language Inference (NLI), the benchmarks used for model selection, and the specific architectures built for these deductive tasks.

2.9.1 Challenges in NLP

Deploying NLP frameworks against municipal archives quickly exposes the mechanical limits of standard Language Models. These architectures must simultaneously overcome the geographic bias built into their pre-training data and the mathematical flaws of relying purely on lexical overlap.

The Global North Bias There is a noticeable lack of research addressing the noisy, low-resource reality of local government data in the Global South. As Ranathunga et al. [115] point out, the vast majority of Legal NLP benchmarks, like LexGLUE [24] and CaseHOLD, are trained exclusively on pristine, digitally native Western legal text. These highly optimized models degrade drastically when applied to Philippine Local Government Unit (LGU) archives. These datasets are filled with scanned PDFs suffering from Optical Character Recognition (OCR) errors, archaic American colonial phrasing, untranslated Spanish legal terms, and local "Taglish." Rust et al. [124] demonstrated that standard WordPiece tokenizers shatter these unrecognized words into meaningless sub-tokens, destroying the model's ability to infer legal intent. Fixing this localized morphological noise requires abandoning low-capacity tokenizers. Instead, developers must deploy models equipped with massive Byte-Pair Encoding (BPE) vocabularies (e.g., 128,000+ tokens) that can absorb linguistic quirks without triggering Out-of-Vocabulary (OOV) errors. Furthermore, He et al. [59] note that utilizing models with Disentangled Attention separates the actual content of a corrupted word from its noisy positional data. This allows the architecture to track logic accurately even when the text is heavily degraded by OCR artifacts.

Shortcomings of NLP in Lexical Similarity While early NLP systems excel at finding topical overlap, they remain structurally deficient for Semantic Conflict Detection. Traditional NLP heavily prioritizes Lexical Similarity—measuring the surface-level token intersection between documents using algorithms like TF-IDF or shallow word embeddings. Bowman et al. [18] explain that this paradigm is fundamentally unsuited for detecting legal contradictions because lexical metrics are entirely blind to deontic logic and syntactic negation. For example, an ordinance stating, "The city permits the sale of fireworks," and a later one asserting, "The city strictly prohibits the sale of fireworks," share near-perfect Lexical Similarity scores. A standard NLP similarity model will cluster them as highly related, completely missing the absolute policy collision happening beneath the vocabulary.

2.9.2 From NLP to Natural Language Inference

Historically, checking legislative consistency relied on Symbolic NLP (hand-written rules) and, later, Statistical NLP. In Statistical NLP, models learned probabilistic weights for specific inputs, like Term Frequency-Inverse Document Frequency (TF-IDF) or N-grams [73]. These models depended heavily on human-engineered features to define grammar rules and categories. While they allowed for soft, probabilistic decisions, they consistently struggled with the deep lexical and semantic ambiguities inherent in legal drafting.

The field largely transitioned to Neural NLP models in the mid-2010s [112]. Unlike statistical systems that treat features as separate entities, neural networks operate end-to-end, mapping raw text directly to output predictions through hidden layers. These architectures capture deeper linguistic nuances than traditional statistical methods by learning complex relationships during optimization. However, while machine learning excels at finding patterns, it can overly mimic its training data, making domain-specific pre-training essential for high performance in rigorous legal tasks.

Within this neural paradigm, Natural Language Inference (NLI) has emerged as the definitive benchmark for evaluating a model's capacity for actual reasoning. Yang et al. [164] define NLI as determining the logical relationship between two text sequences: a "premise" and a "hypothesis." The model must classify whether the hypothesis is true (Entailment), false (Contradiction), or unrelated (Neutral) based solely on the premise. This specific task exposes the core flaw of earlier statistical NLP, which mistakenly equated high word overlap with logical entailment [164]. Modern neural approaches fix this by capturing deep compositional semantics. As Le Hong et al. [80] point out, this allows modern models to resolve complex linguistic phenomena—like coreferences and modality—that rigid rule-based systems historically failed to grasp.

2.9.3 NLI Benchmarks

Identifying the optimal Cross-Encoder for the inference stage requires isolating models with robust, pre-trained deductive logic. Rather than relying on arbitrary accuracy metrics, this study uses established Natural Language Inference (NLI) leaderboards to filter candidates. While these benchmarks primarily evaluate Western jurisprudence, Chalkidis et al. [24] note that a model's baseline performance on these standardized tests is the most reliable indicator of its capacity to parse dense statutory syntax.

GLUE and SuperGLUE Wang et al. introduced the General Language Understanding Evaluation (GLUE) [156] and its more rigorous successor, SuperGLUE [155], as foundational baselines for model reasoning. These frameworks evaluate a Transformer's ability to handle diverse linguistic challenges, such as Recognizing Textual Entailment (RTE) and resolving complex coreferences. While a high SuperGLUE score confirms baseline cognitive capacity, it does not independently guarantee success in the legal domain due to the severe vocabulary shifts present in statutory drafting.

LexGLUE To evaluate specialized legal logic, candidate models are measured against the Legal General Language Understanding Evaluation (LexGLUE) benchmark.

Chalkidis et al. [24] designed LexGLUE specifically to test a model's ability to navigate "legalese," where common words assume highly context-dependent meanings (e.g., "executed" denoting a signed contract rather than a penal death). A model's performance on LexGLUE's entailment tasks acts as a critical proxy for its ability to mathematically weigh legal conditions and exceptions. This proves the architecture has the baseline deductive framework necessary to handle the noisy, uncodified environment of Philippine municipal archives.

2.9.4 Candidate Model Architectures for NLI

The following inference models were selected based on their capacity for deep semantic reasoning and robustness on established logic benchmarks:

XLM-RoBERTa Conneau et al. [31] introduced XLM-RoBERTa as a foundational model for multilingual representation, trained on 2.5TB of text across 100 languages. Nie et al. [102] recognize it for its impressive zero-shot cross-lingual transfer capability, significantly outperforming earlier models on benchmarks like XNLI and XTREME. This architecture is vital for Davao City, allowing the system to project logical relationships across English and Cebuano to ensure local translations remain consistent with the source text.

RoBERTa-ANLI Adversarial NLI (ANLI) is a rigorous dataset designed to intentionally "break" models by exploiting their reliance on simple lexical heuristics [102]. RoBERTa models trained on ANLI demonstrate significantly higher robustness against "trick" inputs, such as premise-hypothesis pairs with high word overlap but contradictory meanings. This ensures the legal conflict detection system performs genuine logical inference instead of just recognizing surface patterns.

FLAN-T5 FLAN-T5 is an instruction-tuned encoder-decoder architecture leveraging over 1,800 diverse task templates to unlock reasoning capabilities for arbitrary instructions. Niklaus et al. [103] detail its specialized legal adaptation, FLawN-T5, which achieved a 50 percent improvement on the LegalBench suite by training on the massive LawInstruct dataset. This framework is highly effective because it can generate an explicit "explanation trace" for its logical predictions, helping human reviewers understand exactly why an ordinance was flagged as a conflict.

DistilBART DistilBART is a compressed variant of the BART architecture, using a "shrink and fine-tune" paradigm to reduce parameters while retaining language understanding. Shleifer and Rush [133] note it is 60 percent faster than full-scale counterparts and excels at zero-shot text classification, treating labels as natural language hypotheses for entailment checking. Its high throughput makes it an excellent candidate for real-time checking during active legislative drafting sessions, where low latency is critical.

DeBERTa-v3 DeBERTa-v3 achieved a landmark milestone as the first single model to surpass human performance on the SuperGLUE benchmark (scoring 89.9 vs. human 89.8). He et al. [59] explain that its Disentangled Attention mechanism processes token content and relative position information using separate vectors. This allows it to capture the subtle syntactic nuances necessary to distinguish legal "rights" from "obligations." By resolving the gradient "tug-of-war" between pre-training objectives, DeBERTa-v3 offers superior sample efficiency and precision for complex Semantic Entailment tasks.

2.10 Ground Truth

In supervised machine learning, a Ground Truth (or Gold Standard) dataset provides the correct, human-verified labels used to train and test models. During training, the neural network uses this dataset as an answer key. It constantly compares its own predictions against these true labels, calculating the difference (or loss) and adjusting its internal weights to make fewer mistakes over time [49]. Without a highly accurate set of target labels, the model will essentially learn the wrong rules, leading to a phenomenon where it memorizes flawed data instead of learning actual reasoning patterns.

Beyond just teaching the model, the ground truth is the only objective way to measure success. Researchers typically split this data into training, validation, and testing sets. The test set acts as the final exam, kept completely hidden from the model until the very end. Evaluating the model against this unseen ground truth allows researchers to calculate standard performance metrics like precision, recall, and F1-score. Without this baseline, it is mathematically impossible to prove that a model generalizes well to new data or works as intended in a real-world scenario [17].

2.10.1 Ground Truth in Natural Language Inference

In Natural Language Inference (NLI), ground truth datasets define the logical relationship between two pieces of text. Annotators read a source text (the Premise) and a claim (the Hypothesis), then categorize their relationship into one of three distinct labels. If the premise guarantees the hypothesis is true, it is labeled as Entailment. If they contradict each other, it is labeled as Contradiction. If there is not enough information to prove either, the pair is labeled Neutral [164].

Most traditional NLI datasets, such as the widely used SNLI (Stanford Natural Language Inference) corpus, are built using crowdsourced workers who write simple variations of everyday sentences [17]. While this approach is cheap and easily generates hundreds of thousands of training pairs, the resulting datasets are incredibly basic and often embed unintended annotation artifacts [56]. They usually feature short, single-sentence premises with obvious logical connections. Training a model exclusively on these simple datasets leaves it entirely unprepared for the dense, multi-layered logic required in specialized fields.

2.10.2 Annotation Procedures and Reliability Metrics

Turning raw text into a useful NLI dataset requires strict guidelines to minimize human bias and prevent annotators from relying on subjective legal interpretations. In standard NLI, guidelines dictate simple factual agreement. However, legal conflict detection requires evaluating normative relationships based on deontic logic. Drawing from the frameworks established by Agotnes et al. [3] and Li et al. [83], the annotation guidelines strictly instruct experts to map the text into obligations (must do), permissions (may do), and prohibitions (cannot do). A "Contradiction" is only flagged if the local ordinance mandates an action the national statute prohibits, or prohibits an action the national statute explicitly permits. If a local ordinance merely adds a regulatory step (e.g., a local processing fee) to a national permit without banning the core activity, the guidelines strictly dictate a "Neutral" classification. This rigid deontic framework guarantees that the dataset measures actual legal preemption rather than just topical friction.

To evaluate the effectiveness of these strict guidelines, researchers must utilize Inter-Rater Reliability (IRR) metrics [95] rather than mathematically flawed percentage agreements. Raw percentage agreement fails to account for statistical probability, where in a multiple-choice classification task, there is an inherent likelihood that annotators will occasionally select the same label by pure random guessing.

The two primary statistical formulas used to subtract this random chance are Cohen's Kappa [29] and Fleiss' Kappa [41]. Cohen's Kappa evaluates agreement between exactly two raters using the following formula:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

where p_o represents the relative observed agreement between the two individual raters, and p_e represents the hypothetical probability of chance agreement.

When a dataset requires a multi-expert consensus—such as utilizing three or more domain experts to establish a majority rule—Fleiss' Kappa becomes the mandatory statistical metric. Fleiss generalized Cohen's formula to accommodate a fixed, larger number of multiple raters:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

In this generalized equation, $\bar{P}$ represents the mean observed agreement across all pairs of raters, while $\bar{P}_e$ represents the expected chance agreement across the entire pool. The resulting κ score is interpreted using the universally accepted statistical matrix established by Landis and Koch [79]:

While objective tasks routinely achieve κ scores above 0.90, applying this expectation to the legal domain is mathematically unrealistic. Bowman et al. [17], in their foundational study for the Stanford Natural Language Inference (SNLI) corpus, established that evaluating semantic relationships is inherently subjective. Their benchmark study hit an approximate performance ceiling with an overall Fleiss' Kappa of 0.70. They explicitly demonstrated that achieving near-perfect agreement (> 0.80) in textual inference usually indicates overly simplistic or fabricated test data.

Table 1: Interpretation of Fleiss' Kappa (κ) Scores

Kappa (κ) Score	**Interpretation**
< 0.00	Poor agreement
$0.01 - 0.20$	Slight agreement
$0.21 - 0.40$	Fair agreement
$0.41 - 0.60$	Moderate agreement
$0.61 - 0.80$	Substantial agreement
$0.81 - 1.00$	Almost perfect agreement

2.10.2.1 Mitigating Sub-Threshold Reliability

While achieving a "Substantial" Inter-Rater Reliability score ($\kappa > 0.61$) validates the clarity of the annotation guidelines, failing to meet this threshold does not inherently invalidate the dataset or halt the experimental pipeline. Klebanov and Beigman [76] establish that a sub-threshold κ score typically indicates systematic ambiguity within the text itself—categorized as "hard" instances—rather than incompetence by the annotators. Their research demonstrates that it is still entirely possible to extract a reliable, low-noise subset from the instances where annotators do achieve consensus.

When researchers encounter this sub-threshold reliability, the mitigation strategy depends entirely on the size of the dataset. For large-scale studies, including the disaster-tweet classification by Ablazo [1], the standard procedure is data omission. Beigman et al. [76] explicitly advocate for filtering out difficult, borderline cases from the gold standard, arguing that evaluating algorithms on inherently ambiguous text introduces statistical noise akin to random coin-flips. In these massive datasets, items that fail to reach majority consensus are simply discarded because the volume of remaining "easy" data easily absorbs the loss.

However, applying this standard omission strategy to a specialized, few-shot legal dataset mathematically destroys the test set's Minimum Detectable Effect. Furthermore, in municipal legislation, these "hard" cases often represent the exact jurisdictional friction the logic engine needs to learn. To preserve dataset volume without introducing benchmarking noise, researchers in specialized domains employ Expert Adjudication rather than omission. Gao et al. [46] establish that in these scenarios, a hierarchical resolution protocol is activated where a senior domain expert reviews the conflicting annotations. The senior expert resolves the ambiguity based on overarching jurisprudential principles and manually establishes the Gold Standard label, treating the initial low κ metric as a measure of text difficulty rather than a rigid experimental blocker.

2.10.3 Ground Truth in the Legal Domain

When applying NLI to the legal domain, the standard approach of comparing short, isolated sentences quickly breaks down. Statutory phrasing is notoriously complex, filled with conditional clauses, exceptions, and domain-specific vocabulary. Consequently, a hypothesis in a legal context rarely aligns perfectly with a single sentence in the premise.

Instead, proving an entailment or contradiction often requires the model to connect evidence scattered across multiple paragraphs or pages.

To address this, researchers have developed specialized legal datasets that reflect real-world document structures. For instance, the Competition on Legal Information Extraction/Entailment (COLIEE) dataset constructs its ground truth by pairing civil code statutes with legal bar exam queries, requiring models to identify specific legal provisions before determining entailment [114]. Similarly, the ContractNLI dataset by Koreeda & Manning [77] engineers its gold standard around non-disclosure agreements. Rather than giving the model pre-cut sentence pairs, ContractNLI forces the system to evaluate hypotheses against entire legal documents, requiring annotators to both classify the logical relationship and extract the specific supporting text span.

Furthermore, unlike general-domain datasets crowdsourced through platforms like Amazon Mechanical Turk, legal ground truth relies heavily on domain experts. Constructing datasets like CaseHOLD [168], which evaluates legal holding statements, requires law students and legal professionals to ensure the labels accurately reflect actual jurisprudence. The push towards comprehensive legal benchmarks, such as LexGLUE [24], demonstrates that evaluating legal NLP models demands rigorously structured, expert-annotated data rather than automated or amateur labeling.

This difficulty is magnified when dealing with local or municipal legislation. While national or federal laws generally follow standardized drafting conventions, municipal texts are highly heterogeneous, exhibiting significant variations in terminology, structure, and overall organization across different jurisdictions [19]. Because of this localized variance, municipal legal documents have historically been neglected in NLP research, hampered primarily by a severe lack of annotated ground truth datasets [20]. Standard legal corpora fail to capture the specific legislative templates, localized governance terminology, or idiosyncratic cross-referencing styles used in municipal laws, such as Philippine local ordinances. Without existing resources that reflect these structural and linguistic nuances, evaluating models on local legislation requires building highly specialized, expert-annotated datasets from the ground up.

A critical architectural decision in dataset construction is determining the exact granularity of the extracted legal premise. While the hypothesis (the local draft) is evaluated as a multi-sentence span to capture context, the premise (the national statute) must be strictly isolated at the sentence level. In Philippine statutory drafting, a single paragraph frequently contains a general rule, multiple conditional modifiers, and distinct exceptions. Holzenberger et al. [61] demonstrate that feeding an entire statutory paragraph into an NLI model as a premise dilutes the model's attention mechanism, causing it to lose track of specific variable constraints. By forcing human annotators to isolate the national rule down to its exact sentence boundary, the resulting Ground Truth prevents the candidate logic engines from getting distracted by surrounding, non-governing text, ensuring the classification mathematically targets the exact point of semantic collision.

The difficulty of dataset construction is severely magnified by the "expert bottleneck" inherent to legal informatics. As Zhong et al. [169] explicitly note, the complex terminology required for legal texts makes manual labeling highly time-consuming and difficult to scale, as it strictly requires domain experts rather than crowdsourced workers. Because constructing massive datasets for localized municipal law is practically unfeasible, researchers must rely on constrained, highly curated Gold Standards. To compensate for this limited data volume, the current NLP standard mandates treating legal conflict detection as a few-shot learning scenario. Gao et al. [45] validate this approach, proving that transformer models can be successfully fine-tuned on specialized datasets containing merely hundreds of examples, provided the evaluation framework rigorously controls for overfitting.

2.10.4 Multi-Sentence Context and Cross-Document Limits

Traditional Natural Language Inference (NLI) evaluates text using a strict 1-to-1 paradigm, comparing a single premise sentence directly against a single hypothesis sentence. However, this highly granular approach fundamentally breaks down in the legal domain due to two specific edge cases. First, intra-document context: a primary rule in one section of an ordinance is frequently modified or negated by an exception buried several paragraphs later (the "1v2" or multi-sentence problem). Second, cross-document dependencies: a contradiction may only emerge when synthesizing provisions from two entirely separate statutes simultaneously.

To address the first edge case, modern legal frameworks utilize Asymmetric Span-Level Pairing. Yin et al. [165] critique traditional sentence-level datasets, proving that capturing contextual exceptions requires evaluating multi-sentence blocks rather than isolated lines. As demonstrated by Koreeda and Manning in the ContractNLI dataset [77], the premise remains a single, focused legal rule, while the corresponding hypothesis is extracted as a multi-sentence chunk. This asymmetric structure allows the model's attention mechanism to accurately evaluate a primary rule against a distant exception contained within the same local provision.

While asymmetric spans successfully capture intra-document context, resolving the second edge case—cross-document multi-hop reasoning—remains computationally prohibitive. Identifying semantic preemption that spans multiple independent laws currently represents the bleeding edge of NLP research and is not yet fully explored for local governance. Recent frameworks have attempted to solve this, but they introduce severe architectural overhead. For instance, Lu et al. [89] address cross-document retrieval by programmatically mapping Entity Graphs to mine evidence paths between laws, a process that is highly resource-intensive and unscalable for unstructured LGU archives. Alternatively, Shen et al. [132] proposed the HopWeaver framework, which relies on massive generative Large Language Models (LLMs) to synthesize complex reasoning paths across documents.

Both of these state-of-the-art solutions immediately violate the strict hardware constraints of consumer-grade LGU workstations. Consequently, practical, edge-deployable systems must intentionally bound their theoretical scope to single-premise, asymmetric span evaluations, deliberately avoiding the quadratic memory scaling required for multi-document fusion.

2.10.5 Train/Test Partitions for Small Datasets

When evaluating NLI models on constrained datasets, researchers often default to k-fold cross-validation to maximize testing coverage. However, Vabalas et al. [146] mathematically demonstrated that applying cross-validation to datasets under 1,000 items introduces severe optimistic bias, causing models to perform significantly worse in real-world deployment than their test scores suggest. To rigorously control for overfitting in few-shot legal scenarios, contemporary evaluation frameworks must abandon k-fold approaches. Instead, they require strict, static Train/Validation/Test partitions (e.g., 70/15/15) to ensure the evaluation metrics remain entirely untainted by training exposure.

Beyond merely preventing optimistic bias, utilizing a static partition allows for rigorous validation monitoring during the fine-tuning phase. In datasets with limited volume, neural networks are highly susceptible to memorizing the training data rather than learning underlying logical patterns. Goodfellow et al. [49] establish that permanently isolating a validation set allows architectures to employ early stopping protocols, halting the training process the moment the model's accuracy on unseen data begins to degrade. Furthermore, as Zhang et al. [166] note regarding few-sample fine-tuning, this static separation ensures strict document-level boundaries, guaranteeing that clauses from a specific municipal draft do not accidentally bleed across the training and testing sets to artificially inflate performance metrics.

2.10.6 Small Test Sets and Minimum Detectable Effect

A standard critique of static splits on small datasets is that the resulting test set lacks the statistical power to reliably compare candidate architectures. However, Card et al. [21] established that a test set's validity is fundamentally determined by its Minimum Detectable Effect (MDE) rather than arbitrary size thresholds. The MDE dictates that if the expected accuracy gap between two models is small, a massive test set is required to prove the difference is not just luck. Conversely, because fine-tuning foundational models on a highly novel, localized municipal domain yields massive margins of performance improvement, the required MDE drops significantly. Consequently, even a constrained test set of approximately 50 pairs is mathematically sufficient to definitively detect the optimal logic engine without sacrificing statistical significance [45].

The practical application of MDE fundamentally shifts how model performance is evaluated in specialized domains. In general NLP research, massive test sets are required because researchers are often evaluating models against benchmarks like GLUE to prove marginal, incremental gains of one percent to two percent over previous state-of-the-art architectures [156]. However, applying a base foundation model—which possesses no prior knowledge of Philippine local governance for example—to a highly specialized task generally results in baseline performance barely above random chance [22]. Zheng et al. [168] demonstrate that in the legal domain, fine-tuning foundational architectures on expertly curated pairs triggers a massive, double-digit jump in classification accuracy compared to zero-shot baselines. Because this expected performance delta is so large, the system does not need thousands of examples to prove the improvement is statistically valid, such that the 50-item test set is perfectly calibrated to mathematically capture this distinct leap in the model's reasoning capacity.

2.11 Explainable Artificial Intelligence

In legal informatics, a "black box" prediction is unacceptable, regardless of its accuracy. A system that flags a conflict between two ordinances must provide the reasoning (*ratio decidendi*) behind its classification to genuinely aid legislative analysis. Without clear justification, human reviewers cannot trust or legally defend the algorithm's output. Consequently, the proposed architecture integrates Explainable AI (XAI) not as an afterthought, but as an intrinsic component of the inference process. This approach prioritizes visualizing the model's actual decision path over generating simplified text summaries.

2.11.1 Intrinsic Explainability Mechanisms

The architecture relies on Intrinsic Explainability via Attention Mechanisms rather than Surrogate Models like LIME or SHAP. These external explainers are computationally expensive and inherently prone to approximation errors because they attempt to learn a separate, simplified linear model around the main prediction. Instead, Attention Weights, or the numerical values the model assigns to specific tokens during inference, can be directly extracted to generate "soft" explanation heatmaps. This native extraction ensures that the explanation perfectly matches the mathematical operations the network actually performed. By bypassing secondary approximations, the system guarantees a much higher degree of structural transparency.

While using attention as a direct proxy for explanation has sparked vigorous debate, it remains the most practical metric for text-based analysis. Jain and Wallace [68] famously argued that Attention Weights do not always correlate perfectly with feature importance. However, Wiegreffe and Pinter [160] provide a crucial counter-argument, contending that Attention Mechanisms offer a "faithful" representation of the model's internal prioritization when rigorously tested. In the context of Semantic Conflict Detection, this specific property allows the system to highlight the exact phrases in Ordinance A (e.g., "absolute ban") and Ordinance B (e.g., "conditional permit") that triggered the logical contradiction. This creates a verifiable visual link between the input text and the output classification, delivering transparency without the massive latency cost of running a separate explanation model.

Complementing this visual transparency is the critical implementation of Confidence Calibration. Deep Learning models, particularly modern Transformers, frequently exhibit extreme overconfidence in incorrect predictions when processing out-of-domain text. As Guo et al. [55] demonstrate, neural networks are inherently poorly calibrated, meaning their unadjusted Softmax probabilities are mathematically deceptive and fail to reflect true predictive uncertainty. To build a reliable risk assessment metric, the proposed system actively mitigates this flaw by applying post-hoc calibration, specifically Temperature Scaling, to the network's pre-Softmax logits. Systematically rescaling these logits before normalizing them successfully aligns the output confidence scores with the actual likelihood of correctness. This crucial transformation replaces illusory confidence with mathematically sound metrics, allowing users to safely filter results between a "clear conflict" (e.g., a calibrated 99 percent probability) and a "potential ambiguity" (e.g., a 55 percent probability).

2.11.2 Edge Deployment Mechanisms

The operational reality of a Local Government Unit (LGU) demands strict adherence to hard hardware and latency limits. Post-hoc explainability methods operate by repeatedly perturbing input data, which requires hundreds or thousands of forward passes per document just to approximate feature importance [118, 91]. Recent empirical evaluations by Kumar et al. [78] confirm that permutation-based explainers increase inference latency by orders of magnitude, making them computationally prohibitive for lengthy legal texts. Furthermore, Wang et al. [157] highlight that deploying heavy post-hoc XAI pipelines on local Edge Servers routinely degrades throughput entirely below acceptable thresholds for real-time analysis. Consequently, adopting Intrinsic Explainability methods, such as extracting native Attention Weights and Softmax logits, is the only viable path to achieve transparency without crashing consumer-grade local hardware.

2.11.3 Metrics for Explanation Quality

Evaluating explainability requires carefully balancing two often-competing metrics: Plausibility and Faithfulness. Plausibility measures how convincing and human-readable the explanation appears to the end user. Conversely, Faithfulness measures how accurately the explanation reflects the model's actual internal reasoning process. Jacovi and Goldberg [67] warn that optimizing purely for Plausibility often leads to misleading explanations that actively mask model biases or hallucinations. In this study, priority is strictly and exclusively given to Faithfulness to ensure legal reliability.

Therefore, the evaluation framework entirely rejects "smoothed" or "generative" explanations that might rewrite the law just to sound simpler. These auto-generated summaries introduce a dangerous layer of interpretation that can subtly alter legislative intent. Instead, the system presents raw Attention Scores directly derived from the mathematical inference layers. While these heatmaps may be noisier to read, they truthfully reveal exactly which tokens the model utilized to reach a verdict of Entailment or Contradiction. This transparent approach aligns perfectly with the rigorous standards of legal scholarship, where the accuracy of the citation always takes precedence over the elegance of the prose.

This strict adherence to Faithfulness directly informs the system's rejection of Generative Explainable AI. Recent trends in legal tech utilize Large Language Models (LLMs) to automatically generate plain-English summaries explaining why a conflict occurred. However, Linna [85] warns that generative models are notoriously prone to "hallucinations," frequently fabricating plausible but entirely fake legal citations or misrepresenting statutory intent when summarizing complex provisions. Jacovi and Goldberg [67] classify these text summaries as highly "plausible" but dangerously "unfaithful," as the generated text does not reflect the actual mathematical operations the model used to classify the text. By restricting the XAI output to raw, intrinsic Attention Heatmaps, the proposed architecture functions as a strict "white-box" system. It does not attempt to explain the law; it visually proves exactly which tokens triggered the contradiction flag, entirely eliminating the risk of generative hallucinations in the final legal audit.

2.11.4 XAI within the Dual-Stage Pipeline

Integrating Explainable AI into a Retrieve-Then-Entail architecture requires deliberately isolating the explanation generation to the second stage of the pipeline. The initial Information Retrieval phase functions purely as a high-throughput, opaque filter designed to handle massive corpus volume. As Anand et al. [7] detail, attempting to extract feature importance during this initial retrieval phase is computationally wasteful and provides little analytical value to the end user. Conversely, the Cross-Encoder stage performs the rigorous Natural Language Inference (NLI) needed to formally confirm a semantic conflict. This makes the secondary stage the most needed and most logical site for generating explainability metrics.

Within this second stage, the system would leverage Raw Computational Transparency instead of outsourcing the explanation to an external text generator. Rather than relying on a Generative LLM to summarize the contradiction, which often prioritizes human plausibility over factual faithfulness [67], the architecture extracts the raw Attention Weights and pre-Softmax logits directly from the Cross-Encoder's internal layers. For the end-user interface, this raw computational data must be mapped cleanly back to the source text to be useful. By overlaying the highest-weighted tokens as visual heatmaps directly onto the text of the ordinances, the system provides immediate, visual proof of the conflict's exact origin [152]. This interface strategy ensures that legislative reviewers do not have to blindly trust a black-box prediction; they can literally see the mathematical decision path the model took to classify the relationship as a contradiction.

2.12 Theoretical Framework

As explored in the earlier sections, the Retrieve-then-Entail architecture has emerged as an ideal theoretical paradigm within the domain of legal informatics, particularly for handling complex statutory reasoning and conflict detection. As demonstrated in the proceedings of the Twelfth International Competition on Legal Information Extraction and Entailment (COLIEE 2025) [48], state-of-the-art systems strategically bifurcate the problem into two distinct computational phases. The initial phase demands a high-recall information retrieval mechanism to rapidly sift through massive legal corpora, while the subsequent phase requires high-precision natural language inference to determine exact logical relationships between the retrieved texts. This theoretical framework establishes a two-phase architecture. It combines a highly scalable, dense Information Retrieval (IR) pipeline to isolate relevant statutory context, followed by a lightweight, encoder-based Natural Language Inference (NLI) pipeline for rapid conflict detection. This structure adapts the methodologies of Team AIIR Lab for statute law retrieval and Team KIS for legal rationale extraction.

2.12.1 Statute Law Retrieval by Team AIIR

The theoretical foundation for the first phase of this system is grounded in a Dense Information Retrieval (IR) architecture, heavily adapted from the top-performing, encoder-based methodologies utilized in modern legal text processing, specifically drawing from the AIIR Lab's approach to statute law retrieval [162]. The primary theoretical objective of Phase 1 is to bridge the semantic gap between a newly submitted document and a massive corpus of existing laws (e.g., 25,000+ statutory documents) without relying on computationally expensive, real-time text generation or manual graph construction. To achieve this, the framework is divided into four core theoretical components: Data Augmentation, Semantic Vectorization, Corpus Pre-computation, and Live Retrieval.

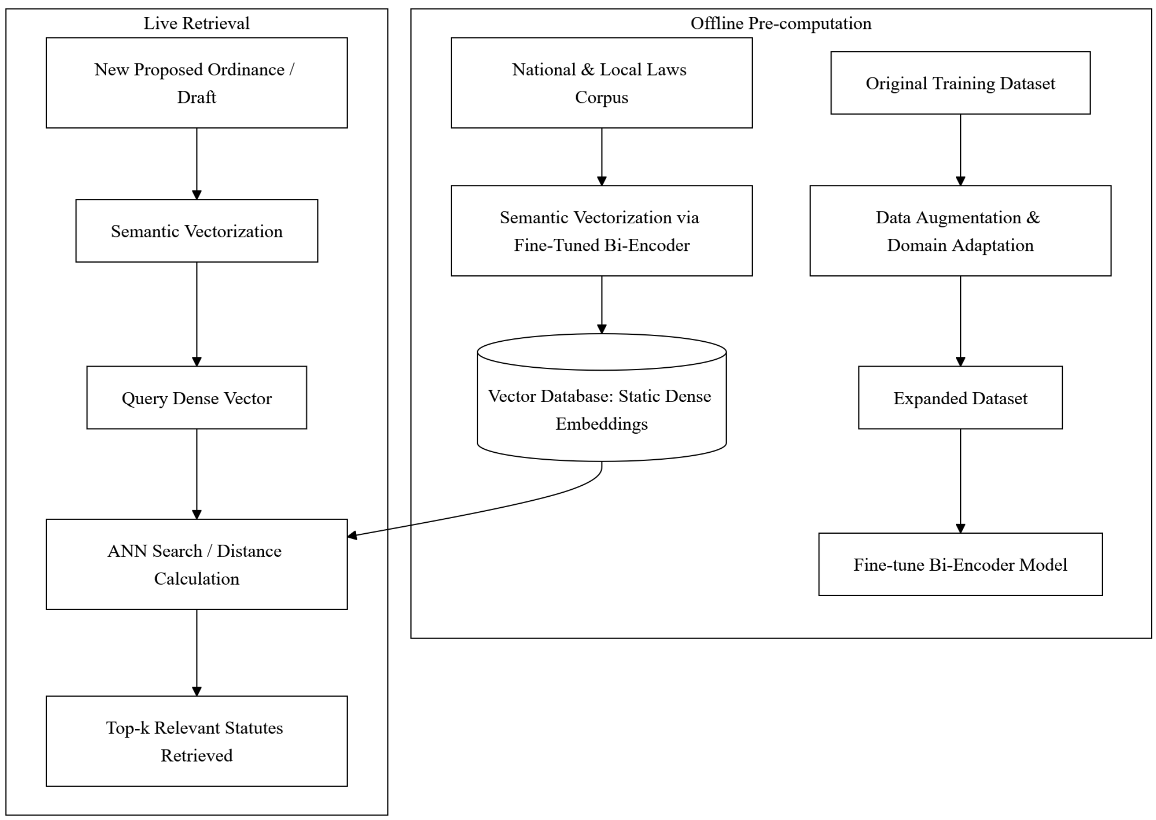

Figure 1: Phase 1 - Statute Law Retrieval (Bi-Encoder Pipeline). Adapted from AIIR Lab's retrieval methodology.

Data Augmentation and Domain Adaptation. By training on this artificially expanded dataset, the retrieval model learns to recognize the same core legal concepts regardless of whether they are expressed in complex statutory jargon or simplified modern text, drastically improving the system's semantic robustness.

Semantic Vectorization via Bi-Encoder. The core engine of Phase 1 is a Bi-Encoder Architecture (such as the all-mpnet-base-v2 model used by AIIR Lab), which maps text into a dense and high-dimensional mathematical space. Unlike cross-encoders, which must read the query and the document simultaneously and are too slow to run across thousands of documents in real-time, a bi-encoder processes the query and the candidate documents independently. The augmented training dataset from the previous step is used to fine-tune this bi-encoder. During fine-tuning, the model learns to adjust its embedding space so that vectors representing a query and its truly relevant legal articles are pulled closer together, while irrelevant pairs are pushed further apart. Optimal model performance is theoretically achieved by evaluating the embeddings using distance metrics such as cosine similarity, Euclidean distance, and Manhattan distance.

Corpus Pre-computation and Storage. In order to satisfy the constraint of executing searches across thousands of documents efficiently, the framework employs an offline pre-computation strategy. Before the system is ever deployed live, the fine-tuned bi-encoder processes the entire corpus of existing national and local laws. It converts every single law, article, and provision into a static, dense vector embedding. These embeddings are stored in a highly scalable vector database. Because the vectorization of the corpus is decoupled from the live search process, the system can effortlessly scale to tens of thousands of documents without requiring heavy real-time GPU compute.

Live Retrieval and Distance Calculation. The final component represents the live user interaction. When a new text (e.g., a proposed ordinance) is submitted to the system, the bi-encoder instantly converts this single document into a dense vector using the same mathematical dimensions as the pre-computed corpus.The system then performs an approximate nearest neighbor (ANN) search, calculating the mathematical distance (e.g., cosine similarity) between the newly submitted vector and the 25,000+ stored statutory vectors. Because this relies entirely on mathematical distance calculation rather than generative text decoding, the system retrieves the top-k most relevant statutes in milliseconds. This purely dense retrieval methodology has been proven to achieve high recall and efficiency, seamlessly narrowing down a massive corpus to a highly focused subset of relevant candidates, perfectly priming the data for the second phase of fine-grained logical analysis.

2.12.2 Statute Law Retrieval by Team JNLP

This framework outlines a highly optimized, three-stage IR pipeline designed to extract relevant statutory articles from a massive legal corpus. Modeled after Team JNLP's methodology for statute law retrieval which achieved the highest F2 score [48], this architecture focuses on progressively narrowing down the search space through three main phases: Pre-Retrieval, Model Fine-Tuning, and Result Ensemble.

2.12.2.1 Pre-Retrieval

The primary objective of the first phase is to drastically reduce the expansive search space while maintaining a near-perfect recall rate. This is achieved through a two-step filtering process that drops low-relevance articles.

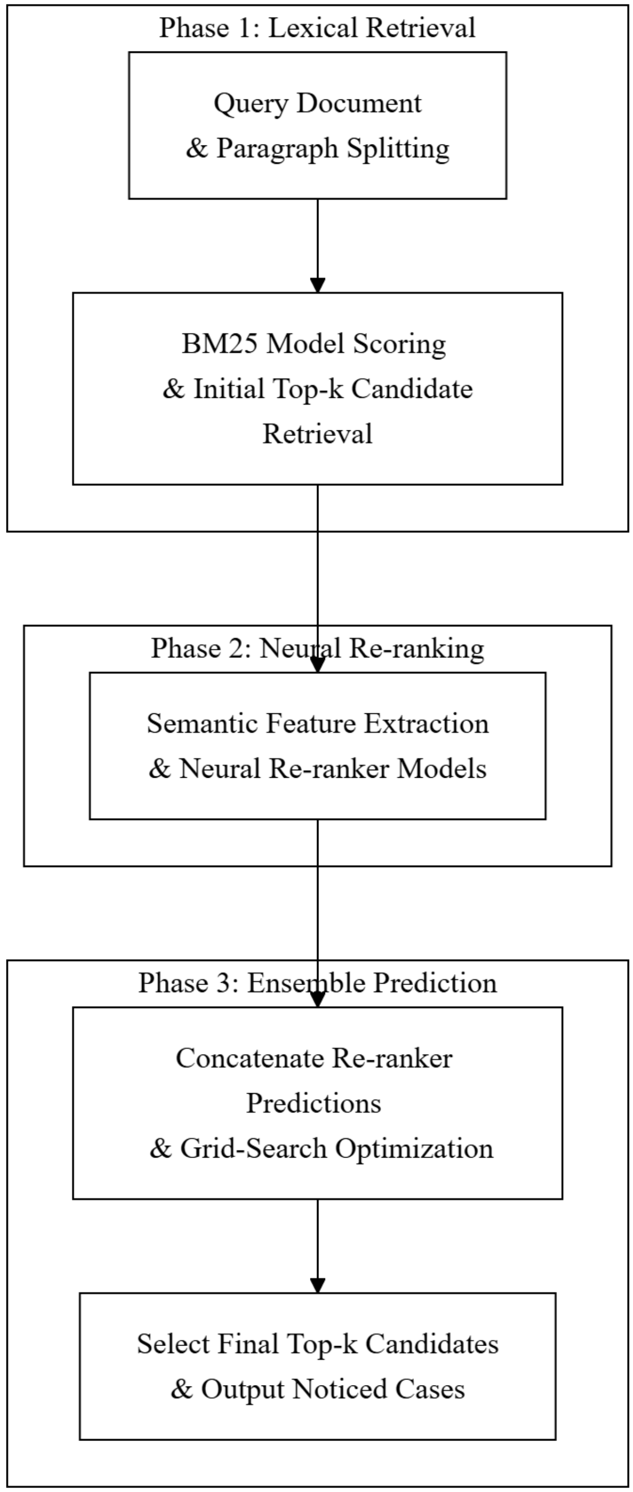

Figure 2: Phase 1 - Statute Law Retrieval (Bi-Encoder Pipeline). Adapted from JNLP's retrieval methodology.

Distance-Based Heuristic Filtering. Dense text vectors for both the proposed query and the corpus articles are extracted using a robust embedding model (e.g., BGE-M3). Instead of relying on direct cosine similarity, the framework calculates the L1 distance between the query vectors and all article vectors, organizing these distances into a predetermined number of bins (e.g., 76 bins). A Gradient Boosting classifier is then trained on these binned distance features to predict relevance. To counter severe class imbalance, the positive (relevant) query-article pairs are heavily over-sampled (e.g., replicated 300 times). The system retains only the top 100 candidate articles with the highest predicted probability.

Passage Re-ranking and Pruning. The initial pool of 100 candidates is further refined to remove the least relevant documents. A specialized passage re-ranker model (i.e. RankLLaMA) evaluates the candidates and prunes the bottom 50, successfully narrowing the search space down to the top 50 most relevant articles without causing a catastrophic drop in overall recall.

2.12.2.2 Model Fine-Tuning

Once the search space is constrained to the top 50 candidates per query, the framework applies a suite of state-of-the-art Large Language Models (LLMs) to perform fine-grained binary classification (determining if a query-article pair is relevant or not).

Parameter-Efficient LLM Adaptation. Because fine-tuning billion-parameter models is computationally prohibitive, the framework employs Parameter-Efficient Fine-Tuning (PEFT) techniques, specifically LoRA and QLoRA. These techniques allow massive causal language models and text embedding models to learn the deep semantic relationships of the legal text with minimal computational overhead.

Class Imbalance Mitigation. The dataset fed into this stage is inherently imbalanced, as only one or two articles out of the 50 candidates are actually relevant. To ensure the models learn to properly identify positive matches, an up-sampling strategy is applied again during training, replicating the positive samples (e.g., 3 times) to balance the ratio of relevant to irrelevant pairs.

2.12.2.3 Result Ensemble

To maximize the robustness of the retrieval system and mitigate the biases of any single LLM, the final phase aggregates the predictions from the batch of models fine-tuned in the previous phase.

Weighted Sum Aggregation. Rather than relying on simple majority voting, the system calculates the final relevance of a legal article by computing a weighted sum of the probability scores outputted by the various fine-tuned LLMs.

Hyperparameter Optimization. The weight assigned to each individual model's prediction is not static, as it is dynamically tuned and optimized using an automated hyperparameter optimization framework, namely Optuna. This mathematical ensembling ensures the highest possible retrieval accuracy by perfectly balancing the strengths of the diverse models in the pipeline.

2.12.3 Statute Law Textual Entailment by Team KIS

The theoretical foundation for the second phase of the system functions as a Natural Language Inference (NLI) pipeline, shifting from broad document retrieval to precise logical evaluation. To maintain high execution speeds and minimize computational overhead, this phase utilizes a lightweight, non-generative encoder architecture, specifically modeled after the ModernBERT-based rationale extraction methodology by KIS [70].

This phase is composed of four theoretical components:

Unified Input Construction. Traditional encoder models face bottlenecks due to short sequence limits, which force long legal documents to be heavily segmented. This framework overcomes that limitation by leveraging the extended context window of modern encoders, which can support up to 8,192 tokens. The retrieved existing law (the premise) and the newly proposed provisions (the hypotheses) are concatenated into a single, continuous text string. Special tokens are utilized to cleanly separate the different legal segments within this unified sequence, ensuring the model understands the boundaries of the premise and the hypothesis.

Token-Level Formulation. To evaluate specific potential conflicts within the proposed text, the framework applies a granular formatting strategy. Special separation tokens (e.g., ¡sep¿) are strategically inserted immediately preceding each individual provision or claim being evaluated within the unified string. This transforms the raw legal text into a highly structured mathematical input tailored for targeted classification.

Simultaneous Binary Classification. The unified sequence is processed by a highly efficient, lightweight encoder (e.g., a 130-million parameter model). Because it is an encoder-only architecture, it completely bypasses the computational overhead of decoding and generating new text. Instead, it performs direct mathematical classification. The model treats each strategically placed $< sep >$ token as an independent classification head. It simultaneously outputs a binary probability score for each provision, indicating whether the proposed text logically conflicts with or is entailed by the existing retrieved law.

Probability Aggregation and Ensembling. In order to prevent overfitting and ensure the highest degree of reliability in its legal reasoning, the framework concludes with an ensemble decision mechanism. Multiple versions of the encoder are trained using diverse hyperparameter configurations (such as varying epochs and learning rates) or different subsets of the training data. During live execution, the predicted probability scores from the top-performing models in the ensemble are aggregated. If the averaged probability across the ensemble exceeds a predefined threshold (e.g., > 0.5), the system outputs a definitive classification flagging a semantic conflict.

2.12.4 Theoretical Significance

This part summarizes how the theories discussed in this chapter will be applied to the development of the conflict detection system. While the previous sections explained these

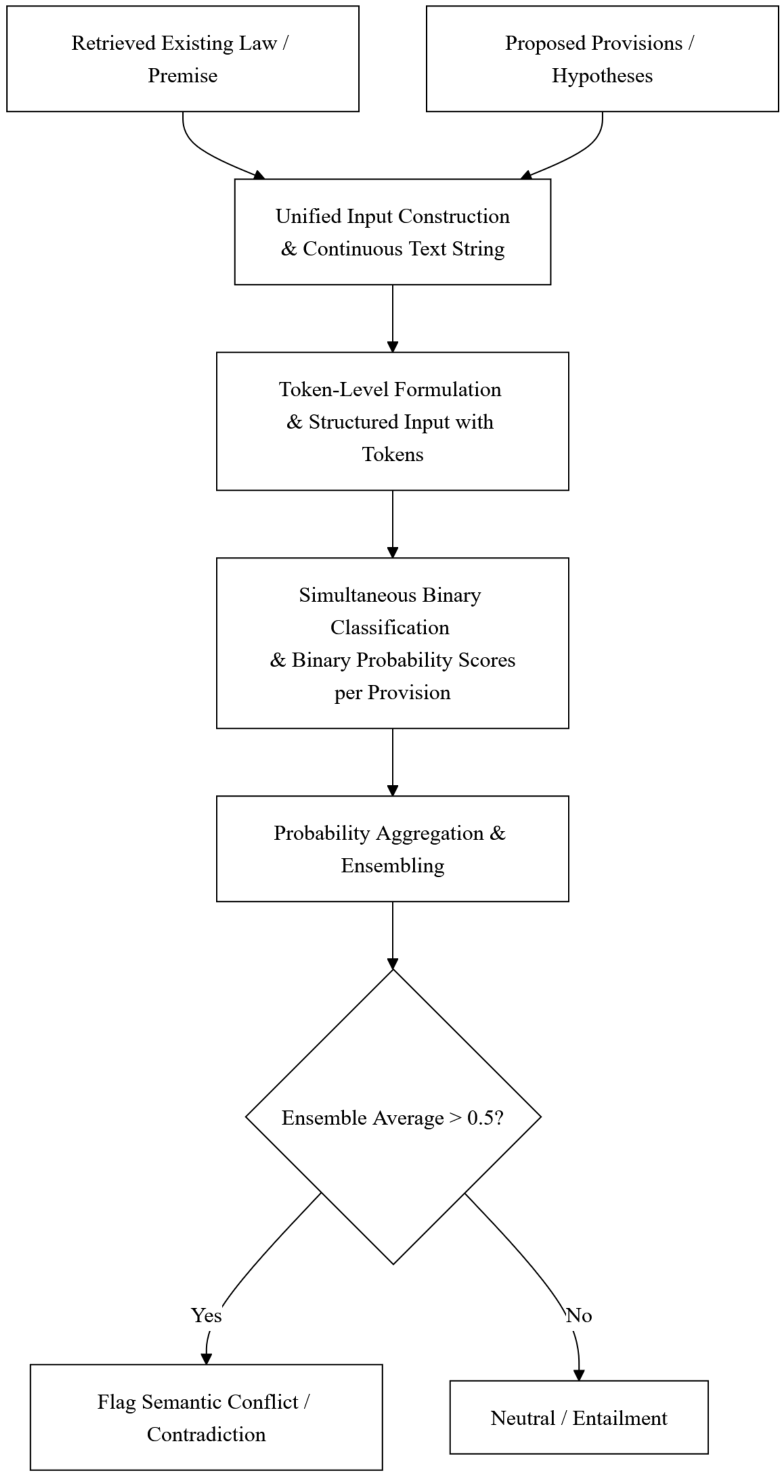

Figure 3: Phase 2 - Statutory Textual Entailment (NLI Pipeline). Adapted from Team KIS's ModernBERT rationale extraction methodology.

models as general research, this synthesis itemizes the specific parts of each framework that will serve as the engine for the Davao City system.

Dense Retrieval Methodology. The study adopts the dense retrieval methodology from the **AIIR Lab** to handle the initial processing of laws [162]. Specifically, the system uses their approach to "Data Augmentation" to fix OCR errors in scanned Davao City ordinances. By following their strategy of "Semantic Vectorization," the system will turn raw legal text into mathematical coordinates (embeddings) that can be searched quickly on the proponents' local hardware [162].

Ensemble Learning Strategy. To ensure the system is both fast and accurate, the study applies the ensemble strategy used by **Team JNLP** [47]. This study adopts their two-step "Lexical and Neural" search process for Stage 1. First, a simple keyword search (BM25) will remove thousands of unrelated laws. Second, their method of "Weighted Sum Aggregation" will be used to combine search scores, ensuring the system picks the most relevant candidates without needing expensive cloud servers [47].

Textual Entailment. For the actual logic check in Stage 2, the study implements the **Team KIS framework** for textual entailment [70]. This study borrows their "Unified Input Construction" method, which combines the draft and the existing law into one long string (up to 8,192 tokens) for the AI to read. Following their "Binary Classification" strategy, the model will judge multiple rules at the same time to see if they logically clash, which is the most efficient way to detect conflicts on standard computers [70].

Synthesis. By combining these three frameworks, the proponents are not just building a search tool, but an automated "diagnostic triage" engine. The AIIR Lab framework provides the clean data, the JNLP framework provides the search speed, and the KIS framework provides the deep logic reasoning. This integrated approach allows the Davao City Sangguniang Panlungsod to perform ex-ante audits that were previously too slow or too difficult to do manually.

Chapter 3

Methodology

This chapter explains the methodology used to build and evaluate the Coarse-to-Fine Semantic Conflict Detection System. It describes how Philippine legal texts are collected and cleaned, and how the two-stage Retrieve-then-Entail architecture is organized. The chapter also details the creation of the ground truth dataset and the selection of candidate models. Finally, it outlines the evaluation metrics and hardware constraints, which are chosen to match the computing capacity of an average Local Government Unit (LGU).

3.1 Conceptual Framework

This section maps how data flows through the proposed system. The architecture follows a basic Input-Process-Output (IPO) logic, broken into a two-step "Coarse-to-Fine" pipeline. By separating the task of searching for documents from the task of checking their logic, the system avoids the hardware bottlenecks that usually cause large language models to fail. The steps below outline how a raw document is processed into a final conflict alert.

3.1.1 Phase 1: Pre-Planning

This initial phase focuses on establishing the system's foundation by preparing the offline database and selecting the best logic model.

Corpus Processing. The system first compiles a knowledge base by gathering 25,432 national laws and local scanned ordinances. Because local municipal archives contain optical character recognition (OCR) errors from scanning, the text undergoes cleaning and normalization. The cleaned texts are then converted into numerical vectors (embeddings) and saved offline in a pre-computed vector database.

Model Training & Selection. To ensure the system evaluates legal logic accurately, candidate Cross-Encoder models are tested against a curated, expert-annotated Ground Truth dataset. The model that achieves the highest accuracy and F1-Score in detecting contradictions is selected as the logic engine for the live system.

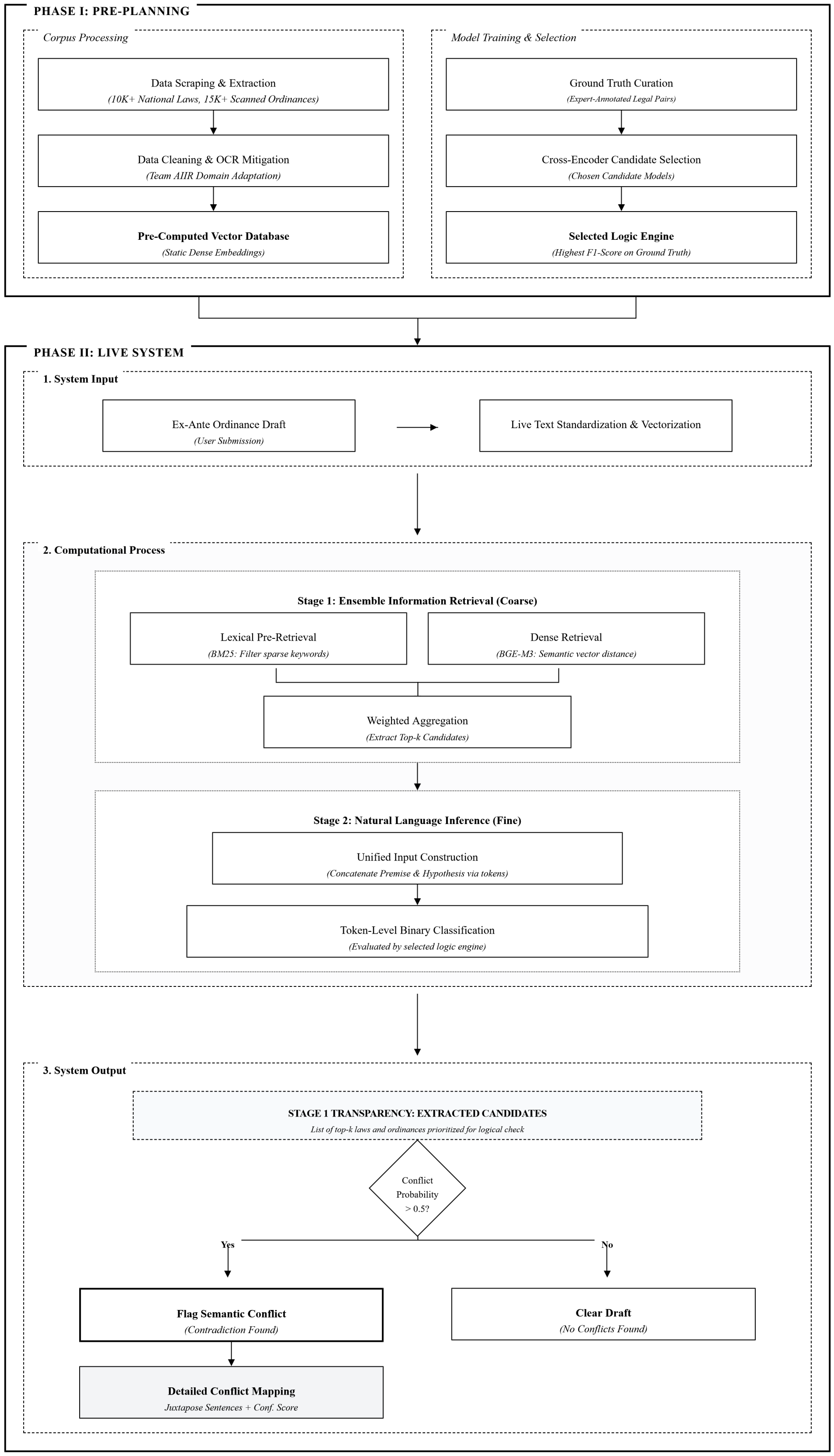

Figure 4: Conceptual Framework

3.1.2 Phase 2: Live System

The live system handles the actual consistency checking and is structured around an Input-Process-Output (IPO) pipeline.

System Input. The live workflow begins when the user submits a new, ex-ante ordinance draft into the system. The system immediately cleans and formats this submitted text so it can be converted into vectors and compared against the pre-computed database.

Computational Process. This part involves two main stages: information retrieval to find relevant laws and textual entailment to perform conflict detection.

Stage 1: Information Retrieval. The system first acts as a search engine to quickly narrow down the large database. It combines a keyword search (BM25) adapted from Team AIIR [162] with a dense retrieval model (Bi-Encoder) inspired by Team JNLP [48]. The keyword search identifies exact statutory citations and numerical penalty amounts, while the Bi-Encoder measures conceptual similarity in vector space. Candidate positions from both retrievers are merged through reciprocal rank fusion and hierarchy safeguards to extract only the top-k most relevant statutory candidates [32]. This hybrid candidate pool provides a focused group of provisions for the logic check without overburdening local computer memory.

Stage 2: Natural Language Inference. The pipeline then moves to logical reasoning using the selected NLI model, following the framework developed by Team KIS [70]. The system joins the proposed draft provisions and the retrieved laws into a single input sequence. Following the classification strategy of Team KIS [70], the lightweight encoder skips text generation and performs direct classification. It evaluates the combined sequence using separation tokens as classification points, producing a probability score for each provision to determine if a conflict exists.

System Output. In the final step, the system delivers a clear result based on the calculated conflict probability. If no conflicts are detected, the system clears the draft. If a contradiction is found, the system displays a conflict report identifying the conflicting sections. To keep the process transparent, an Explainable AI (XAI) attention heatmap highlights the specific words that triggered the conflict alert.

3.2 Research Design and Implementation

This section outlines the methodology for data collection, system engineering, and evaluation. It explains the protocols used to build the pipeline from raw text gathering to the final prototype demonstration.

3.2.1 Research Design

This study uses an applied research design to build a practical software tool for local governance. The researchers use an experimental systems development approach to solve the delays and difficulties of manual ordinance review through automated detection. This setup connects principles from legal informatics and natural language processing to address a real local need. By focusing on practical transformer-based models, the design connects theoretical computer science with everyday government office work. The experiment directly tests whether a retrieve-then-entail pipeline can operate reliably in a local legislative setting.

The experimental setup tests the retrieval and classification modules separately to measure search speed and accuracy independently. Building the system step-by-step allows the researchers to adjust the information retrieval and natural language inference stages using local legal texts. This engineering process ensures that the finished tool accounts for the writing styles found in both city ordinances and national statutes. In this way, applied research methods help turn modern NLP concepts into a usable tool for the Davao City Sangguniang Panlungsod. As a result, the findings establish a clear benchmark for automating pre-enactment legal checks on local legislation.

3.2.2 Corpus Acquisition

Building the legal database requires collecting text from different public repositories so that the system includes as many relevant laws as possible. These sources provide the national statutes and city ordinances needed to check draft legislation for consistency. To keep the language consistent and make vector processing simpler, the system gathers only the English-language versions of all legal texts. This choice matches Philippine legal rules, which give priority to English text if a conflict arises with local language translations. National statutes and local ordinances are gathered using separate automated scripts designed for the format of each source.

3.2.2.1 Corpus Composition

The Lawphil Project serves as the main source for national statutory text because its documents are provided in clean HTML, follow a consistent layout, and offer broad historical coverage [9]. Automated scraping scripts browse through chronological and alphabetical indexes to collect Republic Acts, Batas Pambansa, Commonwealth Acts, Acts, Executive Orders, and Presidential Decrees. Using a custom web extraction pipeline built by the proponents in early 2026, this process collected 25,432 unique legal records enacted between 1900 and 2026. The Official Gazette is used as a secondary check to verify enactment dates and historical amendments [52]. All collected national laws are converted into standardized JSON records with metadata, such as dates of approval, titles, and section numbers.

Collecting national laws exclusively from Lawphil helps avoid the disorganization common to public legal websites. Although the Official Gazette and congressional websites also host statutory records, their web pages frequently change layout and often provide older laws only as image-only PDF scans. In contrast, Lawphil keeps a standard HTML structure across all historical eras, clearly separating introductory preambles from operative legal sections. Scraping from this single repository ensured uniform text formatting across all 25,432 laws, preventing strange formatting differences from skewing vector embeddings during Stage 1 retrieval. The resulting collection provides a clean, consistent text foundation for semantic analysis and conflict detection.

The Legislative Information Support System Program (LISSP) serves as the main source for Davao City municipal legislation [27]. Following formal consultations with the Sangguniang Panlungsod (SP) staff, the proponents received direct access to these municipal records. The resulting local archive contains approximately 1,660 digitized ordinances passed between 1946 and 2025. Unlike national statutes that come in digital HTML format, the local ordinances are mostly scanned PDF files. These scanned files lack selectable text, creating optical character recognition (OCR) errors that must be cleaned before language processing.

3.2.2.2 Corpus Biases

A close look at the collected national laws reveals a large imbalance across legal topics. Over 40% of the 25,432 national statutes consist of routine administrative actions, such as converting local schools, renaming roads, adjusting municipal boundaries, or granting broadcast franchises. In contrast, major regulatory areas that interact with city powers, such as public health, land zoning, and local taxation, make up less than 15% of the total national collection. Searching through this raw distribution without filtering would cause search algorithms to return irrelevant administrative laws instead of actual regulatory rules. To keep search results useful, the system must separate these routine administrative laws from substantive regulatory provisions.

The municipal archive also shows an uneven time distribution and gaps in older records. For ordinances passed before 2021, city record-keeping mainly focused on digitizing major "Landmark Ordinances," leaving ordinary regulations from earlier decades in physical paper storage [127]. As a result, the pre-2021 local records represent a selective collection of prominent laws rather than a complete legislative record. While post-2021 records from the digital LISSP database are much more complete, some numerical gaps remain due to missing committee drafts and unindexed attachments. These gaps mean that the available municipal archive contains more recent ordinances than older ones.

The physical state of the scanned city ordinances adds both visual and linguistic noise. Older ordinances scanned at low resolution suffer from faded print, ink smudges, and crooked pages. The paper copies also contain non-text items such as government stamps, official signatures, and handwritten notes that confuse sentence splitting tools. At the language level, local ordinances sometimes include regional terms and administrative Taglish in their introductory sections [40]. These informal drafting habits differ noticeably from the formal, standardized English found in national laws.

In real-world legislation, direct contradictions between laws are very rare. Lawmakers generally try to write rules that work alongside existing higher laws rather than directly opposing them [43]. As a result, actual legal conflicts make up less than 1% of random pairs formed between municipal ordinances and national statutes. Taking random pairs of text from the unedited collections would lead to an overwhelming majority of unrelated pairs. A machine learning model trained on such unbalanced data could score 99% nominal accuracy simply by guessing "Neutral" every time, while remaining unable to detect real legal conflicts.

3.2.2.3 Corpus Analysis

Across the five main types of legislation with sequential numbers, the collected corpus includes 19,683 statutes out of an estimated 20,246 historical laws, achieving a coverage rate of 97.22%. When combined with the 5,749 Executive Orders from fifteen presidential administrations, the compiled archive holds 25,432 out of about 26,451 total historical legal documents, reaching a national recall rate of 96.15%. Table 2 provides a breakdown of the national corpus, comparing the collected documents with official legislative totals for each statutory type. Figure 5 shows the share of each legal instrument in the archive, illustrating the large volume of executive orders alongside congressional acts.

Table 2: Comparative Statutory Coverage and Composition of the Philippine National Legal Archive ($N = 25,432$)

ID	Statutory Instrument	Corpus	Official Total	Coverage Rate	Corpus Share	Historical Era and Governing Authority
01	Republic Acts (RA)	11,963	12,313	97.16%	47.04%	Post-war and Fifth Republic legislation (1946–present)
02	Executive Orders (EO)	5,749	6,205*	92.65%	22.61%	Presidential administrative issuances (1901–2026)
03	Acts	4,257	4,275	99.58%	16.74%	American Insular Government enactments (1900–1935)
04	Presidential Decrees (PD)	1,844	2,036	90.57%	7.25%	Martial law legislative decrees enacted by executive fiat (1972–1986)
05	Batas Pambansa (BP)	886	889	99.66%	3.48%	Unicameral parliamentary legislation under the 1973 Constitution
06	Commonwealth Acts (CA)	733	733	100.00%	2.88%	Transitional governance enactments under the 1935 Constitution
Sequential Subtotal		**19,683**	**20,246**	**97.22%**	**77.39%**	**All numbered legislative enactments of Congress**
Total Archive		**25,432**	**26,451**	**96.15%**	**100.00%**	**Comprehensive corpus extracted from the Lawphil Project**

*Historical total for Executive Orders represents the aggregate sum of maximum numbered issuances across all fifteen presidential administrations (1935–2026) and colonial governorships (1901–1935).

The coverage rates confirm that the national legal archive is nearly complete. Commonwealth Acts show full 100.00% coverage across all 733 enactments, while American-era Acts (99.58%) and Batas Pambansa (99.66%) miss only 18 and 3 enactments, respectively. For Republic Acts, the 11,963 collected records cover 97.16% of the 12,313 official total, with missing entries mainly being local bills or recent laws still being indexed. For

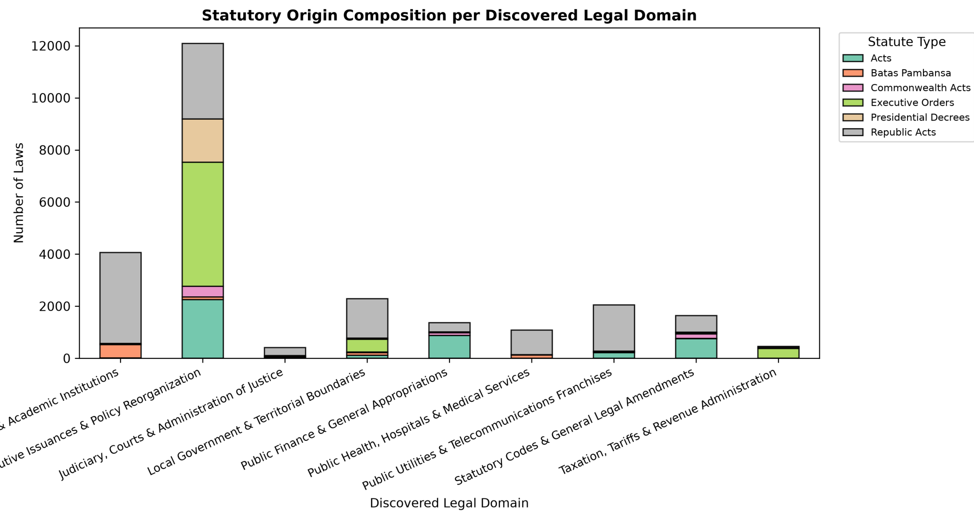

Figure 5: Composition of the Philippine National Statutory Corpus by Legislative Instrument ($N = 25,432$).

Presidential Decrees, the 1,844 gathered records represent 90.57% of the 2,036 numbered decrees from the Martial Law period, where uncollected entries were unpublished directives until the Supreme Court required full publication in *Tañada v. Tuvera* [137]. For Executive Orders, the 5,749 collected records represent 92.65% of the 6,205 numbered orders issued across fifteen presidential terms, offering an extensive record of administrative policy.

This breakdown also explains why executive and administrative orders account for nearly half (48.01%) of the subsequent topic groups. Executive Orders (22.61%) and Presidential Decrees (7.25%) together make up 7,593 documents, representing nearly one-third of the entire national archive. When combined with colonial-era civil administration Acts and agency reorganizations, administrative directives form the largest single category of Philippine legal text. In language processing, this large volume of administrative text poses a challenge during search, as queries can easily match routine bureaucratic language rather than municipal regulations. This reality shows why automated topic grouping is needed to keep local ordinance checks focused on relevant laws.

To trace how the six statutory instruments evolved across Philippine jurisprudence, the proponents categorized statutory enactments across six chronological eras. Each historical era established distinct legislative authorities, ranging from colonial commissions to modern constitutional congresses. Figure 6 charts the chronological succession and aggregate volume of statutory instruments from 1900 through 2026. The distribution reflects the shifting primacy of legislative acts, presidential decrees, and administrative executive orders across political transitions. Examining these enactment waves illustrates how historical governance structures directly shaped the composition of the national legal corpus.

The temporal breakdown reveals that Philippine lawmaking expanded during periods of national administrative reorganization. Early colonial governance relied entirely

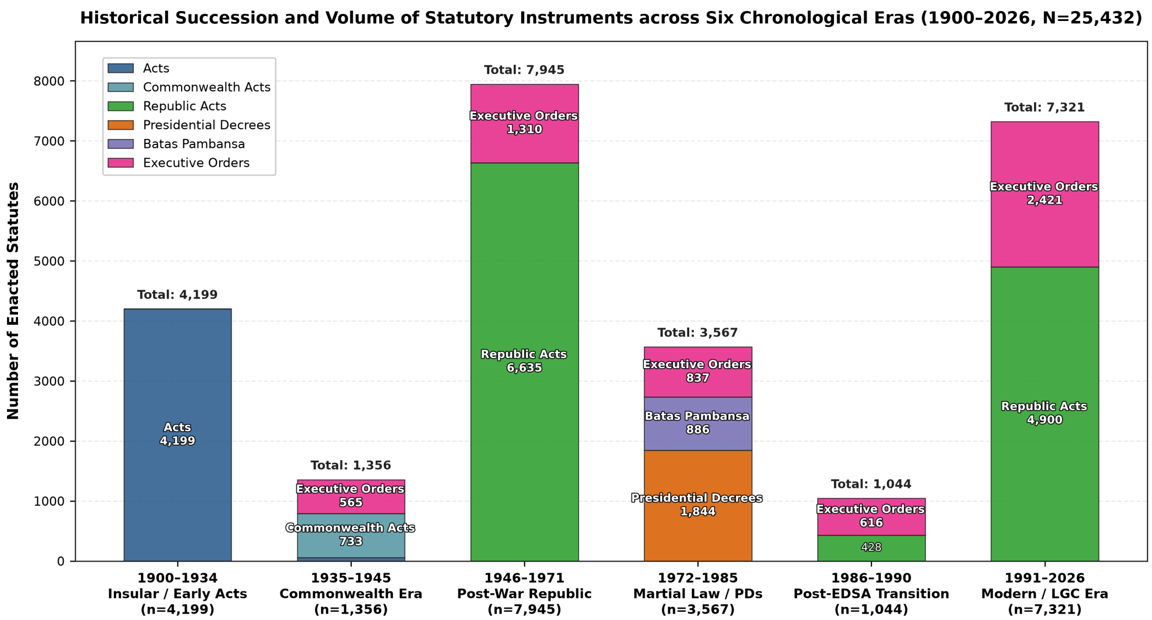

Figure 6: Historical Succession and Volume of Statutory Instruments across Six Chronological Eras (1900–2026, $N = 25,432$).

on 4,199 foundational Acts during the Insular period (1900–1934). Centralized presidential lawmaking predominated during the Martial Law period through 1,844 Presidential Decrees and 886 Batas Pambansa parliamentary enactments (1972–1985). In the modern era governed by Republic Act No. 7160 (1991–2026), statutory production stabilized into a dual stream of 4,900 congressional Republic Acts and 2,421 executive policy orders. Because vocabulary standards shifted markedly across these legislative transitions, search models require chronological awareness to avoid misinterpreting historical statutory phrasing.

3.2.3 Preprocessing and Filtering

Because the raw downloaded files vary widely in formatting, they require thorough cleaning, grouping, and text normalization. While national statutes are already available as clean digital text, the scanned Davao City ordinances contain most of the OCR errors in the dataset. Since these local records come from image-based PDFs, extracting the text often produces misread characters and broken lines. The pipeline uses rule-based filters to find and correct these errors specifically within the local ordinance files. This targeted cleaning ensures that the subsequent vector conversion process is not disrupted by broken words or garbled text.

3.2.3.1 Domain Categorization

During preliminary exploratory research, the proponents conducted initial machine learning experiments on the national statutory archive and identified severe dataset imbalances. In the national collection of 25,432 statutes, routine executive issuances and administrative reorganizations comprise nearly half of all historical enactments, while specialized regulatory domains remain relatively small. Searching for a local regulatory ordinance across this raw, unfiltered collection causes severe search drift because dense retrieval models readily match repetitive administrative phrasing across unrelated legal areas. To resolve these dataset quirks, the proponents formalized a two-stage topic modeling architecture that organizes the statutory archive into cohesive, well-separated subject areas. This structured categorization prevents routine administrative directives from crowding the candidate shortlist during coarse retrieval, establishing a balanced foundation for subsequent conflict checks.

The topic classification pipeline groups documents automatically while running smoothly on standard local computers. First, the 25,432 statutory texts are cleaned by removing HTML tags and routine metadata. The pipeline removes a custom list of 49 common legal stopwords, such as "hereby," "enacted," and "republic," which appear in almost every law without helping identify its specific topic. Second, the cleaned texts are converted into term frequency-inverse document frequency (TF-IDF) feature vectors, capped at a vocabulary of 25,000 single words and two-word phrases. To keep calculations fast on local hardware, the pipeline uses Truncated Singular Value Decomposition (SVD) to project the sparse word vectors into 64 latent semantic dimensions [35], followed by standard L_2 vector normalization.

Grouping legal texts directly into a small number of broad topics risks mixing very different types of laws together. To avoid this problem, the normalized 64-dimensional vectors are first divided into $K = 28$ smaller micro-clusters using MiniBatch K-Means clustering [130]. This initial step allows repetitive laws, such as school conversions or telecom franchises, to gather into their own separate clusters. Next, the pipeline adapts the class-based TF-IDF (c-TF-IDF) method from Grootendorst [54] to find the most representative keywords for each group. By treating all text within a micro-cluster as a single combined document, c-TF-IDF measures how important a word is to that specific cluster compared to the rest of the collection:

$$c\text{-}TF\text{-}IDF_{i,c} = \frac{|t_{i,c}|}{|c|} \times \log\left(1 + \frac{A}{\sum_k |t_{i,k}|}\right) \tag{3.1}$$

where $|t_{i,c}|$ represents the absolute frequency of word i within cluster c, $|c|$ denotes the total word count of cluster c, and A indicates the mean word count per cluster across the full collection.

In order to connect these mathematical clusters to real legal subjects, the extracted c-TF-IDF keywords are matched against a curated Philippine legal taxonomy. This taxonomy reflects the main local government duties and regulatory powers outlined in the Local Government Code of 1991 [50]. Micro-clusters that share the same general legal topic are merged into consolidated macro domains. To remove minor edge cases, the pipeline requires each macro domain to contain at least 1.0% of the corpus ($N \geq 254$ statutes). Micro-clusters that fall below this cutoff are reassigned to their nearest semantic center, ensuring each macro domain has enough documents for reliable search indexing:

$$\mathbf{C}_m = \frac{\sum_{i \in \mathcal{D}_m} \mathbf{z}_i}{\left\|\sum_{i \in \mathcal{D}_m} \mathbf{z}_i\right\|_2}, \quad P(m \mid \mathbf{z}_i) = \frac{\exp(\tau \cdot \cos(\mathbf{z}_i, \mathbf{C}_m))}{\sum_j \exp(\tau \cdot \cos(\mathbf{z}_i, \mathbf{C}_j))} \tag{3.2}$$

where $\mathbf{z}_i$ represents the normalized 64-dimensional SVD vector of document i, $\mathbf{C}_m$ denotes the normalized centroid of macro domain m, and $\tau = 6.0$ functions as the inverse temperature scaling parameter governing prediction confidence.

Running this topic grouping workflow across all 25,432 national statutes organizes the historical collection into eight Consolidated Macro Legal Domains. Table 3 lists the final distribution, showing document counts, percentages, and the most descriptive keywords for each domain. Figure 7 illustrates the relative size of each discovered macro domain.

Table 3: Consolidated Macro Legal Domains of the Philippine National Statutory Corpus ($N = 25,432$)

ID	Consolidated Macro Domain	Statutes	Share	Top Salient *c-TF-IDF* Keywords
00	Executive Issuances & Policy Reorganization	12,210	48.01%	executive, commission, committee, policy, code
01	Education & Academic Institutions	3,492	13.73%	high school, elementary school, campus, college
02	Local Government & Territorial Boundaries	2,134	8.39%	barangay, municipality, boundary, sitio, territorial
03	Public Utilities & Telecom Franchises	2,041	8.03%	franchise, broadcast, telecommunications, radio
04	Public Health, Hospitals & Medical Services	1,890	7.43%	hospital, bed capacity, medical center, infirmary
05	Statutory Codes & General Legal Amendments	1,873	7.36%	administrative code, revised, amend, article, penalty
06	Public Finance & General Appropriations	1,335	5.25%	appropriation, treasury, funds, expenditures
07	Taxation, Tariffs & Revenue Administration	457	1.80%	tariff, customs code, internal revenue, excise
Total Consolidated National Corpus		**25,432**	**100.00%**	**All Enacted Statutes (1900–2026)**

The topic distribution reflects the historical patterns of Philippine lawmaking. Executive issuances, administrative orders, and presidential decrees make up nearly half of the entire statutory collection, reflecting centralized authority during past political transitions. Common municipal subjects, including school infrastructure (13.73%) and public utility franchises (8.03%), form large distinct clusters with repetitive drafting patterns. In contrast, key regulatory topics like taxation (1.80%) and public health (7.43%) occupy smaller, highly focused portions of the database. Separating these regulatory clusters keeps routine administrative orders from crowding the candidate list during Stage 1 retrieval.

To examine the geometric separation of these legal topics, the proponents projected the 64-dimensional SVD document vectors into a two-dimensional latent semantic

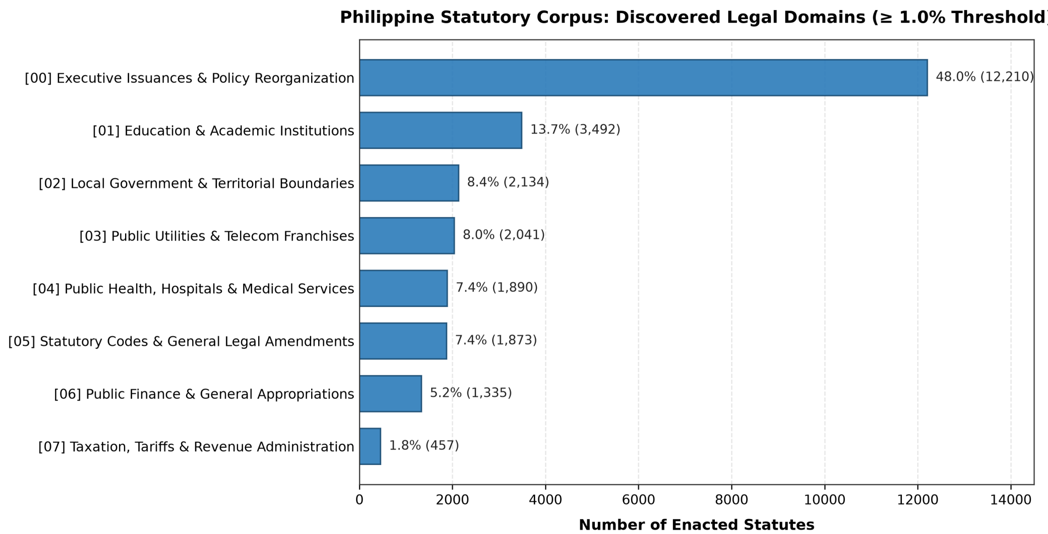

Figure 7: Distribution of the 25,432 Philippine National Statutes across the Consolidated Macro Legal Domains ($\geq$ 1.0% Corpus Threshold).

landscape. Dimensionality reduction maps document similarities onto coordinate axes, allowing direct visual inspection of cluster cohesion. Figure 8 illustrates this topic landscape across a representative stratified sample ($N = 3,500$), highlighting the core clusters and their spatial distributions.

The two-dimensional semantic landscape reveals two distinct structural properties in Philippine statutory text. First, specialized regulatory domains project outward as distinct, isolated rays along the periphery of the vector space. Enactments establishing academic institutions and telecommunications franchises form tight, cohesive rays with minimal dispersion. In contrast, administrative issuances and executive orders cluster densely near the origin, comprising almost half of the entire archive. This dense central mass explains why unconstrained semantic searches frequently drift toward routine bureaucratic orders unless restrained by hierarchical safeguards.

The coordinate axes reflect the two primary orthogonal directions of vocabulary variation discovered through singular value decomposition. The horizontal axis measures operative statutory specificity against general administrative phrasing, where higher positive values indicate detailed technical mandates rather than procedural orders. Conversely, the vertical axis captures functional divergence between social infrastructure such as public schools and commercial infrastructure such as telecommunications franchises. Although the dense cluster on the left appears visually compacted on a two-dimensional plane, the underlying MiniBatch K-Means clustering was performed in 64 dimensions where mathematical boundaries remain well-separated. Projecting 64 dimensions onto a flat plane inevitably collapses higher-order distinctions, yet this dense central mass visually demonstrates why unconstrained semantic retrieval suffers from search drift without domain filtering.

To confirm that the mathematical clusters reflect substantive legal categories rather than shared administrative boilerplate, the proponents examined the most salient vocab-

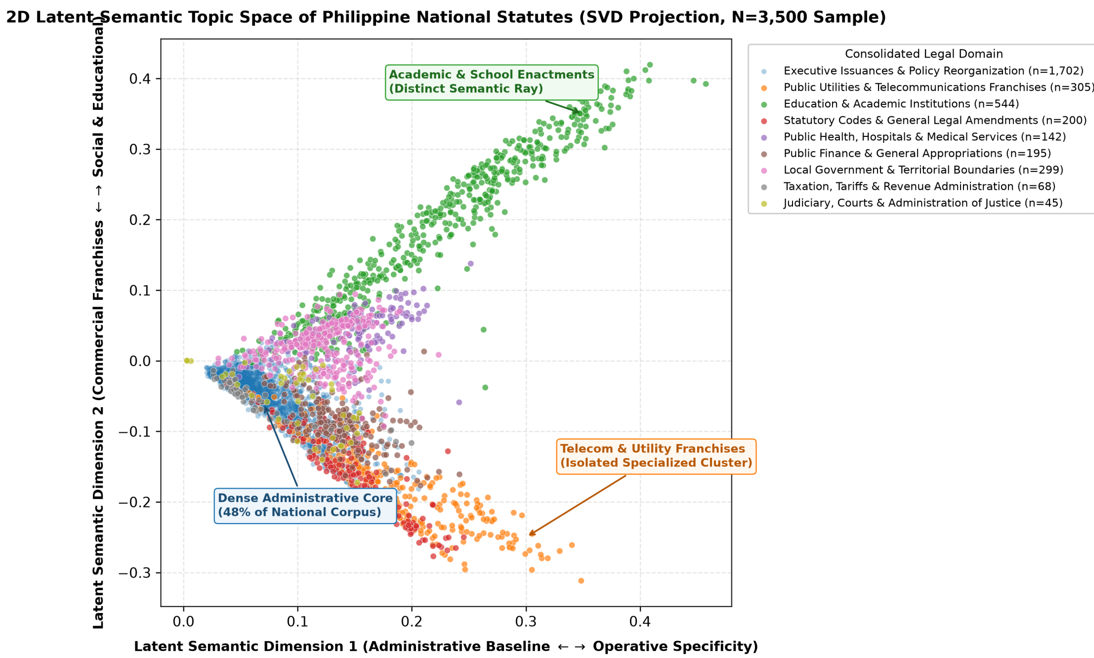

Figure 8: Two-Dimensional Latent Semantic Topic Space of the Philippine National Statutory Corpus (SVD Projection, $N = 3,500$ Representative Sample).

ulary terms discovered by the model. Words that appear frequently across all documents fail to differentiate legal topics, requiring class-based weightings to isolate subject matter. Figure 9 displays the top class-based TF-IDF (c-TF-IDF) keywords across all eight consolidated macro domains.

The vocabulary profiles confirm that each macro domain is defined by operative legal terminology unique to its statutory field. Specialized clusters such as public health are dominated by concrete institutional terms, including "hospital," "bed capacity," and "medical center." Similarly, territorial boundary enactments display spatial cartographic markers such as "sitio," "barrio," and directional bearings. General administrative codes feature broader amendment verbs and statutory cross-references. These distinct vocabulary distributions confirm that the unsupervised clustering algorithm captures authentic legal subject matter across diverse legislative eras.

To examine how individual lawmaking authorities contribute to these eight subject areas, the proponents mapped the six statutory instruments across the consolidated macro domains. In Philippine legislative history, different legal instruments serve distinct constitutional functions, ranging from broad congressional acts to presidential executive directives. Figure 10 presents this cross-allocation matrix, detailing both absolute enactment counts and percentage distributions for each statutory source. This cross-tabulation reveals an explicit division of legal responsibility across Philippine statutory instruments. Analyzing this allocation confirms that regulatory subjects concentrate within specific legislative formats rather than distributing evenly across the archive.

The cross-allocation matrix reveals that congressional enactments and executive orders govern entirely separate legal spheres. Republic Acts account for nearly all spe-

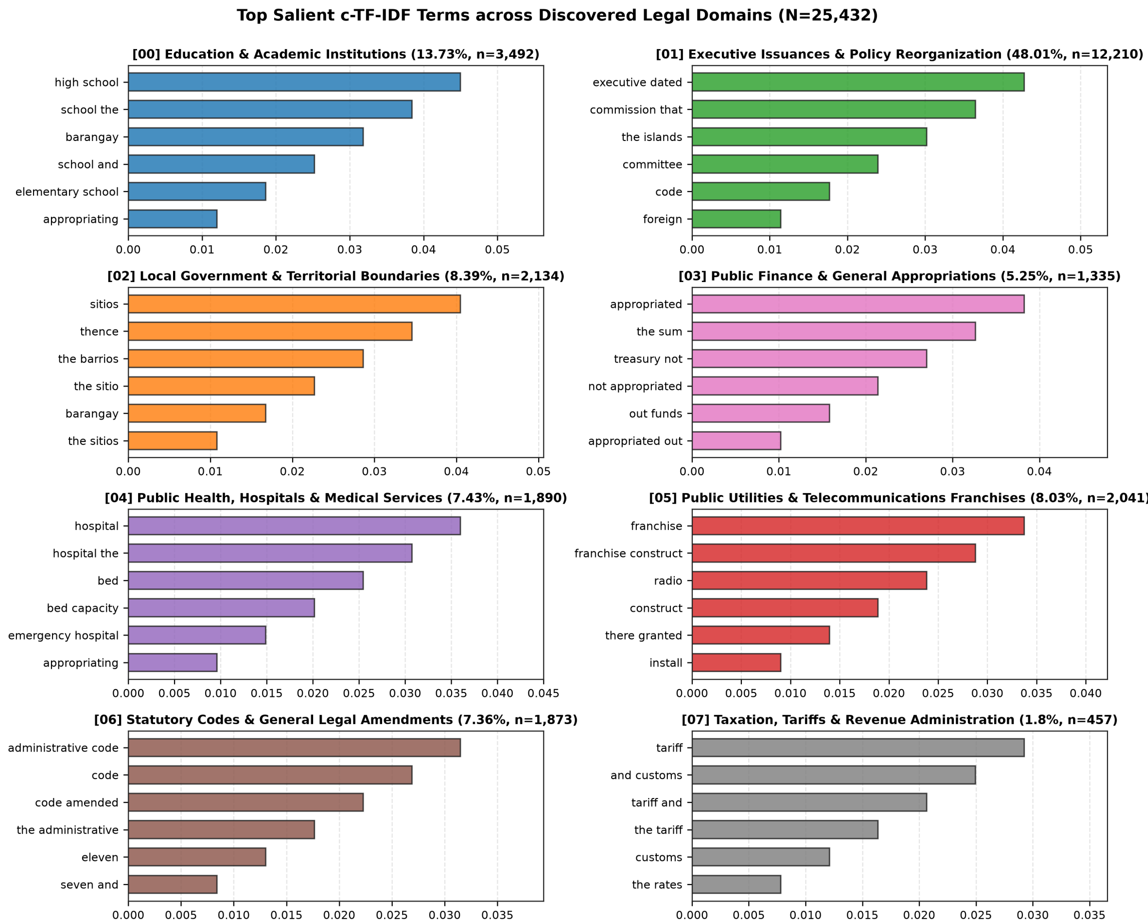

Figure 9: Top Salient c-TF-IDF Keyword Profiles across the Eight Consolidated Macro Legal Domains ($N = 25,432$).

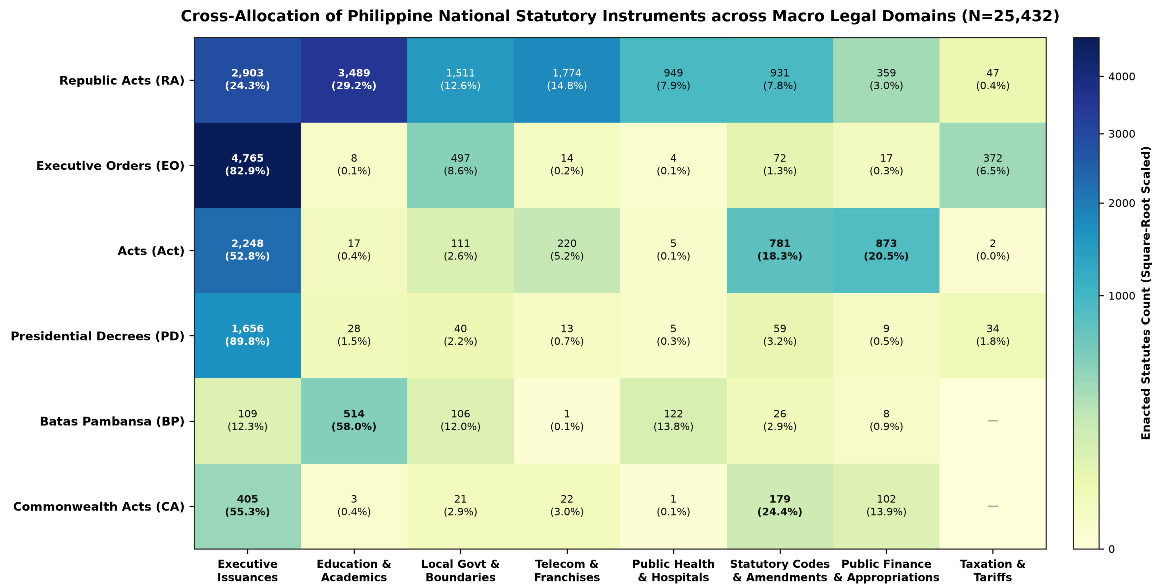

Figure 10: Cross-Allocation Matrix of the Six National Statutory Instruments across the Eight Consolidated Macro Legal Domains ($N = 25,432$).

cialized regulatory measures, containing 99.9% of academic enactments and 86.9% of telecommunications franchises. In contrast, Executive Orders concentrate heavily within administrative policy (82.9%) and presidential tariff adjustments (81.4%). Because municipal ordinances regulate local community behavior rather than internal executive operations, matching draft local clauses against executive orders produces high rates of irrelevant candidate retrievals. These empirical disparities validate using statutory source indicators and domain filters during Stage 1 retrieval to keep candidate shortlists focused on genuine regulatory mandates.

To confirm that the topic pipeline discovered authentic legal boundaries rather than grouping surface vocabulary, the proponents evaluated the model using a stratified 80/20 train/test split. The 80/20 proportion follows the standard Pareto partition for large-scale legal collections exceeding 20,000 documents [2], withholding 20% of the corpus (5,087 statutes) to test generalizability on unseen enactments while training the model on the remaining 80% (20,345 statutes). In statistical testing, an out-of-sample holdout of 5,087 legal texts provides high statistical power with a margin of error below $\pm 1.4\%$, making repetitive cross-validation computationally redundant as vocabulary frequencies and SVD components stabilize asymptotically at this volume. Stratified sampling preserves the natural historical proportions of the eight macro domains, ensuring that smaller regulatory categories like taxation (1.80%, $N = 457$) maintain adequate representation in the test set ($N_{\text{test}} \approx 91$) without class starvation. Table 4 presents the out-of-sample performance scores, default random baselines, literature targets, and empirical verdicts across all evaluation criteria.

The supervised benchmark results confirm that the discovered macro domains generalize reliably to unseen legislation. In an eight-domain classification setting, the uniform proportional chance criterion established by Morrison [97] yields a default random

Table 4: Empirical Holdout Validation Metrics, Baseline Comparisons, and Literature Targets for the Topic Modeler ($N_{\text{train}} = 20,345, N_{\text{test}} = 5,087$)

Validation Metric / Benchmark Test	**Observed**	**Default Baseline**	**Target Range**	**Empirical Verdict**
Out-of-Sample Supervised Accuracy	97.37%	12.50%	$\geq$ 85.0%	Strong Generalization
Out-of-Sample Macro F1-Score	96.26%	12.50%	$\geq$ 80.0%	Balanced Discrimination
Mean Test Cosine Confidence	0.5508	0.1250	$>$ 0.4000	Strong Dominant Probability
High-Confidence Predictions (≥ 0.50)	56.97%	$<$ 20.0%	$>$ 50.0%	Clear Majority Cluster
Holdout Cosine Silhouette Score	0.1602	$\leq$ 0.0000	0.1000–0.2000	Solid Text Separation
Holdout Davies-Bouldin Index	2.4496	$>$ 4.0000	2.0000–3.0000	Compact Cluster Scatter
Explicit Statutory Title Probes (*Unambiguous Indicator Subsets,* $N = 1,259$)				
Telecom Franchise Pattern	97.8%	12.50%	$>$ 90.0%	High Precision ($N = 360$)
Hospital Bed Capacity Pattern	96.6%	12.50%	$>$ 90.0%	High Precision ($N = 203$)
School Infrastructure Pattern	92.2%	12.50%	$>$ 90.0%	High Precision ($N = 645$)
Judiciary & Court Pattern	72.5%	12.50%	$>$ 70.0%	Acceptable ($N = 51$, embedded laws)

baseline accuracy of only 12.50% (1/8). In contrast, the validation classifier reached an out-of-sample accuracy of 97.37% and a Macro F1-Score of 96.26%, surpassing the 85.0% target threshold typically expected for reliable document categorization [2]. To provide an external sanity check beyond learned cluster labels, the evaluation tested four explicit statutory title probes ($N = 1,259$ total) where legislative intent is unmistakable from the statutory catchline. The model achieved 97.8% precision on telecommunications franchises ($N = 360$), 96.6% on hospital bed capacity acts ($N = 203$), and 92.2% on school infrastructure laws ($N = 645$). The lower precision on court reorganization acts (72.5%, $N = 51$) occurs because historical judicial reforms were frequently enacted as embedded chapters within general administrative codes rather than separate court charters.

The probabilistic confidence metrics evaluate how decisively the model assigns unseen statutes to their respective legal domains. In an eight-domain distribution, equal uncertainty across all classes results in a default probability baseline of 0.1250 (12.50%) under standard multinomial chance models [97]. Across all 5,087 holdout statutes, the model achieved a Mean Test Cosine Confidence of 0.5508 (55.08%), exceeding the typical 0.4000 operational target and demonstrating that the top assigned category captured more than half of the total probability mass. Furthermore, 56.97% of all unseen statutes were classified with high certainty ($\geq$ 0.50 probability), well above the majority target threshold of 50.0%. The remaining 43.03% of statutes received confidence scores between 0.25 and 0.49 because Philippine legislation frequently addresses multiple regulatory areas within a single enactment. For example, general appropriations bills and administrative reorganizations distribute their probability across public infrastructure and local governance while still correctly assigning the dominant primary domain.

The geometric clustering metrics evaluate whether the 64-dimensional SVD vector space forms well-separated and compact legal clusters. The holdout evaluation yielded a Cosine Silhouette Score of 0.1602 against the foundational baseline established by Rousseeuw [122], where values near or below zero indicate overlapping, unseparated data points. While synthetic low-dimensional benchmarks often target silhouette scores above 0.50, text-mining literature demonstrates that high-dimensional document clustering rarely exceeds 0.20 due to shared vocabulary across legal topics [2]. In natural language processing, a positive silhouette score between 0.10 and 0.20 provides solid evidence of meaningful cluster cohesion and separation. Similarly, the model achieved a Davies-Bouldin Index of 2.4496, well within the target range of 2.0000 to 3.0000 formulated by Davies and Bouldin [34] and substantially below the diffuse clustering threshold of 4.0000. These geometric measurements prove that the topic model establishes stable, mathematically separated legal boundaries before documents are indexed for coarse retrieval.

3.2.3.2 Preprocessing and Chunking

To determine effective chunking boundaries, the proponents analyzed document and section lengths across the national statutory corpus ($N = 25,432$). Full national enactments span a wide spectrum of text lengths, ranging from brief presidential decrees to massive statutory codes containing hundreds of articles. Direct document-level matching causes extreme context dilution because long enactments far exceed the 512-token maximum sequence length of standard transformer models [162]. Dividing national legislation into natural section units aligns with legal drafting structures where each section forms a self-contained normative command. Figure 11 illustrates the character length distribution of full national statutes alongside the word count granularity of individual section provisions.

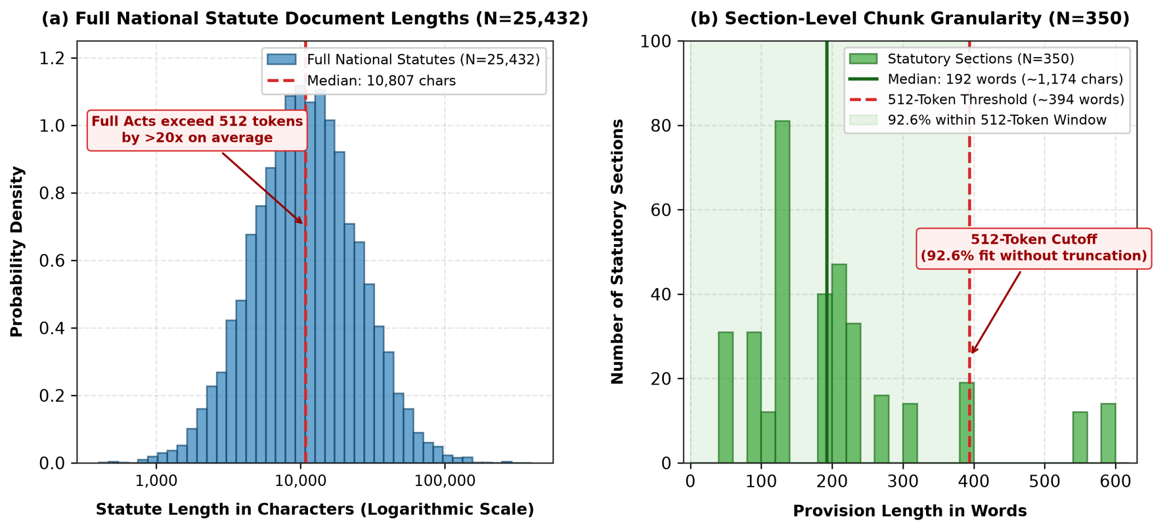

Figure 11: Document Length Distributions and Section-Level Chunk Granularity across the Philippine National Statutory Corpus ($N = 25,432$).

The empirical distributions demonstrate why section-level chunking is necessary for effective statutory retrieval. Across the 25,432 national statutes, full enactments exhibit a median length of 10,807 characters, with comprehensive codifications exceeding one hundred thousand characters. In contrast, isolated statutory sections display a median length of 193 words (1,174 characters), with exactly 92.6% of sections containing fewer than 394 words. Because 394 words translate to approximately 512 subword tokens, over nine-tenths of national provisions fit completely within standard transformer attention windows without truncation. Segmenting legislation at natural section boundaries preserves complete legal rules while eliminating the need for arbitrary sliding windows that fragment statutory conditions.

After establishing the need for granular text units, the preprocessing pipeline cleans local ordinance texts and national statutory provisions. Custom regular expressions find and remove repetitive introductory preamble paragraphs and official signature blocks from city ordinances. Unlike ordinary documents, legal acts follow an organized structure divided into numbered chapters and sections. The researchers divide statutes at the section level rather than using fixed word-count windows [162]. In Philippine legal drafting, each section acts as a complete rule, stating a main legal command along with its specific conditions and exceptions.

Splitting text by an arbitrary word count can easily cut a legal rule in half, separating exceptions from the main requirement. When sub-clauses are separated this way, they lose the context of who must follow or enforce the rule. To preserve complete legal meaning, the preprocessing pipeline adds a structural metadata header to each chunk before saving it in the vector database. Every extracted statutory section begins with a header containing the statute title, section number, and section title:

$$\mathbf{x}_{\text{chunk}} = [\text{Statute Title, Section } S_i\text{: Catchline}] \parallel \mathbf{t}_{\text{verbatim}} \tag{3.3}$$

where $\mathbf{t}_{\text{verbatim}}$ denotes the exact text extracted from Lawphil, and $\parallel$ represents text joining. This prefix helps the Stage 2 Cross-Encoder immediately recognize the governing authority and official context of the provision.

The practical need for these headers is shown by sections within composite legal codes. In Republic Act No. 7160, Section 447 sets penalty limits for municipalities, while Section 458 sets distinct penalty limits for city councils. Without adding the section title and number, downstream Cross-Encoders can confuse different levels of local government and misjudge municipal powers. Adding the structural metadata ensures that the model evaluates each legal clause under its correct administrative level. As a result, the vector embeddings retain both the exact wording of the clause and its broader statutory context.

3.2.3.3 Exception Handling

A key challenge in conflict detection is telling the difference between a real legal clash and an authorized legal exception. In regulatory laws, general prohibitions and permitted exceptions often appear in the same section using conditional phrasing. Because statutory sections average between 200 and 500 words, an entire section fits within the 512-token input limit of candidate Cross-Encoders. The model's attention mechanism can read the general rule and its internal exceptions together, preventing the system from incorrectly flagging valid local exceptions as conflicts. This structure helps keep the legal relationship between general rules and local exceptions clear.

When a general ban and an exception appear in separate sections of a statute, the pipeline relies on the two-stage search process. An ordinance clause that drafts an authorized exception often uses words that match both the general ban and the exception section. The Stage 1 hybrid retriever searches both keyword and vector indexes, helping bring both candidate sections into the top-k candidate shortlist. Stage 2 then evaluates each candidate section individually, allowing the system to identify the authorizing provision and confirm that the local rule is valid. This multiple-candidate review prevents premature conflict warnings on complex laws that place their exceptions in separate sections.

In Philippine statutory construction, related enactments and provisions governing the same subject matter must be interpreted together as a harmonious whole under the doctrine of *in pari materia* [4]. Under this established legal rule, an isolated subsection cannot be understood in a vacuum without the overarching context of its governing section and related statutory exemptions. If an ordinance draft relies on an authorized statutory exemption, evaluating a specific clause like a penalty rule without retrieving the parent exception can trigger an incorrect conflict warning. The system addresses this challenge by prepending hierarchical catchlines to granular sub-clauses during preprocessing and using hybrid retrieval to pull both general bans and specific exception clauses into the candidate shortlist. The Cross-Encoder can then evaluate the draft against the authorizing exception clause and assign an Entailment classification, confirming that the local measure operates within valid legislative authority.

3.2.4 Ground Truth Benchmark Construction

Evaluating candidate Cross-Encoder models in Stage 2 requires a high-quality ground truth dataset. Unlike the broad search phase, the Natural Language Inference (NLI) model needs expert-verified examples to train and measure its classification accuracy. Because standard NLP benchmarks do not include Philippine local laws, a custom dataset had to be created for Davao City. This curated collection provides a clear benchmark for identifying legal conflicts between draft local ordinances and national statutes. In this way, the ground truth dataset connects basic language processing with real-world legal analysis.

3.2.4.1 Span Pairing

To construct a representative ground truth benchmark, candidate statutory premises must reflect the full breadth of Philippine national law. During the exploratory phase of this research, the proponents conducted preliminary machine learning experiments on the national statutes, uncovering severe class imbalances and repetitive drafting formulas across historical eras. These exploratory findings directly informed the consolidated topic modeling architecture detailed in Section 3.2.3.1, which organizes the corpus into eight validated Macro Legal Domains such as local governance, public health, and taxation. The proponents then sampled national statutory sections across these eight domains to establish the fixed premises for the evaluation benchmark. To evaluate multi-sentence legal commands without breaking internal qualifications, the pipeline uses Asymmetric Span-Level Pairing, coupling each statutory premise with a multi-sentence ordinance hypothesis block [165]. This asymmetric structure (1 premise vs N hypothesis sentences) ensures that local exceptions, cross-references, and penalty conditions are evaluated together against the national law [77].

To make the evaluation as realistic as possible, the drafted test sentences closely mirror the writing style of Davao City ordinances. The proponents analyzed municipal laws enacted between 2021 and 2025 to identify standard drafting phrases and administrative office names. Each drafted clause uses standard enacting syntax, such as: "*SECTION X. [TITLE]. — It shall be unlawful for any person, natural or juridical, within the territorial jurisdiction of Davao City, to...*" The draft sentences also name real local enforcement bodies, such as the City Transport and Traffic Management Office (CTTMO), the City Health Office (CHO), the City Environment and Natural Resources Office (CENRO), and the Business Bureau of Davao City. Operational units such as the Anti-Vices Task Force (AVTF) and the Davao City Police Office (DCPO) are also included to reflect genuine local enforcement.

The synthetic ordinance clauses are also linked to real administrative districts in Davao City. Draft rules specify geographic areas from urban centers like Poblacion and Talomo to suburban districts like Buhangin and Toril. Local offices such as the City Disaster Risk Reduction and Management Office (CDRRMO) and the City Social Welfare and Development Office (CSWDO) are included to match actual city administration. This realistic drafting prevents evaluators from reviewing artificial example sentences that differ from real legal texts. As a result, the 15 Sangguniang Panlungsod legal researchers evaluate draft provisions that look like real measures introduced on First Reading.

While the system uses multi-sentence spans to understand local context, it does not attempt cross-document multi-hop reasoning (detecting conflicts that appear only by combining two or more separate national laws). Resolving dependencies across multiple separate laws is an active research challenge in natural language inference. Current systems that attempt this require either complex knowledge graphs [89] or very large generative language models [132]. Running those multi-premise models would exceed the consumer-grade hardware limits set for local government deployment. For this reason, the ground truth benchmark and the inference system focus strictly on single-premise evaluations.

3.2.4.2 Class Balancing and Adversarial Injection

In standard machine learning on tabular data, researchers often handle uneven classes using methods like the Synthetic Minority Over-sampling Technique (SMOTE) [25]. However, the proponents avoid continuous vector oversampling for this legal text pipeline. SMOTE creates artificial examples by interpolating between numerical feature vectors ($x_{\text{new}} = x_i + \lambda(x_{zi} - x_i)$). In high-dimensional language models representing discrete words, averaging or blending token vectors creates artificial representations that confuse attention layers and carry no real legal meaning. Simple word replacement also fails with legal texts because statutory words have exact legal definitions; for example, swapping mandatory words like "shall" with permissive words like "may" fundamentally changes a legal duty.

To build a balanced, reliable Ground Truth dataset, the researchers combined historical legal records with expert-guided text adjustments. First, the researchers gathered documented legal clashes from local government records, including mayoral veto messages, repealed ordinances, and explicit statutory amendments [66]. Second, following the evaluation methods of Koreeda and Manning [77] and Nie et al. [102], legal annotators introduced conflicting rules into real statutory provisions. Annotators reversed permissions, changed numerical zoning limits, and added conflicting penalties into ordinance clauses. This expert-guided process produced a balanced distribution across the Entailment, Neutral, and Contradiction classes in the 350 Ground Truth pairs, preventing class imbalance without creating unnatural legal phrasing.

3.2.4.3 Jurisprudential Difficulty Tiers

Building an effective evaluation benchmark requires pairs with varying levels of difficulty. If all contradiction examples were simple penalty ceiling violations or obvious word swaps, models could score well by memorizing simple keyword patterns. Following the evaluation designs of Nie et al. [102] and Koreeda and Manning [77], the proponents organized the contradiction pairs into three distinct difficulty tiers. This setup ensures that the benchmark evaluates both clear numerical violations and subtle legal disagreements. Table 5 outlines the target proportions and legal examples for each difficulty tier.

Table 5: Stratified Jurisprudential Difficulty Tiers for the 350-Pair Ground Truth Evaluation Benchmark

Difficulty Tier	**Target Share**	**Legal Conflict Mechanism**	**Representative Statutory Example**
Tier 1: Surface & Quantitative	30%	Direct penalty ceiling violations and explicit changes in mandatory words (*shall* converted to *may*).	Local draft imposes a PHP 15,000 fine for traffic obstruction when RA 7160 Sec. 458 caps city penalties at PHP 5,000.
Tier 2: Preemption & Carve-Outs	40%	Exceeding local regulatory powers under *Magtajas*, and blanket local bans that omit statutory exceptions.	Draft ordinance enacts a total ban on commercial freight transit without exempting national agricultural and medical logistics.
Tier 3: Latent & Paraphrastic	30%	Subtle conflicts where the draft redefines statutory terms or creates incompatible rules without using the exact statutory words.	Draft anti-vagrancy ordinance assigns apprehended minors to detention cells, violating non-custodial diversion under RA 9344 without citing the law.

Dividing the contradictions into 30% Tier 1, 40% Tier 2, and 30% Tier 3 adapts the evaluation setup used by Koreeda and Manning [77] for legal inference. In municipal law, jurisdictional clashes and unhonored statutory exemptions (Tier 2) represent the majority of real-world conflicts encountered during committee reviews. Allocating 30% to Tier 1 provides a basic test to verify that candidate models recognize direct numerical limits and mandatory commands. The remaining 30% in Tier 3 tests the model's ability to detect subtle legal disagreements where the local ordinance avoids using the exact words of the national statute. This balance prevents the benchmark from being too easy and sets up a realistic test for the error analysis in Chapter 4.

The legal reasoning behind Tier 3 contradictions is grounded in Supreme Court rulings such as *City of Manila v. Laguio, Jr.* [141] and *Magtajas v. Pryce Properties* [140]. Under these rulings, a local council cannot pass a total ban under the guise of zoning, nor may it create procedures that take away rights protected by national law. To enforce this distinction during testing, the proponents checked all 104 Tier 3 draft sentences to confirm that none contained numbers or monetary amounts. Without explicit numerical clues, Tier 3 requires the Cross-Encoder to evaluate the underlying meaning of the text, such as detecting when an anti-vagrancy draft detains minors in violation of Republic Act No. 9344. This design ensures that the benchmark tests genuine language comprehension rather than simple word matching.

3.2.4.4 Annotation Workflow

Classifying the sampled text pairs requires legal background to ensure accurate labels. The Davao City Sangguniang Panlungsod currently has 24 elected councilors, with eight councilors representing each of the city's three districts [170]. Discussions with SP staff showed that each councilor is assisted by two dedicated legal researchers, forming a team of 48 legal professionals. Because reviewing local legislation is their primary daily responsibility, the researchers recruited 15 of these active legal researchers through availability sampling across the council offices. These annotators were given the sample pairs along with clear guidelines explaining the requirements for Entailment, Contradiction, and Neutral labels.

Following the consensus guidelines adapted from Ablazo [1], a label is only accepted into the final Ground Truth if at least two out of the three assigned reviewers agree. To measure reviewer consistency, the researchers calculate Fleiss' Kappa for Inter-Rater Reliability (IRR). In each three-member sub-panel, the reviewer with the longest service in legislative drafting serves as the Lead Annotator. When all three annotators choose different labels—resulting in no majority agreement—the pair is not discarded. Instead, following the adjudication approach described by Gao et al. [46], a designated Senior Legal Researcher from the SP steps in as Adjudicator to make the final determination.

3.2.4.5 Annotation Codebook

To help the 15 legal researchers apply consistent criteria, the proponents developed a standardized Legal Annotation Codebook. Adapting principles from Koreeda and Manning [77] and Italiani et al. [66], the codebook turns the three-class NLI framework into clear legal decision rules. Each evaluated pair consists of a single national statute section as the premise and a multi-sentence municipal ordinance clause as the hypothesis. Annotators assign one of three labels: Contradiction, Entailment, or Neutral. Table 6 summarizes the working definitions, legal triggers, and example patterns used in the annotation guidelines.

The decision rules for Contradiction directly apply the consistency test established in the Magtajas Doctrine [140]. Because local governments have delegated rather than unlimited legislative powers under Republic Act No. 7160 [50], any local draft that exceeds national statutory limits is legally invalid (*ultra vires*). Annotators look for shifts in mandatory language, such as turning a strict command ("shall") into an optional choice ("may"), or removing an exemption that national law protects. In the test samples, numerical rules are audited against national limits, including tax caps and penalty ceilings. These criteria ensure that conflict warnings highlight actual legal problems rather than minor wording choices.

The guidelines follow a structured process to resolve disagreements among the three panel reviewers. When two or three reviewers agree, the majority label is accepted into the benchmark. In cases where all three reviewers select different categories, the item is sent to the Senior Legal Researcher for final review [46]. The adjudicator reviews the explanations against the codebook to determine whether the difference came from text ambiguity or an overlooked detail. This review step ensures that every item in the 350-pair Ground Truth reaches an authoritative label while preserving the original votes to calculate Fleiss' Kappa.

Table 6: Sangguniang Panlungsod Ground Truth Legal Annotation Codebook and Decision Rules

Logical State	Operational Legal Definition	Primary Legal and Structural Triggers	Representative Statutory Pattern
Contradiction	The municipal ordinance prohibits what national statute permits, permits what statute prohibits, or exceeds statutory ceilings [140].	Clashes in mandatory words (*shall* vs. *may*); penalty limit violations (RA 7160 Sec. 458); territorial or franchise preemption.	National statute caps ordinance fines at PHP 5,000; draft ordinance mandates a PHP 15,000 penalty.
Entailment	The municipal ordinance directly executes, complies with, or derives its regulatory authority from the national premise without adding conflicting mandates.	Compliant devolved implementation; literal statutory adoption; regulatory enforcement within prescribed legislative boundaries.	Local draft establishes a municipal health board strictly adhering to the composition mandated by RA 7160.
Neutral	The national statute and municipal ordinance address the same broad legal domain, but govern independent obligations without mutual conflict or derivation.	Shared topic without direct legal clash; complementary administrative scopes; distinct non-overlapping legal subjects.	National statute establishes hospital licensing standards; local draft regulates municipal market sanitary permits.

3.2.4.6 Batched Annotation Allocation

To prevent reviewer fatigue while keeping the results statistically sound, this study adapts the batched review framework developed by Ablazo [1]. In that earlier study, each item required three independent evaluations to determine a majority label. Managing a collection of 12,054 short texts required over 36,000 individual reviews. By dividing this work among 250 students, the workload was reduced to about 145 items per person. This demonstrated how batched distribution can maintain a three-vote overlap without placing an unrealistic burden on reviewers.

This study applies the same mathematical setup to the legal domain. Because reviewing multi-sentence legal provisions requires much more concentration than reading short messages, the dataset is smaller, but the distribution logic is identical. The workload for the legal reviewers is calculated as follows:

$$
\begin{aligned}
N &= 350 \text{ premise-hypothesis pairs} \\
k &= 3 \text{ independent votes per pair} \\
P &= 15 \text{ legal annotators}
\end{aligned}
$$

$$
\begin{aligned}
W &= N \times k \\
W &= 350 \times 3 \\
W &= 1,050 \text{ total annotations}
\end{aligned}
$$

$$
\begin{aligned}
\text{Work per person} &= \frac{W}{P} \\
\text{Work per person} &= \frac{1,050}{15} \\
\text{Work per person} &= \mathbf{70} \text{ pairs per annotator}
\end{aligned}
$$

To organize this workflow, the dataset is divided into five equal sets of 70 pairs each. The 15 researchers are grouped into five sub-panels of three members, with each sub-panel reviewing one 70-pair set. This arrangement ensures that every pair receives three independent reviews while keeping each reviewer's workload manageable.

To reduce fatigue during the review process, the proponents provided structured reference materials. Reviewing statutory pairs requires significant effort, as reviewers must compare detailed legal clauses against local drafts. Evaluators received spreadsheets with complete statutory text to ensure they did not have to read isolated sub-clauses out of context. Each reviewer also received a one-page summary sheet explaining the *Magtajas* decision steps and statutory penalty limits. These reference guides helped reviewers work consistently across their 70-pair evaluation sets.

3.2.4.7 Inter-Rater Reliability

To measure the consistency of the Ground Truth labels, the proponents calculate Fleiss' Kappa (κ). While raw percentage agreement shows how often reviewers picked the same label, it does not account for agreement that occurs by chance. Fleiss' Kappa isolates genuine agreement by mathematically subtracting the probability of random guessing from the final score.

This metric is used instead of Cohen's Kappa because Cohen's formula is limited to exactly two raters, whereas this study assigns three reviewers to each pair. The resulting score is interpreted using the guidelines from Landis and Koch [79], where a κ score between 0.61 and 0.80 indicates "Substantial Agreement."

Using this target in natural language inference is supported by Bowman et al. [18] in their work on the Stanford Natural Language Inference (SNLI) benchmark. Evaluating text relationships involves human judgment; in their study, Bowman et al. reached an overall Fleiss' Kappa of 0.70 (scoring 0.77 for Contradiction, 0.72 for Entailment, and 0.60 for Neutral). They showed that achieving very high agreement (> 0.80) in language inference is rare and usually happens only with overly simple test sentences. Because

Philippine local ordinances are more complex than general English sentences, expecting agreement higher than 0.70 is unrealistic. Therefore, this study targets a minimum κ threshold of 0.61, which confirms substantial agreement among the legal reviewers.

3.2.5 Experimental Setup and Hardware Constraints

The system runs on consumer-grade hardware to match the computing capacity of an average Local Government Unit. Model training, vectorization, and inference are all conducted on a local workstation equipped with a standard processor and graphics memory. This hardware setup reflects the typical technological environment of Davao City legislative offices. By testing the pipeline on everyday office hardware, the research confirms that the tool can be deployed without requiring expensive cloud computing resources. The system must operate smoothly within these limits to demonstrate practical value.

These hardware boundaries rule out the use of large generative language models. Running generative models with billions of parameters would exceed local memory limits and cause long processing delays. Instead, the pipeline uses lightweight, encoder-only models optimized for quick classification. This constraint allows the system to rely on fast vector calculations rather than expensive text generation. As a result, hardware limitations directly support the choice of a two-stage Retrieve-then-Entail pipeline.

3.2.6 Model Implementation and Pipeline Architecture

The live conflict detection workflow uses a two-stage architecture designed to combine fast searching with detailed logical analysis.

Stage 1: Information Retrieval. The live pipeline begins when a user uploads a draft municipal ordinance. The system first runs the BM25 algorithm to quickly search the statutory database for exact lexical matches. This sparse search captures specific statutory citations, agency acronyms, and numerical penalty limits. At the same time, a dense Bi-Encoder maps the draft provision and the statutes into a continuous semantic space, measuring cosine distance to capture legal meaning that keyword matching overlooks [116]. Rather than relying on fragile score normalization, the pipeline combines the two retrievers using Reciprocal Rank Fusion (RRF) to generate an ordered shortlist of the top-k candidate provisions [32]:

$$\mathrm{RRF}(d) = \sum_{m \in \mathcal{M}} \frac{1}{k_{\mathrm{rrf}} + \mathrm{rank}_m(d)} \tag{3.4}$$

where $\mathcal{M} = \{\mathrm{BM25}, \mathrm{Dense}\}$ represents the candidate retrieval models, and $\mathrm{rank}_m(d)$ denotes the ordinal position of document d produced by model m. The smoothing constant $k_{\mathrm{rrf}} = 60$ follows the standard calibration formulated by Cormack et al. [32] to prevent high-ranking outliers from dominating the fused score. Standard linear score interpolation ($\alpha S_{\mathrm{BM25}} + (1-\alpha) S_{\mathrm{Dense}}$) requires min-max normalization across unbounded BM25 distributions, which fluctuates wildly across diverse queries. In contrast, Reciprocal Rank Fusion operates entirely on ordinal rankings, making the combination robust against differences in score scales. The resulting fused score ranks provisions that demonstrate both lexical precision and semantic relevance at the top of the candidate pool.

Stage 2: Natural Language Inference. After receiving the top-k shortlist from Stage 1, the pipeline moves to logical evaluation using the Natural Language Inference (NLI) module. The system formats an input sequence for each candidate provision, pairing the retrieved national law (the premise) with the draft ordinance (the hypothesis). To guide the model's attention, the pipeline places a $[CLS]$ token at the start of the sequence and a $[SEP]$ token between the two texts. This structure helps the model distinguish the existing law from the proposed draft. The combined text is then processed by the selected Cross-Encoder model. The model's attention mechanism compares every word in the premise against every word in the hypothesis to evaluate their relationship. A classification layer outputs calibrated softmax probabilities labeling the relationship as Entailment, Contradiction, or Neutral. If the contradiction probability crosses the designated threshold, the Explainable AI (XAI) module highlights the conflicting text for user review.

3.2.6.1 Retrieval Safeguards and Failure Modes

Using a two-stage retrieve-then-entail setup introduces potential search challenges when comparing draft ordinances against large statutory collections. If the coarse search misses the governing national law, the inference model cannot detect the conflict regardless of its analytical capability. Legal conflicts often appear through indirect wording, where local drafts use permissive phrasing or leave out specific statutory terms. Standard search tools can also miss isolated prohibitions tucked inside long, otherwise compliant legislative codes. To protect the pipeline against these retrieval errors, the system incorporates five architectural safeguards across data preparation and search. Table 7 summarizes these statutory challenges alongside their operational countermeasures.

These safeguards directly resolve critical failure modes uncovered during preliminary retrieval experiments across the 25,432 national statutes. In particular, attempting a naive hard filter based on predicted macro domains caused candidate Recall@50 to collapse from 43.14% down to 13.43%. This severe degradation occurs because foundational Philippine enactments, such as Republic Act No. 7160, function as omnibus codes that span local administration, disaster response, and municipal taxation simultaneously. Hard-filtering by a single predicted topic immediately prunes the governing code whenever a draft clause touches another administrative area. Replacing hard filtering with a Hierarchy Priority Safeguard shields Republic Acts from the 12,210 administrative executive orders (48% of the corpus), increasing the Mean Reciprocal Rank from 0.1521 to 0.2217 (+45.8% improvement).

To evaluate how candidate retrieval strategies behave in an unconstrained production environment, the proponents benchmarked baseline architectures against the entire national statutory database ($N = 25,432$). This live haystack test evaluates whether sparse lexical matching can isolate governing national laws amidst thousands of administrative issuances and historical enactments. The 350 ground truth queries were submitted directly to the full 25,432-document corpus without manual pre-filtering or provision isola-

Table 7: Architectural Retrieval Safeguards Against Real-World Preemption Failure Modes in Stage 1

Statutory Challenge	Architectural Countermeasure	Operational Implementation Logic
Isolated Statutory Clauses	Structural Section Chunking ($\sim$ 200–300 token windows)	Segments multi-chapter enactments into discrete provisions so isolated prohibitions rank independently without vector dilution.
Indirect or Evasive Wording	Dense Bi-Encoder Semantic Vectors (`all-MiniLM-L6-v2`)	Maps conceptual intent into 384-dimensional latent space, retrieving relevant statutes despite divergent local phrasing.
Numerical Penalty Limits	Sparse BM25 Lexical Matching with Sublinear Scaling	Matches exact statutory citations, numerical penalty limits, and administrative agency acronyms directly.
Rank Variance in Shortlist	Top-k Candidate Safety Buffer ($k = 50$ provision shortlist)	Retains 50 candidate provisions for Stage 2 review, ensuring relevant statutes ranked below the top ten are evaluated.
Omnibus Statute Trap & Administrative Clutter	Hierarchy Priority Safeguard & Soft Domain Priors	Prioritizes primary legislative statutes (Republic Acts) over administrative orders, while applying gentle soft priors instead of destructive hard filtering.

tion. Table 8 reports the empirical retrieval sensitivity and ranking metrics across the full national corpus. These empirical findings demonstrate how unconstrained corpus noise affects retrieval accuracy before provisions reach the Stage 2 inference engine.

Table 8: Empirical Retrieval Sensitivity Benchmark across the Full National Statutory Database ($N = 25,432$)

Stage 1 Retrieval Strategy	Latency	R@5	R@10	R@20	R@50	MRR@50
1. Baseline BM25 (Unconstrained)	329.6 ms	21.43%	29.71%	35.71%	43.14%	0.1521
2. BM25 + Hierarchy Priority	300.1 ms	29.14%	34.86%	39.43%	43.14%	0.2217
3. BM25 + Hard Domain Filter	297.5 ms	4.86%	7.71%	10.29%	13.43%	0.0467

Note: Evaluated on all 350 ground truth queries searching the entire 25,432 national statute corpus on a single local CPU. Latency values represent mean per-query retrieval runtime.

The full-corpus benchmark highlights the severe vulnerability of unconstrained lexical search in large legal archives. Pure BM25 achieves only 29.71% Recall@10 and 43.14% Recall@50, meaning more than half of the governing national statutes fail to reach the candidate shortlist. Applying a hard domain filter further degrades performance, dropping Recall@50 from 43.14% to 13.43% and collapsing MRR to 0.0467 because foundational omnibus laws are discarded. In contrast, incorporating the Hierarchy Priority Safeguard shields Republic Acts from administrative noise, raising the Mean Reciprocal Rank from 0.1521 to 0.2217 (+45.8% precision improvement). These full-database results prove that sparse search alone cannot serve as a reliable standalone retriever, establishing the necessity of dense semantic representations and rank fusion.

Separating Stage 1 retrieval from Stage 2 inference is essential when running on standard office hardware. Feeding full-length statutes directly into a Cross-Encoder requires quadratic computation ($O((L_{\text{ord}} + L_{\text{stat}})^2)$), which quickly exhausts local memory on long legislative texts. Full-text inputs also dilute model attention, burying specific penalty clauses under thousands of routine procedural words. The hybrid retriever in Stage 1 performs a rapid coarse-to-fine filter ($25,432 \to 50 \to 1$), removing over 99.8% of unrelated laws in milliseconds. This separation allows the Cross-Encoder to focus its attention entirely on high-probability candidate provisions.

3.2.6.2 Stage 1 Candidate Retrieval Ablation Benchmark

To empirically determine the optimal Stage 1 retrieval pipeline, the researchers conducted an ablation benchmark evaluating nine candidate configurations against the 350 curated ground truth queries. The evaluation compares sparse lexical retrieval, dense semantic vector matching, and hybrid fusion mechanisms across the three stratified difficulty tiers. Measuring retrieval latency on a single standard CPU workstation confirms whether each configuration meets the computational constraints of local government offices. Table 9 presents the empirical performance metrics across candidate retrieval architectures. Figure 12 illustrates the candidate Recall@k progression curves alongside the performance breakdown across jurisprudential difficulty tiers.

Table 9: Empirical Performance Comparison of Candidate Stage 1 Retrieval Architectures ($N = 350$)

Candidate Retrieval Architecture	**Latency**	**R@5**	**R@10**	**R@20**	**MRR**	**Tier 3 R@5**
1. Pure BM25 Baseline (AIIR)	1.39 ms	79.7%	88.6%	97.4%	0.6113	74.0%
2. BM25 + Hard Domain Filter	1.23 ms	99.1%*	100.0%*	100.0%*	0.7921*	99.0%*
3. BM25 + Soft Domain Prior (+20%)	1.43 ms	87.7%	94.6%	99.7%	0.7122	79.8%
4. Pure Dense Bi-Encoder (MiniLM)	1.33 ms	87.1%	97.1%	100.0%	0.7388	91.3%
5. Hybrid JNLP (Weighted Sum $\alpha = 0.5$**)**	1.23 ms	82.6%	91.7%	98.9%	0.6409	77.9%
6. Hybrid Dense-Biased ($\alpha = 0.7$)	1.20 ms	84.0%	94.0%	99.4%	0.6656	78.8%
7. Hybrid BM25-Biased ($\alpha = 0.3$)	1.22 ms	80.9%	90.6%	98.3%	0.6202	76.0%
8. Reciprocal Rank Fusion (RRF $k = 60$**)**	1.26 ms	88.6%	96.0%	99.7%	0.7040	88.5%
9. Hybrid + Soft Domain Prior ($\alpha = 0.5$)	1.21 ms	89.7%	95.7%	99.7%	0.7328	84.6%

Note: *Configuration 2 demonstrates the artificial isolation effect; when evaluated across the full 25,432 corpus, hard filtering prunes omnibus statutes and causes overall Recall@50 to collapse to 13.43%. Latencies represent mean per-query CPU runtime.

The empirical benchmark results highlight a critical semantic gap between lexical

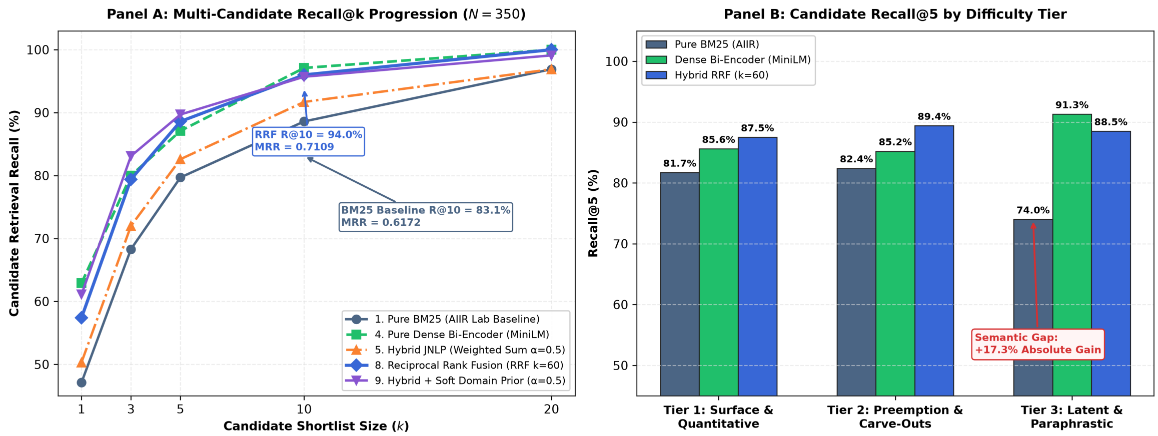

Figure 12: Stage 1 Candidate Retrieval Performance across Shortlist Thresholds and Jurisprudential Difficulty Tiers ($N = 350$). Panel A traces the Recall@k progression curves across candidate models. Panel B highlights the semantic gap in Tier 3 (Latent and Paraphrastic), where dense semantic vectors achieve a +17.3% absolute gain over sparse BM25.

and dense retrieval in legal preemption analysis. While pure BM25 achieves a respectable 81.7% Recall@5 on Surface and Quantitative queries (Tier 1), its performance drops to 74.0% on Latent and Paraphrastic queries (Tier 3) due to statutory vocabulary divergence. In contrast, the Dense Bi-Encoder achieves 91.3% Recall@5 on Tier 3, providing a +17.3% absolute gain by capturing underlying semantic intent even when municipal drafts use sophisticated legislative framing. Reciprocal Rank Fusion successfully combines both approaches, reaching an Overall Recall@10 of 96.0% and an Overall Recall@20 of 99.7% while maintaining a Mean Reciprocal Rank of 0.7040. Furthermore, all evaluated configurations execute in under 1.5 milliseconds on a standard local CPU, confirming that hybrid retrieval delivers high candidate recall without exceeding local government hardware constraints.

3.2.7 Empirical Model Selection and Overfitting Control

Before deploying the system, the researchers evaluate candidate Cross-Encoder models to find the most accurate architecture. Candidate models are tested against the curated Ground Truth dataset. To fit local hardware limits, the candidate pool is restricted to lightweight and memory-efficient models. Despite their smaller size, these models have proven track records in language reasoning tasks. Testing them against a standardized, expert-reviewed benchmark reveals which model best recognizes conflicts in local legislation.

To keep the model evaluation fair and reproducible, the 350 Ground Truth pairs are divided using a static 70/15/15 train, validation, and test split. This split provides 245 training pairs, 52 validation pairs, and 53 test pairs. Working with this sample size places the evaluation in a few-shot learning setting, which works well for pretrained language models [45]. Although k-fold cross-validation is commonly used, Vabalas et al. [146]

showed that fixed train-test splits are preferable for datasets under 1,000 items to avoid overly optimistic results and manage overfitting. A test set of 53 items also meets the statistical sample size guidelines from Card et al. [21], ensuring that the evaluation has enough power to detect real performance differences.

The choice of 350 total pairs is based on statistical power analysis for multi-class evaluations [30]. Under Cohen's power framework, detecting a medium effect size ($w = 0.30$) across three classes at a significance level of $\alpha = 0.05$ requires at least $N = 108$ items for 80% power ($(1-\beta) = 0.80$). Using $N = 350$ pairs provides a statistical power exceeding 0.98, ensuring good coverage across all eight legal domains. These eight domains reflect the basic public services and regulatory powers defined in Section 17 and Section 458 of Republic Act No. 7160. As a result, 350 pairs provide sufficient data for inter-rater validation while covering a realistic cross-section of local government law.

3.2.7.1 Overfitting Control

Because fine-tuning deep transformer models on a small dataset can lead to memorizing training phrasing, the researchers use several regularization techniques. First, model optimization uses the AdamW algorithm with weight decay ($\lambda = 0.01$), as developed by Loshchilov and Hutter [87]. This weight decay keeps network parameters small and prevents individual weights from growing disproportionately large. Second, dropout layers with a rate of $p = 0.1$ are applied across attention layers and the classification head [136]. By turning off a random subset of neurons during training passes, dropout encourages the model to learn broader semantic features rather than depending on specific word patterns.

Third, training uses early stopping based on validation results [111]. The system measures validation loss and the validation Macro F1-Score after every training epoch. If the validation Macro F1-Score does not improve over four consecutive checks (patience = 4), training stops automatically. The system then restores the model checkpoint that achieved the lowest validation loss, preventing the model from fitting too closely to the training data. Finally, the learning rate warms up linearly over the first 10% of training steps before decreasing steadily, which helps keep gradient updates stable during the initial phase of fine-tuning.

3.2.8 Evaluation Metrics

Evaluating the two-stage pipeline requires separate metrics for each operational phase. The coarse retrieval phase in Stage 1 prioritizes candidate recall over final classification, while the inference phase in Stage 2 requires high precision to avoid false conflict alerts.

3.2.8.1 Stage 1 Metrics

Measuring search performance in Stage 1 requires ranking metrics rather than standard classification accuracy [93]. Because the purpose of the coarse retriever is to find candidate laws rather than make the final decision, standard accuracy does not reflect search quality. If the relevant national statute is left out of the top-k shortlist, the downstream Cross-Encoder cannot detect the conflict. For this reason, the retrieval stage is evaluated primarily using Recall at rank k (Recall@k) at thresholds of $k \in \{10, 30, 50\}$. This metric measures the percentage of test queries where the correct national law appears within the candidate shortlist:

$$\text{Recall@}k = \frac{1}{|Q|}\sum_{q=1}^{|Q|} \mathbf{1}\left(\text{rank}(d_q^*) \leq k\right) \tag{3.5}$$

where $|Q|$ is the total number of test queries, d_q^* is the correct national statute paired with query q, and $\mathbf{1}(\cdot)$ is an indicator function that equals one when the target law ranks within the top k candidates.

To measure how high the target law ranks within the candidate list, the researchers use Mean Reciprocal Rank (MRR@k) [93, 120]. While Recall@k treats all positions within the top-k list the same, MRR@k gives higher scores to systems that place the governing statute near the top of the list:

$$\text{MRR@}k = \frac{1}{|Q|}\sum_{q=1}^{|Q|} \frac{1}{\text{rank}(d_q^*)} \cdot \mathbf{1}\left(\text{rank}(d_q^*) \leq k\right) \tag{3.6}$$

In this formula, a target statute retrieved at rank one receives a reciprocal score of 1.0, while a statute retrieved at rank ten receives 0.10. The 350 curated premise-hypothesis pairs serve as the query benchmark for testing Stage 1. By using the 350 draft clauses as queries against the 25,432-statute collection, the proponents measure how often candidate retrieval models locate the paired statutory provision. Calculating Recall@10, Recall@30, and Recall@50 confirms whether the coarse retriever supplies a dependable candidate pool for subsequent natural language inference.

3.2.8.2 Stage 2 Metrics

The classification performance of the models is measured using standard evaluation metrics. Precision is calculated as the ratio of true positive contradiction alerts to the total number of alerts produced by the system. Recall is calculated as the ratio of true positive contradiction alerts to the total number of actual contradictions in the Ground Truth. These metrics show how well the model avoids false alarms while still finding genuine legal conflicts. High precision ensures users are not distracted by incorrect warnings, while high recall ensures critical conflicts are not missed.

To provide a single balanced score for comparing models, the system calculates the harmonic mean of precision and recall, known as the F1-Score. The F1-Score penalizes large gaps between precision and recall, ensuring that the chosen model does not favor one metric over the other. The Cross-Encoder that achieves the highest F1-Score on the annotated pairs is selected for the final pipeline. These evaluations confirm the system's reliability before it is deployed.

3.2.9 Prototype Development

The completed pipeline is deployed as a single-page web application for end-user testing. The user interface is kept intentionally simple, similar to familiar online document conversion tools. Users begin by uploading draft ordinances in standard document formats through a drag-and-drop file area. After submission, the interface shows an active progress indicator while the two-stage search and inference pipeline checks the text against the database. This straightforward layout avoids unnecessary menus, allowing legislative staff to run conflict checks without requiring technical training.

When the analysis finishes, the interface displays a clear results dashboard. To maintain transparency in Stage 1 retrieval, a dedicated section lists the top-k statutory provisions identified by the search. This module allows users to see which national laws were selected before inference took place. When a conflict is detected, the dashboard provides a side-by-side view comparing the specific draft sentence with the conflicting national provision. The report includes a confidence percentage from the inference model, showing why the section was flagged as a contradiction.

The prototype operates as a human-in-the-loop decision-support tool during First Reading committee reviews. Instead of replacing human legal judgment, the tool highlights potential conflicts to help legal staff review drafts faster. An Explainable AI module generates attention heatmaps that highlight the specific phrases that contributed to the conflict score. Sangguniang Panlungsod legal researchers can review these highlighted phrases to decide whether the draft clause represents an invalid conflict or an allowed local adaptation. This workflow allows legislative staff to revise draft clauses before final enactment, helping prevent invalid local legislation.

3.2.10 Methodological Constraints and Future Directions

While the data preparation and training steps address many data challenges, several constraints remain. First, restricting the dataset to English-language texts excludes local Cebuano terms and Taglish expressions that occasionally appear in committee discussions. Second, the municipal collection has more records from 2021 to 2025 because earlier ordinances were only digitized if considered landmark laws. Third, evaluating pairs as a single national premise against an ordinance block (1-vs-N) means the tool does not detect complex conflicts that involve two or more separate national laws. These remaining limitations serve as baseline conditions for the study and provide clear directions for the Future Work recommendations in Chapter 5.

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